Programare Python - Curs 5 2023-2024

Functii (continuare)

Accesul la variabilele definite in afara functiilor variabile globale

Python permite funcțiilor să acceseze variabile definite în afara lor (din domeniul global sau din domenii exterioare). Aceste variabile pot fi citite direct în interiorul funcției. Dacă însă se face o atribuire asupra unei astfel de variabile în interiorul funcției, Python o consideră implicit variabilă locală. Pentru a modifica variabila globală este necesară declarația global, iar pentru variabile dintr-un domeniu exterior (dar nu global) se folosește nonlocal. În cazul obiectelor mutabile, conținutul lor poate fi modificat fără global, deoarece nu se schimbă referința variabilei.

```
m = 32


def findme(a):
  print(m)

findme(12)
```

```
32
```

Daca variabila este declarata dupa apelul functiei, aceasta nu va fi disponibila ca referinta in interiorul acesteia.

```
def findme2(a):
  print(mmm)

findme2(12)
mmm = 32
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_1471/1992285719.py in <cell line: 0>()
      2   print(mmm)
      3
----> 4 findme2(12)
      5 mmm = 32

/tmp/ipykernel_1471/1992285719.py in findme2(a)
      1 def findme2(a):
----> 2   print(mmm)
      3
      4 findme2(12)
```

```
      5 mmm = 32

NameError: name 'mmm' is not defined
```

variabilele globale nu pot fi modificate in interiorul functiei ca si cum ar fi o variabila normala.

```
mm = 32


def findme(a):
  mm = 100
  print(mm)


findme(12)
print('afara: ', mm)

100
afara:  32
```

Python determină dacă o variabilă este locală, globală sau nonlocală în momentul definirii funcției (la analiză/compilare), parcurgând codul pentru a identifica atribuiri și declarații, nu în timpul execuției. Urmatorul cod va produce o eroare, pentru ca instructiunea de atribuire `m=21` forteaza tratarea variabilei m in interiorul functiei ca fiind locala si la momentul la care se executa print() ea nu este definita.

```
m = 32


def findme(a):
  print(m)
  m=21

findme(12)

---------------------------------------------------------------------------
UnboundLocalError                         Traceback (most recent call last)
/tmp/ipykernel_1471/2265294586.py in <cell line: 0>()
      6   m=21
      7
----> 8 findme(12)

/tmp/ipykernel_1471/2265294586.py in findme(a)
      3
      4 def findme(a):
----> 5   print(m)
      6   m=21
      7
```

```
UnboundLocalError: cannot access local variable 'm' where it is not
associated with a value
```

Pentru a modifica variabila globala in itneriorul functiei trebuie sa semnalam acest lucru. Aceasta se face, putin contraintuitiv, local (ca si in PHP de exemplu)

```
m = 32


def changemeingl(a):
  global m
  print(id(m))
  m = 21
  print(id(m))


print(id(m))
changemeingl(12)
print(id(m))
print(m)
```

```
11646344
11646344
11645992
11645992
21
```

Argumente pozitionale si argumente denumite (keyword)

Multe functii in Python permit trimiterea unor argumente specificand direct valoarea acesteia (sau o variabila care o detine) sau specificand o eticheta a argumentului, semnul egal si valoareea argumentului. De exemplu in urmatorul apel al functii print:

```
print(2, 1, 3, 4, sep=' <-> ')
```

```
2 <-> 1 <-> 3 <-> 4
```

primele argumente (2,1,3,4) se numesc argumente pozitionale (datorita faptului ca ordinea lor conteaza si tot ce le defineste in referintele in care vor fi reluate in itneriorul functiei este pozitia acestora) iar argumentul end=' ' este un argument de tip keyword

```
help(print)
```

```
Help on built-in function print in module builtins:

print(*args, sep=' ', end='\n', file=None, flush=False)
    Prints the values to a stream, or to sys.stdout by default.
```

```
    sep
      string inserted between values, default a space.
    end
      string appended after the last value, default a newline.
    file
      a file-like object (stream); defaults to the current sys.stdout.
    flush
      whether to forcibly flush the stream.
```

Daca privim help-ul functiei print observam ca sunt disponibile mnai multe argumente de tip keyword.

Pentru a defini parametrii de tip keyword intr-o functie trebuie sa fie numiti la momentul definirii acesteia

```
def mykwparfunc(one = 1, two = 0):
  print('one: ', one)
  print('two: ', two)
  print(' --- ')

mykwparfunc(one = 10, two = 12) #apel normal
mykwparfunc(two = 12, one = 10) # apel cu argumente inversate
mykwparfunc(50, 112) # apel cu argumente pozitionale
mykwparfunc(112, 50) # apel cu argumente pozitionale inversate
mykwparfunc(70, two = 212) # apel cu argumente mixte, respectand
pozitia
mykwparfunc()
```

```
one:  10
two:  12
 ---
one:  10
two:  12
 ---
one:  50
two:  112
 ---
one:  112
two:  50
 ---
one:  70
two:  212
 ---
one:  1
two:  0
 ---
```

daca incercam un apel cu argumente mixte dar nu respectam pozitia parametrului pozitional, vom capata o eroare:

```
mykwparfunc(70, one = 212) # apel cu argumente mixte, nerespectand
pozitia
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call
last)
/tmp/ipykernel_1471/1681946840.py in <cell line: 0>()
----> 1 mykwparfunc(70, one = 212) # apel cu argumente mixte,
nerespectand pozitia

TypeError: mykwparfunc() got multiple values for argument 'one'
```

Putem forta apelul cu argumente de tip keyword?

Python 3.8 a introdus o modificare relevanta in acest moment. Inainte de python 3.8 declaratia unei functii era astfel:

```
def functie(parametrii pozitionali sau keyword, *,
parametrii_care_sunt_exclusiv_keyword):
```

dupa python 3.8 declaratia astata astfel:

```
def name(parametrii_exclusiv_pozitionali, /,
parametrii_pozitionali_sau_kw, *, parametrii_exclusiv_kw):

def myallparfunc(pone,ptwo, /, one = 1, two = 0, *, three = 0,
four=0):
  print('pone: ', pone)
  print('ptwo: ', ptwo)

  print('one: ', one)
  print('two: ', two)
  print('three: ', three)
  print('four: ', four)
  print(' --- ')


myallparfunc(12,13)
```

```
pone:  12
ptwo:  13
one:  1
two:  0
three:  0
four:  0
 ---
```

```
myallparfunc(12,13,14,15)
```

```
pone:  12
ptwo:  13
one:  14
two:  15
three:  0
four:  0
 ---

myallparfunc(12,13,14,15,16,17)

---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
/tmp/ipykernel_1471/4148942032.py in <cell line: 0>()
----> 1 myallparfunc(12,13,14,15,16,17)

TypeError: myallparfunc() takes from 2 to 4 positional arguments but 6 were given
```

Observam ca:

parametrii dinainte de / sunt exclusiv pozitionali, acestia sunt obligatorii

- Parametrii din mijloc (dintre / si *) sunt semi-pozitionali care au valori predefinite, functia poate fi apelata cu acesti parametrii ca parametrii pozitionali sau ca parametrii de tip keyword
- Parametrii finali (dupa *) sunt parametrii de tip keyword care nu pot fi apelati decat ca parametrii de tip keywords.
- Ultimii doi parametrii (three, four) trebuie obligatoriu sa fie chemati cu numele lor.

La definirea functiei trebuie tinut cont de tipurile de date necesare si de modul de apelare dorit. Parametrii de tip exclusiv keyword pot fi utilizati pentru a specifica o lista de valori predefinite a caror modificare poate fi dorita sporadic. Aceasta ii face sa poata fi evitati de la majoritatea apelarilor, fara consecinte asupra executiei functiei datorita existentei valorilor predefinite.

Functii cu numar variabil de argumente

Sunt situatii in care nu stim exact cate argumente va primi functia. Putem defini functii care vor primi un numar variabil de argumente

Functii cu numar variabil de parametrii pozitionali

```
def suma(*numere):
    suma = 0
    for num in numere:
        suma = suma + num
    return suma

print(suma(12,34,56))
```

```
print(suma(12))
print(suma(12,34,56,54,67,897,23,54,23))
```

```
102
12
1220
```

Ar fi interesant de vazut exact ce tip are variabila "numbers". Acest tip este tuplul. Putem verifica, rescriind functia astfel incat sa imprime tipul variabilei "numere"

```
def suma_t(*numere):
    print (type(numere))
    suma = 0
    for num in numere:
        suma = suma + num
    return suma

print(suma_t(12,34,56))
print(suma_t(12))
print(suma_t(12,34,56,54,67,897,23,54,23))
```

```
<class 'tuple'>
102
<class 'tuple'>
12
<class 'tuple'>
1220
```

Argumentele trimise catre functie sunt puse intr-un tuplu iar acesta este "despachetat" in interiorul functiei. Despachetarea permite variabile cu stea, variabile care vor acumula tot ce ramane neatribuit in cele dinaintea lor

Functii cu numar variabil de parametrii de tip keyword

O functie poate astepta un numar variabil de parametrii de tip keyword.

```
def calculate_total_cost(price, **kwargs):
    tax_rate = kwargs.get('tax_rate', 0.05)
    discount = kwargs.get('discount', 0)
    shipping_cost = kwargs.get('shipping_cost', 5.0)

    total_cost = price + (price * tax_rate) - discount + shipping_cost
    return total_cost

# Example usage
price = 100.0
total_cost = calculate_total_cost(price, tax_rate=0.08, discount=10.0,
shipping_cost=8.0)
print(f"Total cost: {total_cost}")
```

```
Total cost: 106.0
```

Evident, ca si in cazul parametrilor pozitionali variabili, o lista variabila de parametrii, se poate inlocui cu un singur parametru de tip dictionar

Functii ca "first class citizens"

"first class citizen" inseamna ca functiile au aceleasi "drepturi" ca variabilele. Adica functiile pot fi atribuite variabilelor si pot fi returnate din alte functii.

In exemplul urmator, o functie este folosita ca parametru al altei functii

```
def add_numbers(a, b):
    return a + b

def subtract_numbers(a, b):
    return a - b

def calculate(operation, a, b):
    return operation(a, b)

# Example usage
result1 = calculate(add_numbers, 10, 5)
result2 = calculate(subtract_numbers, 10, 5)
print(f"Result1: {result1}")
print(f"Result2: {result2}")
```

```
Result1: 15
Result2: 5
```

Functiile pot fi atribuite unor variabile si apelate ulterior cu ajutorul acestora

> Add blockquote

```
def add_numbers(a, b):
    return a + b

my_function = add_numbers

result = my_function(5, 10)
print(result)
```

```
15
```

Functii pot returna alte functii:

```
def create_multiplier(n):
    def multiply(x):
        return x * n
    return multiply
```

```
# Example usage
double = create_multiplier(2)
triple = create_multiplier(3)

print(double(5))
print(triple(5))
```

```
10
15
```

LEGB

LEGB este un acronim care arata ordinea in care Python cauta un nume

- Local - functia curenta
- Enclosing - functia imediat superioara, daca exista
- Global - la nivelul modulului
- Builtin - namespace global

LEGB se refera la urdinea cautarii NUMELOR - atribuirile au un set suplimentar de reguli

```
x = "global"

def outer():
    x = "enclosing"

    def inner():
        x = "local"
        print(x)

    inner()

outer()
```

```
global
```

Functii lambda

Functiile lambda sunt mici functii anonime de care este nevoie in anumite operatii (ex: sortare).

Functiile de tip lambda nu pot contine instructiuni de atribuire, blocuri if, for, while, return

```
# declaratia unei functii lambda
lmsalut = lambda : print('Hello World')

# apelul unei functii lambda
lmsalut()
```

```
Hello World
```

Functiile lambda sunt limitate la o singaura expresie, ele sunt menite unor utilizari foarte simple, in special in expresii asociate altor functii.

```
numbers = [1, 2, 3, 4]
squared = list(map(lambda x: x**2, numbers))

print(squared)

[1, 4, 9, 16]
```

Evident, putem obtine acelasi efect folosind o functie denumita

```
# Defining a named function to square all items in a list
def square(x):
    return x**2

squared2 = list(map(square, numbers))

print(squared)

[1, 4, 9, 16]
```

Exemple de utilizare a functiilor lambda

```
#Filtrarea numerelor pare dintr-o lista

numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
filtered_list = list(filter(lambda x: x % 2 == 0, numbers))
print(filtered_list)

[2, 4, 6, 8, 10]

conducatori = [
     { "nume": "Lulu", "conduce": ["bicicleta", "masina"] },
     { "nume": "Judith", "conduce": ["trotineta", "vapor"] },
     { "nume": "Alladin", "conduce": ["trotineta", "role",
"bicicleta"] },
     { "nume": "Heidi", "conduce": ["avion", "role", "bicicleta"] }]



biciclisti = list(filter(lambda x: "bicicleta" in x['conduce'],
conducatori))

print(biciclisti)

[{'nume': 'Lulu', 'conduce': ['bicicleta', 'masina']}, {'nume':
'Alladin', 'conduce': ['trotineta', 'role', 'bicicleta']}, {'nume':
'Heidi', 'conduce': ['avion', 'role', 'bicicleta']}]
```

De cele mai multe ori, in special in cazul listelor, dictionarelor... etc. lambda sunt inlocuite cu expresii de tip comprehension

```
biciclisti = list(filter(lambda x: "bicicleta" in x['conduce'],
conducatori))

print(biciclisti)
```

```
[{'nume': 'Lulu', 'conduce': ['bicicleta', 'masina']}, {'nume':
'Alladin', 'conduce': ['trotineta', 'role', 'bicicleta']}, {'nume':
'Heidi', 'conduce': ['avion', 'role', 'bicicleta']}]
```

care sunt mai flexibile

```
biciclisti = [x["nume"] for x in conducatori if "bicicleta" in
x["conduce"]]

print(biciclisti)
```

```
['Lulu', 'Alladin', 'Heidi']
```

Pentru a obtine doar numele folosind functii lambda trebuie sa crestem complexitatea constructiei:

```
biciclisti = list(
    map(lambda x: x["nume"],
        filter(lambda x: "bicicleta" in x["conduce"], conducatori))
)

print(biciclisti)
```

```
['Lulu', 'Alladin', 'Heidi']
```

utilizarea functiei map pentru a aplica o operatie asupra tuturor elementelor unei liste

```
numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
doubled_list = list(map(lambda x: x * 2, numbers))
print(doubled_list) # [2, 4, 6, 8, 10]
```

```
[2, 4, 6, 8, 10, 12, 14, 16, 18, 20]
```

utilizarea functiei "reduce" pentru a obtine un rezultat sumar pe baza elementelor unei liste

functia reduce asteapta o functie lambda cu doua argumente iar aplicarea acesteia este urmatoarea: primele doua elemente ale listei sunt supuse operatiei din functie, apoi rezultatul operatiei si cel de-al treilea element sunt supuse operatiei, apoi rezultatul si cel de-al patrulea... etc.

```
from functools import reduce

numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
```

```
suma = reduce(lambda x, y: x + y, numbers)
print(suma)
```

```
55
```

Recursivitate

Python permite recursivitate. Celebrul factorial:

```
def factorial(n):
    if n == 0:
        return 1
    else:
        return n * factorial(n-1)

# Example usage
print(factorial(5)) # Output: 120
```

```
120
```

cel mai mare divizor comun

```
def cmmdc(a, b):
    if b == 0:
        return a
    else:
        return cmmdc(b, a % b)


print(cmmdc(24, 36))
```

```
12
```

Closures

Inchiderile (closures) sunt un tip de functii care sunt returnate de alte functii si care contin si mediul in care au fost apelate (variabilele definite la momentul definirii functiei).

O inchidere implica in general existenta unei functii exterioare si a unei functii interioare care va fi returnata.

```
def outer_func(x):
    def inner_func(y):
        return x + y
    return inner_func

add_five = outer_func(5)
```

```
result = add_five(10)
print(result)
```

```
15
```

Variabilele definite in afara functiei interioare sunt retinute si dupa finalizarea executiei functiei outer_func.

```
def outer_func(x):
    alfa = 21
    def inner_func(y):
        return x + y + alfa
    return inner_func

add_five = outer_func(5)

result = add_five(10)
print(result)
```

```
36
```

Decoratori

```
def my_decorator(func):
    def wrapper():
        print("Before function call")
        func()
        print("After function call")
    return wrapper

@my_decorator
def my_function():
    print("Function being called")

my_function()
```

```
Before function call
Function being called
After function call
```

In exemplul urmator putem folosi decoratorul petnru a masura durata executiti unei functii

```
import time
import math

def calculate_time(func):

    def inner1(*args, **kwargs):
        begin = time.time()
        func(*args, **kwargs)
```

```
        end = time.time()
        print("Total time taken in : ", func.__name__, end - begin)

    return inner1

@calculate_time
def factorial(num):

    time.sleep(2)
    print(math.factorial(num))

factorial(10)

@calculate_time
def collatz_sequence(n):
    sequence = [n]
    while n != 1:
        if n % 2 == 0:
            n = n // 2
        else:
            n = 3 * n + 1
        sequence.append(n)
    return sequence

collatz_sequence(27)
```

```
3628800
Total time taken in :  factorial 2.002150297164917
Total time taken in :  collatz_sequence 4.38690185546875e-05
```

Un alt exemplu foarte util este un sistem de cache al rezultatelor unor operatii care dureaza foarte mult.

```
import time

def cache_decorator(f):
    cache = {}
    def wrapper(x):
        if x in cache:
            print(f"din cache: square({x}) = {cache[x]}")
            return cache[x]
        print(f"calculez: square({x})")
        result = f(x)
        cache[x] = result
        return result
    return wrapper


@cache_decorator
def square(x):
```

```
    time.sleep(2)
    return x * x


print(square(8))
print(square(5))
print(square(8))
print(square(3))
print(square(5))
```

```
calculez: square(8)
64
calculez: square(5)
25
din cache: square(8) = 64
64
calculez: square(3)
9
din cache: square(5) = 25
25
```

Exista chiar un o functie pentru aceasta

```
from functools import lru_cache

@lru_cache
def square(x):
    time.sleep(2)
    return x * x

print(square(8))
print(square(5))
print(square(8))
print(square(3))
print(square(5))
```

```
64
25
64
9
25
```

Decoratoarele pot fi inlantuite

```
def printegal(func):
    def inner(*args, **kwargs):
        print("=" * 15)
        func(*args, **kwargs)
        print("=" * 15)
```

```
    return inner


def printminus(func):
    def inner(*args, **kwargs):
        print("-" * 15)
        func(*args, **kwargs)
        print("-" * 15)
    return inner


@printegal
@printminus
def salutare(msg):
    print(msg)

salutare("Hello")
```

```
===============
---------------
Hello
---------------
===============
```

Generatori

Un generator este un tip special de functie care este apelata o singura data si poate returna valori de mai multe ori.

Aceasta se face prin returnarea unui obiect de tip generator care ruleaza functia din nou de cate ori este apelat.

Denumirea de generator vine de la utilizarea acestor functii pentru generarea unor secvente care presupun fie calcule complexe (si generarea intregii liste ar opri programul) fie pot fi extrem de variabile si nu are sens sa se genereze o lista foarte mare degeaba.

Exemplu de generator:

```
def count_up_to(n):
    i = 1
    while i <= n:
        yield i
        i += 1
```

Se observa ca nu exista alte cuvinte cheie implicate in definirea generatorului, acesta este "ghicit" de compilator datorita prezentei cuvantului "yield" in stanga expresiei.

Apelul acestuia se face intr-o prima faze similar cu apelul oricarei functii, apoi se foloseste functia next pentru a obtine restul elementelor

```
mg = count_up_to(10)

print (next(mg))
print (next(mg))
print (next(mg))

1
2
3
```

Daca executam de mai multe ori celula cu print(next) vom obtine valori succesive, generatorul ramane in rulare. Pentru a reinitializa generatorul trebuie sa rulam din nou si celula cu

```
    mg = count_up_to(10)
```

Obiectul generator si metodele lui

```
print(mg.__class__)
print(dir(mg))

<class 'generator'>
['__class__', '__del__', '__delattr__', '__dir__', '__doc__',
'__eq__', '__format__', '__ge__', '__getattribute__', '__gt__',
'__hash__', '__init__', '__init_subclass__', '__iter__', '__le__',
'__lt__', '__name__', '__ne__', '__new__', '__next__', '__qualname__',
'__reduce__', '__reduce_ex__', '__repr__', '__setattr__',
'__sizeof__', '__str__', '__subclasshook__', 'close', 'gi_code',
'gi_frame', 'gi_running', 'gi_yieldfrom', 'send', 'throw']
```

Se observa ca interpretorul a captat automat intentia de a construi un generator si, desi cuvantul return nu exista in corpul functiei, a returnat ceva. Acel ceva este un obiect instantiat din clasa generator.

Metodele apelabile sunt urmatoarele

- **close** Inchide generatorul inainte de vreme. Nu e necesara (cum este la fisiere) si functioneaza prin generarea unei exceptii de tip GeneratorExit. Utila pentru generatori infiniti (fara conditie interioara de oprire)
- **gi_code** Obiectul de cod care contine functia generator (bytecode-ul). Spre deosebire de metoda
- **code** disponibila pentru orice functie,
- **gi_code** permite vizualizarea starii curente a generatorului
- **gi_frame** Referinta catre frame-ul curent din memoria masinii virtuale unde se gaseste generatorul. Similara cu **code**.co_frame dar si aici cu informatii suplimentare necesare pentru rularea generatorului
- **gi_running** Variabila booleana care arata daca generatorul ruleaza in momentul apelarii
- **gi_yieldfrom** Afiseaza obiectul iterator din care se produce yield-ul
- **send** Permite trimiterea unor valori catre generator. Pentru aceasta, generatorul trebuie sa aiba o instructiune de genul x = yield in interior, transformandu-se in corutina

- **throw** metoda determina generatorul sa emita o erare

generator.gi_code()

```
def count_up_to_e2(n):
    try:
      i = 1
      while i <= n:
          yield i
          i += 1
    except GeneratorExit:
        print('Count stopped.')

g2 = count_up_to_e2(5)
print(next(g2))
print(g2.gi_code.co_code)
print(g2.gi_code.co_consts)
print(g2.gi_code.co_varnames)
print(next(g2))
```

```
1
b'\x81\x00z\x17d\x01}\x01|\x01|\x00k\x01r\x16|\x01V\x00\x01\x00|\x01d\
x017\x00}\x01|\x01|\x00k\x01s\x08W\x00d\x00S\x00W\x00d\x00S\x00\x04\
x00t\x00y&\x01\x00\x01\x00\x01\x00t\x01d\x02\x83\x01\x01\x00Y\x00d\
x00S\x00w\x00'
(None, 1, 'Count stopped.')
('n', 'i')
2
```

Corutine

Corutinele sunt generalizari ale subrutinelor si generatorilor.

O subrutina are un punct de intrare si un punct de iesire, este ca un fir.

Un generator are un singur punct de intrare si mai multe puncte de iesire

O corutina are mai multe puncte de intrare si, posibil, mai multe puncte de iesire

```
def coroutine_1():
    print("Coroutina porneste")
    while True:
        value = yield
        print("Coroutina primeste:", value)

coro = coroutine_1()
next(coro)
coro.send(10)
coro.send(20)
coro.send(30)
```

```
Coroutina porneste
Coroutina primeste: 10
Coroutina primeste: 20
Coroutina primeste: 30
```

Corutina poate contine mai mutle intructiuni yield.

```
def coroutine_d():
    print("Coroutina porneste")
    while True:
        value = yield
        print("Coroutina primeste 1:", value)
        value2 = yield
        print("Coroutina primeste 2:", value2)

coro = coroutine_d()
next(coro)
coro.send(10)
coro.send(20)
coro.send(30)
```

```
Coroutina porneste
Coroutina primeste 1: 10
Coroutina primeste 2: 20
Coroutina primeste 1: 30
```

Corutinele de obicei nu returneaza pentru ca aceasta le inchide. Corutinele sunt utilizate de regula pentru a lansa in executie (de obicei asincrona) o serie de alte functii. Dar, corutinele pot executa operatia de yield ca si generatorii cu putina atentie, dupa cum se vede in exemplul urmator:

```
def more_coroutine():
    val = 0
    while True:
        val = yield
        if val == 'exit':
            yield None
            return
        yield val

coro = more_coroutine()
coro.send(None)

for i in range(5):
    result = coro.send(i)
    coro.send(None)
    print(result)

coro.send('exit')
```

```
0
1
2
3
4
```

De ce este nevoie de coro.send(None) imediat dupa result = coro.send(i)? Penmtru ca instructiunea yield val (ultime linie din functia more_coroutine) nu doar returneaza val ci si suspenda executia la o instructiune yield care nu este pregatita sa primeasca date ci doar sa le trimita.

coro.send(None) muta executia la urmatorul yield astfel incat urmatorul coro.send(i) sa inceapa executia de unde ne dorim

Putem (daca ceea ce facem ne permite aceasta) sa folosim o singura instructiune yield care sa primeasca si sa trimita in acelasi timp. Dupa cum se vede in exemplul urmator, aceasta nu mai are nevoie de send(None) pentru ca executia functiei este suspendata exact la instructiunea care (in timpo ce trimite) asteapta date de la functia apelanta. Problema cu aceasta abordare este ca nu putem face mare lucru cu valorile intr-o simpla linie de cod.

```python
def more_coroutine():
    val = 0
    while True:
        val = yield val + 3
        if val == 'exit':
            yield None
            return


coro = more_coroutine()
coro.send(None)

for i in range(5):
    result = coro.send(i)
    print(result)

coro.send('exit')
```

```
3
4
5
6
7
```

Pentru a vedea un caz ceva mai "real" putem sa inspectam (daca suntem pe google collaboratory) unul dintre seturile de date pe care le primim automat la initializarea oricarui notebook. Acestea se gasesc in directorul sample_data

```
!ls -all sample_data
```

```
total 55512
drwxr-xr-x 1 root root     4096 Apr 11 13:21 .
drwxr-xr-x 1 root root     4096 Apr 11 13:21 ..
-rwxr-xr-x 1 root root     1697 Jan  1  2000 anscombe.json
-rw-r--r-- 1 root root   301141 Apr 11 13:21
california_housing_test.csv
-rw-r--r-- 1 root root  1706430 Apr 11 13:21
california_housing_train.csv
-rw-r--r-- 1 root root 18289443 Apr 11 13:21 mnist_test.csv
-rw-r--r-- 1 root root 36523880 Apr 11 13:21 mnist_train_small.csv
-rwxr-xr-x 1 root root      930 Jan  1  2000 README.md
```

toate datele sunt in format csv, putem folosi unul dintre ele pentru a testa corutinele. Intai sa aflam cum arata, tot cu comenzi de sistem. E mult mai simplu decat sa il incarcam in notebook si sa il inspectam.

```
!head -n 20 sample_data/mnist_test.csv
```

```
7,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,84,185,159,151
,60,36,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,222,254,254,254,254
,241,198,198,198,198,198,198,198,198,170,52,0,0,0,0,0,0,0,0,0,0,0,0,67
,114,72,114,163,227,254,225,254,254,254,250,229,254,254,140,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,17,66,14,67,67,67,59,21,236,254,106,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,83,253,209,18,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,22,233,255,83,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,129,254,238,44,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
,0,0,0,0,0,59,249,254,62,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
,0,133,254,187,5,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,9,205,2
48,58,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,126,254,182,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,75,251,240,57,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,19,221,254,166,0,0,0,0,0,0,0,0,0,0,0
,0,0,0,0,0,0,0,0,0,0,0,0,3,203,254,219,35,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,38,254,254,77,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,31,224,254,115,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
,133,254,254,52,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,61,242,2
54,254,52,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,121,254,254,21
9,40,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,121,254,207,18,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0
2,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,116,125,171,255,255,
```

150,93,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,169,253,253,253,253,253
,253,218,30,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,169,253,253,253,213,14
2,176,253,253,122,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,52,250,253,210,32,
12,0,6,206,253,140,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,77,251,210,25,0,0
,0,122,248,253,65,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,31,18,0,0,0,0,20
9,253,253,65,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,117,247,253
,198,10,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,76,247,253,231,63,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,128,253,253,144,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,176,246,253,159,12,0,0,0,0,0,0,0
,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,25,234,253,233,35,0,0,0,0,0,0,0,0,0,0,0
,0,0,0,0,0,0,0,0,0,0,0,0,198,253,253,141,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
,0,0,0,0,0,0,0,0,78,248,253,189,12,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
,0,0,0,0,19,200,253,253,141,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,134,253,253,173,12,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,2
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253,43,20,20,20,20,5,0,5,20,20,37,150,150,150,147,10,0,0,0,0,0,0,0,0,0
,248,253,253,253,253,253,253,253,168,143,166,253,253,253,253,253,253,2
53,123,0,0,0,0,0,0,0,0,0,174,253,253,253,253,253,253,253,253,253,253,2
53,249,247,247,169,117,117,57,0,0,0,0,0,0,0,0,0,0,118,123,123,123,166,
253,253,253,155,123,123,41,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,38,254,109,0,0,0,0,0,0,0
,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,87,252,82,0,0,0,0,0,0,0,0,0,0
,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,135,241,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
,0,0,0,0,0,0,0,0,45,244,150,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,84,254,63,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,20
2,223,11,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,32,254,216,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,95,254,195,0,0,0,0,0,0
,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,140,254,77,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,57,237,205,8,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
,0,0,0,0,0,0,0,0,0,124,255,165,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
,0,0,0,0,0,171,254,81,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
24,232,215,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,120,254,1
59,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,151,254,142,0,0,0
,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,228,254,66,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,61,251,254,66,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,141,254,205,3,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,10,215,254,121,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
,0,0,0,0,5,198,176,10,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,

0,0,0,

,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,179,66,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0
1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,77,254,107,3,0,0,0,0,0,0,0,0,0,0,0,0
,0,0,0,0,0,0,0,0,0,0,0,19,227,254,254,9,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,81,254,254,165,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
,0,0,0,7,203,254,254,73,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
53,254,254,250,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,134,254
,254,180,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,196,254,248,4
8,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,58,254,254,237,0,0,0,0
,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,111,254,254,132,0,0,0,0,0,0,0
,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,163,254,238,28,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,60,252,254,223,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
,0,0,0,0,0,0,0,0,0,79,254,254,154,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,0,0,0,0,163,254,238,53,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
,0,28,252,254,210,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,86,2
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53,117,0,0,0,0,25,118,253,253,253,47,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,28,
242,254,187,104,146,159,220,244,239,254,224,18,0,0,0,0,0,0,0,0,0,0,0,0,0
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,0,0,0,0,0,0,0,0,5,56,56,50,0,0,38,253,253,74,0,0,0,0,0,0,0,0,0,0,0,0,0,
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66,253,253,112,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,87,253,
244,3,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,170,253,198,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,254,253,149,0,0,0,0,0,0,
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0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,74,249,254,254,254
,245,167,167,136,25,80,60,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,104,254,254,
254,254,254,254,254,254,249,254,252,197,113,71,39,0,0,0,0,0,0,0,0,0,0,0,
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3,200,19,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,95,183,228,253,253,253,253,253,
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,37,7,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
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46,254,247,94,0,0,0,0,0,0,0,0,0,0,0,0,0,0,8,217,239,117,22,0,0,0,0,2
26,254,242,197,0,0,0,0,0,0,0,0,0,0,0,0,0,0,14,18,0,0,0,0,0,27,243,
207,46,37,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,99,254,132,0,0
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0,0,0,0,0,0,187,254,83,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
,0,115,176,10,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
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```

Fisierul are o gramada de cifre pe o linie si nimic altceva. Incercam sa vedem (tot cu comenzi de sistem) cat de multe sunt

```
!wc -l sample_data/mnist_test.csv

10000 sample_data/mnist_test.csv
```

comanda ne arata ca avem 10000 de linii in fisier, mult prea multe fata de ce avem noi nevoie

```
import csv
import time

def process_row():
    while True:
    # perform some processing on the row
      row = yield
      time.sleep(1)
      if row is not None:
        cu = 0
        for hh in row:
          cu = cu + int(hh)
        yield cu

def process_csv(input_filename):
    with open(input_filename, 'r') as input_file:

        reader = csv.reader(input_file)
        coro = process_row()
        coro.send(None)        ## avanseaza codul la primul yield
(priming the coroutine)
        cnt = 1
        for row in reader:
           if cnt < 10:
              result = coro.send(row)
              coro.send(None)
              print(str(cnt) + ': ' + str(result))
              cnt = cnt+1


input_filename = 'sample_data/mnist_test.csv'

process_csv(input_filename)
```

```
1: 18461
2: 28852
3: 9872
4: 37014
5: 19241
6: 13856
7: 21188
8: 21071
9: 30739
```

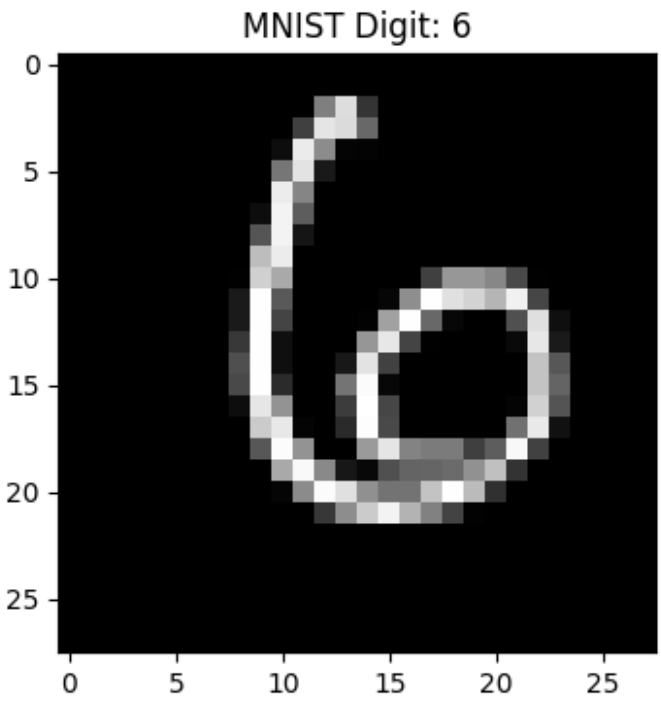
```python
import numpy as np
import csv
import matplotlib.pyplot as plt

def plot_mnist_from_csv(file_path, line_number):
    try:
        with open(file_path, 'r') as file:
            reader = csv.reader(file)
            for i, row in enumerate(reader, start=1):
                if i == line_number:
                    # Convertim elementele randului la numere intregi
                    data = list(map(int, row))
                    label = data[0]  # primul element este labelul
(cifra pe care o reprezinta)
                    image_data = data[1:]  # Urmatoarele elemente sunt
pixeli ai aimginii

                    if len(image_data) != 784:
                        print("Eroare: Imaginea nu are 784 pixeli")
                        return
                    image_size=(4, 4)
                    # redimensionare si afisare
                    image_matrix = np.array(image_data).reshape(28,
28)
                    plt.figure(figsize=image_size)
                    plt.imshow(image_matrix, cmap='gray')
                    plt.title(f'MNIST Digit: {label}')
                    plt.colorbar()
                    plt.show()
                    return
        print(f"Line {line_number} not found in file.")
    except Exception as e:
        print(f"An error occurred: {e}")



file_path ='sample_data/mnist_test.csv'
line_number = 101
plot_mnist_from_csv(file_path, line_number)
```



Expresii regulate

O expresie regulata este o succesiune de caractere (normale sau speciale) care se constituie intr-o structura de cautare (sablon).

O structura de cautare este o succesiune de caractere care poate sa "se potriveasca" cu mai multe siruri de caractere diferite dar care sunt asemanatoare structural

Expresiile regulate sunt implementate in Python intr-un modul extern numit re

Principalele metode sunt:

metoda	Definitie
findall	gaseste toate fragmentele de text care se potrivesc cu patternul. Returneaz o lista
search	Cauta si returneaza sub forma unui obiect de tip Match prima instanta care se potriveste cu patternul
split	Separa un sir de caractere dupa o expresie regulata
sub	Inlocuieste anumite structuri identificate cu ajutorul unei expresii regulate cu alte structuri

```
import re

text = "Numarul meu de telefon este 1234567890 si numarul cardului meu
de credit este 9876543210987654."
```

```
pattern = '\d\d\d\d\d\d\d\d\d\d'

results = re.findall(pattern,  text)

for match in results:
  print (str(type(match)) + ': ' + match)
```

```
<class 'str'>: 1234567890
<class 'str'>: 9876543210
```

```
<>:5: SyntaxWarning: invalid escape sequence '\d'
<>:5: SyntaxWarning: invalid escape sequence '\d'
/tmp/ipykernel_20617/4213832180.py:5: SyntaxWarning: invalid escape
sequence '\d'
  pattern = '\d\d\d\d\d\d\d\d\d\d'
```

```
pattern = r'\d\d\d\d\d\d\d\d\d\d'

results = re.findall(pattern,  text)

for match in results:
  print (str(type(match)) + ': ' + match)
```

```
<class 'str'>: 1234567890
<class 'str'>: 9876543210
```

```
textnm = 'Rick and Nick have decided to visit Dick because he is sick'

pattern = '.ick'

results = re.findall(pattern,  textnm)

for match in results:
  print (str(type(match)) + ': ' + match)
```

```
<class 'str'>: Rick
<class 'str'>: Nick
<class 'str'>: Dick
<class 'str'>: sick
```

Putem separa doar numele proprii? fara sick

```
textnm = 'Rick and Nick have decided to visit Dick because he is sick'

pattern = '[A-Z]ick'

results = re.findall(pattern,  textnm)

for match in results:
  print (str(type(match)) + ': ' + match)
```

```
<class 'str'>: Rick
<class 'str'>: Nick
<class 'str'>: Dick
```

Patternul poate deveni extrem de complicat

https://www.dataquest.io/wp-content/uploads/2019/03/python-regular-expressions-cheat-sheet.pdf

Putem gasi o expresie regulata care sa "se potriveasca" cu orice adresa de e-mail?

(username)@(domainname).(top-leveldomain)

(string1)@(string2).(2+characters)

```
emailad = 'ana.are.mere@domeniu.ro si vasile nu are deloc'

pattern = 'ana'

result = re.search(pattern,  emailad)

if result is not None:
  print(type(result))
  print(dir(result))
  print(result.group())
  print('start:' + str(result.start()))
  print('end:' + str(result.end()))
```

```
<class 're.Match'>
['__class__', '__class_getitem__', '__copy__', '__deepcopy__',
'__delattr__', '__dir__', '__doc__', '__eq__', '__format__', '__ge__',
'__getattribute__', '__getitem__', '__gt__', '__hash__', '__init__',
'__init_subclass__', '__le__', '__lt__', '__module__', '__ne__',
'__new__', '__reduce__', '__reduce_ex__', '__repr__', '__setattr__',
'__sizeof__', '__str__', '__subclasshook__', 'end', 'endpos',
'expand', 'group', 'groupdict', 'groups', 'lastgroup', 'lastindex',
'pos', 're', 'regs', 'span', 'start', 'string']
ana
start:0
end:3
```

caracterul punct (.) tine locul oricarui caracter (mai putin newline) iar * inseamna "repetare de oricate ori".

Astfel, patternul ana.*@ va cuprinde ana si tot ceea ce urmeaza pana la intalnirea caracterului @.

```
emailad = 'ana.are.mere@domeniu.ro si vasile nu are deloc'

pattern = 'ana.*@'
```

```
result = re.search(pattern,  emailad)

if result is not None:
  print(result.group())
  print('start:' + str(result.start()))
  print('end:' + str(result.end()))

ana.are.mere@
start:0
end:13
```

Ce se intampla daca sunt mai multe @ in sir?

```
emailad = 'ana.are.mere@domeniu.ro si @vasile nu are @deloc'

pattern = 'ana.*@'

result = re.search(pattern,  emailad)

if result is not None:
  print(result.group())
  print('start:' + str(result.start()))
  print('end:' + str(result.end()))

ana.are.mere@domeniu.ro si @vasile nu are @
start:0
end:43
```

Se observa ca end este la 43, spre deosebire de 13. Aceasta inseamna ca morotul de expresii regulate s-a dus "cat de departe a putut". Acest comportament se numeste "greedy (lacom)". Toate caracterele generelizatoare (de tip *, +) au un comportament greedy in mod predefinit.

Daca punem un semn de intrebare imediat dupa cuantificatorul * atunci comportamentul acestuia se schimba din greedy (lacom) in lazy (lenes) si identificarea va merge pana la prima aparitie a caracterului @

```
emailad = 'ana.are.mere@domeniu.ro si @vasile nu are @deloc'

pattern = 'ana.*?@'

result = re.search(pattern,  emailad)

if result is not None:
  print(result.group())
  print('start:' + str(result.start()))
  print('end:' + str(result.end()))

ana.are.mere@
start:0
end:13
```

cum obtinem un match pe intreaga adresa de e-mail?

```
emailad = 'ana.are.mere@domeniu.ro si @vasile nu are @deloc'

pattern = '.*?@.*?\s'

result = re.search(pattern,  emailad)

if result is not None:
  print(result.group())

ana.are.mere@domeniu.ro
```

ce probleme are aceasta expresie regulata?

1. Daca adresa de e-mail nu incepe la inceputul sirului?

```
emailad = 'Adresa mea de email este ana.are.mere@domeniu.ro si @vasile
nu are @deloc'

pattern = '.*?@.*?\s'

result = re.search(pattern,  emailad)

if result is not None:
  print(result.group())

Adresa mea de email este ana.are.mere@domeniu.ro
```

Nu putem folosi spatiul pentru ca nu este unul singur. Pentru aceasta situatie au fost inventate seturile de caractere. De exemplu setul [a-zA-Z] va recunoaste orice caracter alfanumeric (unul singur daca nu exista cuantificatori). Daca includem si caractelerel permise in prima parte a adresei de e-mail (cifre, caractere speciale) putem sa ne asiguram ca prim parte (partea locala) este identificata corect

```
emailad = 'Adresa mea de email este ana.are.mere@domeniu.ro si @vasile
nu are @deloc'

pattern = '[a-zA-Z0-9\.]*?@.*?\s'

result = re.search(pattern,  emailad)

if result is not None:
  print(result.group())

ana.are.mere@domeniu.ro
```

Partea din dreapta @ este compusa din grupuri de litere separate cu puncte.

Daca textul are mai multe adrese de e-mail si ne dorim identificarea tuturor, folosim findall

```
emailad = 'Adresele de email inregistrate sunt urmatoarle:
1234567890@numarulcelmaimare.xxx, gentle-Man@iron-self.tv și
almond@caramela.tv. \
De asemenea, sunt inregistrate si anton@big.domain.museum și
carcalete@insecticid.co.jp și in final seful-cel-mai-
mare@supercompanie.com'

pattern = '[a-zA-Z0-9._%-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,6}'

results = re.findall(pattern,  emailad)

for match in results:
  print (str(type(match)) + ': ' + match)
```

```
<class 'str'>: 1234567890@numarulcelmaimare.xxx
<class 'str'>: gentle-Man@iron-self.tv
<class 'str'>: almond@caramela.tv
<class 'str'>: anton@big.domain.museum
<class 'str'>: carcalete@insecticid.co.jp
<class 'str'>: seful-cel-mai-mare@supercompanie.com
```

fie dat textul urmator

"Asadar, vine un podar, destul de murdar, care spune ca a fost grajdar, exprimandu-se lapidar, clar lipsit complet de abecedar, deloc gospodar, certat cu orice calendar, a carui viata e in zadar si spune ca e miliardar, ca a fost caramidar si nu grajdar si ca se considera solidar cu orice om prins de radar."

Separati toate cuvintele care se termina in "dar"

```
dartestxt = "Asadar, vine un podar, destul de murdar, care spune ca a
fost grajdar, exprimandu-se lapidar, clar lipsit complet de abecedar,
\
deloc gospodar, certat cu orice calendar, a carui viata e in zadar si
spune ca e miliardar, ca a fost caramidar si nu grajdar si ca se \
considera solidar cu orice om prins de radar."

pdar = '[a-zA-Z]*?dar'

darresults = re.findall(pdar, dartestxt)

for match in darresults:
  print (str(type(match)) + ': ' + match)
```

```
<class 'str'>: Asadar
<class 'str'>: podar
<class 'str'>: murdar
<class 'str'>: grajdar
<class 'str'>: lapidar
<class 'str'>: abecedar
```

```
<class 'str'>: gospodar
<class 'str'>: calendar
<class 'str'>: zadar
<class 'str'>: miliardar
<class 'str'>: caramidar
<class 'str'>: grajdar
<class 'str'>: solidar
<class 'str'>: radar
```

Split cu ajutorul expresiilor regulate

```
split_term = ' '

phrase = "When Chuck Norris enters a room, he doesn't turn the lights
on, he turns the dark off"

# Split the phrase
result = re.split(split_term,phrase)

print (type(result))
print (result)
```

```
<class 'list'>
['When', 'Chuck', 'Norris', 'enters', 'a', 'room,', 'he', "doesn't",
'turn', 'the', 'lights', 'on,', 'he', 'turns', 'the', 'dark', 'off']
```

Se observă că utilizarea spațiului în split determină taierea unora dintre cuvinte împreună cu virgula. Putem adăuga virgula la caracterele după care se face splitul, sau, mai bine, combinatia de spatiu si virgula:

```
split_term = ' | ,'

phrase = "When Chuck Norris enters a room, he doesn't turn the lights
on, he turns the dark off"

# Split the phrase
result = re.split(split_term,phrase)

print (result)
```

```
['When', 'Chuck', 'Norris', 'enters', 'a', 'room,', 'he', "doesn't",
'turn', 'the', 'lights', 'on,', 'he', 'turns', 'the', 'dark', 'off']
```

Ce se întâmplă dacă există mai multe spații în șir? Se poate adauga plus după spatiu: + inseamnă caracterul anterior prezent o data sau de mai multe ori

```
split_term = ' +|, +'

phrase = "When    Chuck Norris    enters a room, he  doesn't turn the
lights on, he turns the dark off"
```

```
# Split the phrase
result = re.split(split_term,phrase)

print (result)
```

```
['When', 'Chuck', 'Norris', 'enters', 'a', 'room', 'he', "doesn't", 'turn', 'the', 'lights', 'on', 'he', 'turns', 'the', 'dark', 'off']
```

Înlocuirea cu ajutorul expresiilor regulate

```
phrase = "It takes Chuck Norris 20 minutes to watch 60 Minutes"
find = 'minutes'
replace = 'hours'

text_after = re.sub(find, replace, phrase)

print (text_after)
```

```
It takes Chuck Norris 20 hours to watch 60 Minutes
```

re.sub este case senzitive, in afara cazului in care i se comunica altfel. re.sub permite adăugarea unor flag-uri, printre care și re.I sau re.IGNORECASE

Flag	Alias	Inline	Flag Meaning
re.ASCII	re.A	?m	The re.ASCII is relevant to the byte patterns only. It makes the \w, \W,\b, \B, \d, \D, and \S perform ASCII-only matching instead of full Unicode matching.
re.DEBUG	N/A	N/A	The re.DEBUG shows the debug information of compiled pattern.
re.IGNORECASE	re.I	?i	perform case-insensitive matching. It means that the [A-Z] will also match lowercase letters.
re.LOCALE	re.L	?L	The re.LOCALE is relevant only to the byte pattern. It makes the \w, \W, \b, \B and case-sensitive matching dependent on the current locale. The re.LOCALE is not compatible with the re.ASCII flag.
re.MUTILINE	re.M	?m	The re.MULTILINE makes the ^ matches at the beginning of a string and at the beginning of each line and $ matches at the end of a string and at the end of each line.
re.DOTALL	re.S	?s	By default, the dot (.) matches any characters except a newline. The

Flag	Alias	Inline	Flag Meaning
			re.DOTALL makes the dot (.) matches all characters including a newline.
re.VERBOSE	re.X	?x	The re.VERBOSE flag allows you to organize a pattern into logical sections visually and add comments.

```
text_after = re.sub(find, replace, phrase, flags=re.I)

print (text_after)

It takes Chuck Norris 20 hours to watch 60 hours
```

O alta varianta este folosirea unui pattern - înlocuirea primei litere din patternul find cu caracterul special . care înseamnă "orice caracter o singură dată"

```
phrase = "It takes Chuck Norris 20 minutes to watch 60 Minutes"
find = '.inutes'
replace = 'hours'

text_after = re.sub(find, replace, phrase)

print (text_after)

It takes Chuck Norris 20 hours to watch 60 hours
```

Un exemplu ceva mai complicat - identificarea adreselor de e-mail

```
emtext = 'Cineva a scris un email de pe adresa cracanel@memyself.com
către adresele: pompiliu.chininescu@doctorul.ro,
cufurescu_88@toilette.com.ro \
în care menționa dorinta de a vinde un produs îndoielnic. Consider că
acest abuz trebuie pedepsit cu tărie conform politicii de toleranță
ZERO față \
de comunicări nesolicitate. Pentru aceasta, am trimis acest  email si
pe adresele: 1234567890@numarulcelmaimare.xxx, gentle-Man@iron-self.tv
\
și Gladiather@caramela.tv. Dacă considerați necesar, vă rog sa
trimiteți și către venerus@gineco.museum și carcalete@insecticid.co.jp
și dacă e \
absolut necesar si către seful-cel-mai-mare@supercompanie.com.ro.'

pattern = r'[a-zA-Z0-9._%-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,6}'

emails = re.findall(pattern, emtext)
print(emails)


['cracanel@memyself.com', 'pompiliu.chininescu@doctorul.ro',
'cufurescu_88@toilette.com.ro', '1234567890@numarulcelmaimare.xxx',
```

```
'gentle-Man@iron-self.tv', 'Gladiather@caramela.tv',
'venerus@gineco.museum', 'carcalete@insecticid.co.jp', 'seful-cel-mai-
mare@supercompanie.com.ro']
```

In anumite situatii adresele de email trebuie inlocuite cu o versiune anonima

```
pattern = r'[a-zA-Z0-9._%-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,6}'
replace = 'email@domeniu.com'

text_after = re.sub(pattern, replace, emtext)

print(text_after)

Cineva a scris un email de pe adresa email@domeniu.com către adresele:
email@domeniu.com, email@domeniu.com în care menționa dorinta de a
vinde un produs îndoielnic. Consider că acest abuz trebuie pedepsit cu
tărie conform politicii de toleranță ZERO față de comunicări
nesolicitate. Pentru aceasta, am trimis acest  email si pe adresele:
email@domeniu.com, email@domeniu.com și email@domeniu.com. Dacă
considerați necesar, vă rog sa trimiteți și către email@domeniu.com și
email@domeniu.com și dacă e absolut necesar si către
email@domeniu.com.
```

Expresie regulata care detecteaza adresele Ipv4

\d inseamna o cifra

{1,3} inseamna aparitia caracterului anterior intre 1 si 3 ori

```
iptext = "traficul de la adresa 192.168.12.12 catre adresa
193.221.23.125"

ipmatch = r'\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3}'

ips = re.findall(ipmatch, iptext)

print (ips)

['192.168.12.12', '193.221.23.125']
```

expresiile regulate sunt implementate diferit in diferite limbaje de programare - chiar dacă modul de scriere a secvenței de caractere care reprezintă patternul căutat este la fel. Modulul re are urmatoarele metode:

Functie	Exemplu	Descriere
match	result = re.match(pattern, string)	Returnează
search	result = re.search(pattern,	

Functie	Exemplu	Descriere
	string)	
fullmatch	result = re.fullmatch(pattern, string)	
split	result = re.split(pattern, string)	Returneaza o lista de fragmente obtinute in urma splitului
findall		
sub		

```
print (dir(re))
```

```
['A', 'ASCII', 'DEBUG', 'DOTALL', 'I', 'IGNORECASE', 'L', 'LOCALE',
'M', 'MULTILINE', 'Match', 'NOFLAG', 'Pattern', 'RegexFlag', 'S',
'Scanner', 'T', 'TEMPLATE', 'U', 'UNICODE', 'VERBOSE', 'X',
'_MAXCACHE', '_MAXCACHE2', '__all__', '__builtins__', '__cached__',
'__doc__', '__file__', '__loader__', '__name__', '__package__',
'__path__', '__spec__', '__version__', '_cache', '_cache2',
'_casefix', '_compile', '_compile_template', '_compiler',
'_constants', '_parser', '_pickle', '_special_chars_map', '_sre',
'compile', 'copyreg', 'enum', 'error', 'escape', 'findall',
'finditer', 'fullmatch', 'functools', 'match', 'purge', 'search',
'split', 'sub', 'subn', 'template']
```

Requests

Requests este un client HTTP pentru python care permite interactiunea cu serverele web

```
import requests


r = requests.get('https://corporatebs-generator.sameerkumar.website/')
if (r.status_code == 200):
  print(r.json())
```

```
{'phrase': 'Efficiently Cloudify Cloud-based Results'}
```

```
r = requests.get('https://v2.jokeapi.dev/joke/Any?safe_mode')
mydata = r.json()

print(r.json())
```

```
{'error': False, 'category': 'Programming', 'type': 'twopart',
'setup': 'How did the programmer die in the shower?', 'delivery': 'He
read the shampoo bottle instructions: Lather. Rinse. Repeat.',
'flags': {'nsfw': False, 'religious': False, 'political': False,
'racist': False, 'sexist': False, 'explicit': False}, 'safe': False,
'id': 266, 'lang': 'en'}
```

```
print(mydata['category'])
if mydata['type'] == 'single':
    print(mydata['joke'])
elif mydata['type'] == 'twopart':
    print(mydata['setup'])
    print(mydata['delivery'])


flags = []
for key in mydata['flags']:
    if mydata['flags'][key] == True:
        flags.append(key)
if len(flags) > 0:
    print('Flags: ' + ', '.join(flags))
else:
    print('No flags')
```

```
Programming
How did the programmer die in the shower?
He read the shampoo bottle instructions: Lather. Rinse. Repeat.
No flags
```

Preluarea si afisarea imaginilor

```
import skimage
import matplotlib.pyplot as plt
balerina = skimage.io.imread('https://cataas.com/cat/says/hello
%20world!')
balerina_bw = balerina[:,:,0]
plt.imshow(balerina_bw, cmap='gray')
plt.axis('off')
plt.show()
```

Combinare requests + expresii regulate

Scrieti o expresie regulata care extrage toate adresele ipv4 din urmatorul fragment de log, care se gaseste la adresa https://pastebin.com/raw/FCAhtVzF

Puneti aceste adrese intr-o lista unica

```
import requests

r = requests.get('https://pastebin.com/raw/ZEFADn6k')
if (r.status_code == 200):
  textlog = r.text
  print(textlog)
  ipmatch = '\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3}'
  results = re.findall(ipmatch,  textlog)

  ips = set()
  for match in results:
    ips.add(match)

print(ips)
```

```
May  8 23:01:17 isp postfix/qmgr[15768]: DA6F36742E82: removed
May  8 23:01:22 isp postfix/smtpd[24660]: connect from
unknown[193.32.162.188]
May  8 23:01:22 isp dovecot: auth: Warning: auth client 0 disconnected
with 1 pending requests: Connection reset by peer
```

```
May  8 23:01:24 isp postfix/smtpd[28657]: connect from
unknown[185.254.37.189]
May  8 23:01:26 isp postfix/smtpd[28657]: warning:
unknown[185.254.37.189]: SASL LOGIN authentication failed:
UGFzc3dvcmQ6
May  8 23:01:26 isp postfix/smtpd[28657]: lost connection after AUTH
from unknown[185.254.37.189]
May  8 23:01:26 isp postfix/smtpd[28657]: disconnect from
unknown[185.254.37.189]
May  8 23:01:30 isp postfix/smtpd[24660]: lost connection after AUTH
from unknown[193.32.162.188]
May  8 23:01:30 isp postfix/smtpd[24660]: disconnect from
unknown[193.32.162.188]
May  8 23:01:33 isp postfix/smtpd[28657]: connect from
unknown[185.254.37.189]
May  8 23:01:36 isp postfix/smtpd[7789]: connect from
unknown[118.41.204.67]
May  8 23:01:37 isp postfix/smtpd[24660]: connect from
unknown[193.32.162.188]
May  8 23:01:39 isp postfix/smtpd[24660]: warning:
unknown[193.32.162.188]: SASL LOGIN authentication failed:
UGFzc3dvcmQ6
May  8 23:01:39 isp postfix/smtpd[28657]: warning:
unknown[185.254.37.189]: SASL LOGIN authentication failed:
UGFzc3dvcmQ6
May  8 23:01:39 isp postfix/smtpd[28657]: lost connection after AUTH
from unknown[185.254.37.189]
May  8 23:01:39 isp postfix/smtpd[28657]: disconnect from
unknown[185.254.37.189]
May  8 23:01:39 isp postfix/smtpd[24660]: lost connection after AUTH
from unknown[193.32.162.188]
May  8 23:01:39 isp postfix/smtpd[24660]: disconnect from
unknown[193.32.162.188]
May  8 23:01:39 isp dovecot: auth: Warning: auth client 0 disconnected
with 1 pending requests: EOF
May  8 23:01:44 isp postfix/smtpd[7789]: warning:
unknown[118.41.204.67]: SASL LOGIN authentication failed: UGFzc3dvcmQ6

May  8 23:01:45 isp postfix/smtpd[7789]: lost connection after AUTH
from unknown[118.41.204.67]
May  8 23:01:45 isp postfix/smtpd[7789]: disconnect from
unknown[118.41.204.67]
May  8 23:01:47 isp postfix/smtpd[28657]: connect from
unknown[185.254.37.189]
May  8 23:01:49 isp postfix/smtpd[7789]: connect from cpe-75-80-101-
22.san.res.rr.com[75.80.101.22]
May  8 23:01:50 isp postfix/smtpd[7890]: connect from
unknown[193.32.162.188]
May  8 23:01:56 isp postfix/smtpd[7789]: warning: cpe-75-80-101-
22.san.res.rr.com[75.80.101.22]: SASL LOGIN authentication failed:
```

```
UGFzc3dvcmQ6
May  8 23:01:56 isp postfix/smtpd[7890]: warning:
unknown[193.32.162.188]: SASL LOGIN authentication failed:
UGFzc3dvcmQ6
May  8 23:01:56 isp postfix/smtpd[7890]: lost connection after AUTH
from unknown[193.32.162.188]
May  8 23:01:56 isp postfix/smtpd[7890]: disconnect from
unknown[193.32.162.188]
May  8 23:01:57 isp postfix/smtpd[7789]: lost connection after AUTH
from cpe-75-80-101-22.san.res.rr.com[75.80.101.22]
May  8 23:01:57 isp postfix/smtpd[7789]: disconnect from cpe-75-80-
101-22.san.res.rr.com[75.80.101.22]
May  8 23:01:57 isp postfix/smtpd[28657]: warning:
unknown[185.254.37.189]: SASL LOGIN authentication failed:
UGFzc3dvcmQ6
May  8 23:01:57 isp postfix/smtpd[28657]: lost connection after AUTH
from unknown[185.254.37.189]
May  8 23:01:57 isp postfix/smtpd[28657]: disconnect from
unknown[185.254.37.189]
May  8 23:02:05 isp postfix/smtpd[7890]: connect from
unknown[185.254.37.189]
{'185.254.37.189', '193.32.162.188', '75.80.101.22', '118.41.204.67'}
```

```
wit = 'UGFzc3dvcmQ6'

import base64

print (base64.b64decode(wit))
```

```
b'Password:'
```

Există multe liste de API-uri disponibile gratuit. Una dintre ele este aici: https://github.com/public-apis/public-apis

API Metar (vremea pe aeroporturi)

```
r =
requests.get('http://tgftp.nws.noaa.gov/data/observations/metar/statio
ns/LROP.TXT')
if (r.status_code == 200):
    print(r.headers)
```

```
{'Date': 'Sat, 13 Apr 2024 17:33:30 GMT', 'Server': 'Apache', 'X-
Frame-Options': 'SAMEORIGIN', 'X-Content-Type-Options': 'nosniff', 'X-
XSS-Protection': '1; mode=block', 'Last-Modified': 'Sat, 13 Apr 2024
17:04:14 GMT', 'Accept-Ranges': 'bytes', 'Vary': 'Accept-Encoding',
'Content-Encoding': 'gzip', 'Content-Length': '81', 'Keep-Alive':
'timeout=300, max=100', 'Connection': 'Keep-Alive', 'Content-Type':
'text/plain; charset=UTF-8', 'Strict-Transport-Security': 'max-
age=31536000; includeSubdomains; preload'}
```

Content-type este text/plain - ceea ce am primit este un document text (ASCII)

```
print (r.content)
```

Retete de coctail

```
r =
requests.get('https://www.thecocktaildb.com/api/json/v1/1/search.php?
f=a')

data = r.json()

print(data)
```


```
{'drinks': [{'idDrink': '17222', 'strDrink': 'A1',
'strDrinkAlternate': None, 'strTags': None, 'strVideo': None,
'strCategory': 'Cocktail', 'strIBA': None, 'strAlcoholic':
'Alcoholic', 'strGlass': 'Cocktail glass', 'strInstructions': 'Pour
all ingredients into a cocktail shaker, mix and serve over ice into a
chilled glass.', 'strInstructionsES': 'Vierta todos los ingredientes
en una coctelera, mezcle y sirva con hielo en un vaso frío.',
'strInstructionsDE': 'Alle Zutaten in einen Cocktailshaker geben,
mischen und über Eis in ein gekühltes Glas servieren.',
'strInstructionsFR': None, 'strInstructionsIT': 'Versare tutti gli
ingredienti in uno shaker, mescolare e servire con ghiaccio in un
bicchiere freddo.', 'strInstructionsZH-HANS': None,
'strInstructionsZH-HANT': None, 'strDrinkThumb':
'https://www.thecocktaildb.com/images/media/drink/2x8thr1504816928.jpg
', 'strIngredient1': 'Gin', 'strIngredient2': 'Grand Marnier',
'strIngredient3': 'Lemon Juice', 'strIngredient4': 'Grenadine',
'strIngredient5': None, 'strIngredient6': None, 'strIngredient7':
None, 'strIngredient8': None, 'strIngredient9': None,
'strIngredient10': None, 'strIngredient11': None, 'strIngredient12':
None, 'strIngredient13': None, 'strIngredient14': None,
'strIngredient15': None, 'strMeasure1': '1 3/4 shot ', 'strMeasure2':
'1 Shot ', 'strMeasure3': '1/4 Shot', 'strMeasure4': '1/8 Shot',
'strMeasure5': None, 'strMeasure6': None, 'strMeasure7': None,
'strMeasure8': None, 'strMeasure9': None, 'strMeasure10': None,
'strMeasure11': None, 'strMeasure12': None, 'strMeasure13': None,
'strMeasure14': None, 'strMeasure15': None, 'strImageSource': None,
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'dateModified': '2017-09-07 21:42:09'}, {'idDrink': '13501',
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'strVideo': None, 'strCategory': 'Shot', 'strIBA': None,
'strAlcoholic': 'Alcoholic', 'strGlass': 'Shot glass',
'strInstructions': 'Layered in a shot glass.', 'strInstructionsES':
'Coloque todos los ingredientes en un vaso de chupito.',
'strInstructionsDE': 'Schichtaufbau in einem Schnapsglas.',
'strInstructionsFR': None, 'strInstructionsIT': "Versa in ordine di
lettera i vari ingredienti. 1/3 del bicchiere va riempito con
```

l'Amaretto, 1/3 di Baileys e il restante di Cognac.",
'strInstructionsZH-HANS': None, 'strInstructionsZH-HANT': None,
'strDrinkThumb':
'https://www.thecocktaildb.com/images/media/drink/tqpvqp1472668328.jpg
', 'strIngredient1': 'Amaretto', 'strIngredient2': 'Baileys irish
cream', 'strIngredient3': 'Cognac', 'strIngredient4': None,
'strIngredient5': None, 'strIngredient6': None, 'strIngredient7':
None, 'strIngredient8': None, 'strIngredient9': None,
'strIngredient10': None, 'strIngredient11': None, 'strIngredient12':
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'strIngredient15': None, 'strMeasure1': '1/3 ', 'strMeasure2': '1/3 ',
'strMeasure3': '1/3 ', 'strMeasure4': None, 'strMeasure5': None,
'strMeasure6': None, 'strMeasure7': None, 'strMeasure8': None,
'strMeasure9': None, 'strMeasure10': None, 'strMeasure11': None,
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'strMeasure15': None, 'strImageSource': None, 'strImageAttribution':
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19:32:08'}, {'idDrink': '17225', 'strDrink': 'Ace',
'strDrinkAlternate': None, 'strTags': None, 'strVideo': None,
'strCategory': 'Cocktail', 'strIBA': None, 'strAlcoholic':
'Alcoholic', 'strGlass': 'Martini Glass', 'strInstructions': 'Shake
all the ingredients in a cocktail shaker and ice then strain in a cold
glass.', 'strInstructionsES': 'Agite todos los ingredientes en una
coctelera con hielo y entonces cuélelos sobre un vaso enfriado.',
'strInstructionsDE': 'Alle Zutaten in einem Cocktailshaker mit Eis
schütteln und dann in einem kalten Glas abseihen.',
'strInstructionsFR': None, 'strInstructionsIT': 'Shakerare tutti gli
ingredienti in uno shaker e ghiaccio, quindi filtrare in un bicchiere
freddo.', 'strInstructionsZH-HANS': None, 'strInstructionsZH-HANT':
None, 'strDrinkThumb':
'https://www.thecocktaildb.com/images/media/drink/l3cd7f1504818306.jpg
', 'strIngredient1': 'Gin', 'strIngredient2': 'Grenadine',
'strIngredient3': 'Heavy cream', 'strIngredient4': 'Milk',
'strIngredient5': 'Egg White', 'strIngredient6': None,
'strIngredient7': None, 'strIngredient8': None, 'strIngredient9':
None, 'strIngredient10': None, 'strIngredient11': None,
'strIngredient12': None, 'strIngredient13': None, 'strIngredient14':
None, 'strIngredient15': None, 'strMeasure1': '2 shots ',
'strMeasure2': '1/2 shot ', 'strMeasure3': '1/2 shot ', 'strMeasure4':
'1/2 shot', 'strMeasure5': '1/2 Fresh', 'strMeasure6': None,
'strMeasure7': None, 'strMeasure8': None, 'strMeasure9': None,
'strMeasure10': None, 'strMeasure11': None, 'strMeasure12': None,
'strMeasure13': None, 'strMeasure14': None, 'strMeasure15': None,
'strImageSource': None, 'strImageAttribution': None,
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22:05:06'}, {'idDrink': '14610', 'strDrink': 'ACID',
'strDrinkAlternate': None, 'strTags': None, 'strVideo': None,
'strCategory': 'Shot', 'strIBA': None, 'strAlcoholic': 'Alcoholic',
'strGlass': 'Shot glass', 'strInstructions': 'Poor in the 151 first
followed by the 101 served with a Coke or Dr Pepper chaser.',

'strInstructionsES': 'Vierta primero el Bacardi 151, seguido del Wild Turkey 101 y sirva con Coca-Cola o Dr Pepper.', 'strInstructionsDE': 'Gering den 151 gefolgt von der 101, die mit einer Cola oder Dr. Pepper Chaser serviert wird.', 'strInstructionsFR': None, 'strInstructionsIT': 'Versa prima il Bacardi 151 seguito dal 101, servito con una Coca-Cola.', 'strInstructionsZH-HANS': None, 'strInstructionsZH-HANT': None, 'strDrinkThumb': 'https://www.thecocktaildb.com/images/media/drink/xuxpxt1479209317.jpg', 'strIngredient1': '151 proof rum', 'strIngredient2': 'Wild Turkey', 'strIngredient3': None, 'strIngredient4': None, 'strIngredient5': None, 'strIngredient6': None, 'strIngredient7': None, 'strIngredient8': None, 'strIngredient9': None, 'strIngredient10': None, 'strIngredient11': None, 'strIngredient12': None, 'strIngredient13': None, 'strIngredient14': None, 'strIngredient15': None, 'strMeasure1': '1 oz Bacardi ', 'strMeasure2': '1 oz ', 'strMeasure3': None, 'strMeasure4': None, 'strMeasure5': None, 'strMeasure6': None, 'strMeasure7': None, 'strMeasure8': None, 'strMeasure9': None, 'strMeasure10': None, 'strMeasure11': None, 'strMeasure12': None, 'strMeasure13': None, 'strMeasure14': None, 'strMeasure15': None, 'strImageSource': None, 'strImageAttribution': None, 'strCreativeCommonsConfirmed': 'No', 'dateModified': '2016-11-15 11:28:37'}, {'idDrink': '17837', 'strDrink': 'Adam', 'strDrinkAlternate': None, 'strTags': 'Alcoholic,Holiday', 'strVideo': None, 'strCategory': 'Ordinary Drink', 'strIBA': None, 'strAlcoholic': 'Alcoholic', 'strGlass': 'Cocktail glass', 'strInstructions': 'In a shaker half-filled with ice cubes, combine all of the ingredients. Shake well. Strain into a cocktail glass.', 'strInstructionsES': 'Mezclar todos los ingredientes en una coctelera con cubitos de hielo hasta la mitad. Agitar bien. Colar el contenido en un vaso de cóctel.', 'strInstructionsDE': 'In einem Shaker, der halb mit Eiswürfeln gefüllt ist, alle Zutaten vermengen. Gut schütteln. In ein Cocktailglas abseihen.', 'strInstructionsFR': None, 'strInstructionsIT': 'In uno shaker riempito a metà con cubetti di ghiaccio, unire tutti gli ingredienti.Agitare bene.Filtrare in un bicchiere da cocktail.', 'strInstructionsZH-HANS': None, 'strInstructionsZH-HANT': None, 'strDrinkThumb': 'https://www.thecocktaildb.com/images/media/drink/v0at4i1582478473.jpg', 'strIngredient1': 'Dark rum', 'strIngredient2': 'Lemon juice', 'strIngredient3': 'Grenadine', 'strIngredient4': None, 'strIngredient5': None, 'strIngredient6': None, 'strIngredient7': None, 'strIngredient8': None, 'strIngredient9': None, 'strIngredient10': None, 'strIngredient11': None, 'strIngredient12': None, 'strIngredient13': None, 'strIngredient14': None, 'strIngredient15': None, 'strMeasure1': '2 oz ', 'strMeasure2': '1 oz ', 'strMeasure3': '1 tsp ', 'strMeasure4': None, 'strMeasure5': None, 'strMeasure6': None, 'strMeasure7': None, 'strMeasure8': None, 'strMeasure9': None, 'strMeasure10': None, 'strMeasure11': None, 'strMeasure12': None, 'strMeasure13': None, 'strMeasure14': None, 'strMeasure15': None, 'strImageSource': None, 'strImageAttribution': None, 'strCreativeCommonsConfirmed': 'Yes', 'dateModified': '2016-09-

02 11:29:29'}, {'idDrink': '13938', 'strDrink': 'AT&T', 'strDrinkAlternate': None, 'strTags': None, 'strVideo': None, 'strCategory': 'Ordinary Drink', 'strIBA': None, 'strAlcoholic': 'Alcoholic', 'strGlass': 'Highball Glass', 'strInstructions': 'Pour Vodka and Gin over ice, add Tonic and Stir', 'strInstructionsES': 'Vierta el vodka y la ginebra sobre el hielo, agregue la tónica y revuelva.', 'strInstructionsDE': 'Wodka und Gin über das Eis gießen, Tonic hinzufügen und umrühren.', 'strInstructionsFR': None, 'strInstructionsIT': "Versare la vodka e il gin sul ghiaccio, aggiungere l'acqua tonica e mescolare", 'strInstructionsZH-HANS': None, 'strInstructionsZH-HANT': None, 'strDrinkThumb': 'https://www.thecocktaildb.com/images/media/drink/rhhwmp1493067619.jpg', 'strIngredient1': 'Absolut Vodka', 'strIngredient2': 'Gin', 'strIngredient3': 'Tonic water', 'strIngredient4': None, 'strIngredient5': None, 'strIngredient6': None, 'strIngredient7': None, 'strIngredient8': None, 'strIngredient9': None, 'strIngredient10': None, 'strIngredient11': None, 'strIngredient12': None, 'strIngredient13': None, 'strIngredient14': None, 'strIngredient15': None, 'strMeasure1': '1 oz ', 'strMeasure2': '1 oz ', 'strMeasure3': '4 oz ', 'strMeasure4': None, 'strMeasure5': None, 'strMeasure6': None, 'strMeasure7': None, 'strMeasure8': None, 'strMeasure9': None, 'strMeasure10': None, 'strMeasure11': None, 'strMeasure12': None, 'strMeasure13': None, 'strMeasure14': None, 'strMeasure15': None, 'strImageSource': None, 'strImageAttribution': None, 'strCreativeCommonsConfirmed': 'No', 'dateModified': '2017-04-24 22:00:19'}, {'idDrink': '17833', 'strDrink': 'A. J.', 'strDrinkAlternate': None, 'strTags': None, 'strVideo': None, 'strCategory': 'Ordinary Drink', 'strIBA': None, 'strAlcoholic': 'Alcoholic', 'strGlass': 'Cocktail glass', 'strInstructions': 'Shake ingredients with ice, strain into a cocktail glass, and serve.', 'strInstructionsES': 'Agite bien los ingredientes con hielo, cuélelos en una copa de cóctel y sirva.', 'strInstructionsDE': None, 'strInstructionsFR': None, 'strInstructionsIT': 'Shakerare gli ingredienti con ghiaccio, filtrare in una coppetta da cocktail e servire.', 'strInstructionsZH-HANS': None, 'strInstructionsZH-HANT': None, 'strDrinkThumb': 'https://www.thecocktaildb.com/images/media/drink/l74qo91582480316.jpg', 'strIngredient1': 'Applejack', 'strIngredient2': 'Grapefruit juice', 'strIngredient3': None, 'strIngredient4': None, 'strIngredient5': None, 'strIngredient6': None, 'strIngredient7': None, 'strIngredient8': None, 'strIngredient9': None, 'strIngredient10': None, 'strIngredient11': None, 'strIngredient12': None, 'strIngredient13': None, 'strIngredient14': None, 'strIngredient15': None, 'strMeasure1': '1 1/2 oz ', 'strMeasure2': '1 oz ', 'strMeasure3': None, 'strMeasure4': None, 'strMeasure5': None, 'strMeasure6': None, 'strMeasure7': None, 'strMeasure8': None, 'strMeasure9': None, 'strMeasure10': None, 'strMeasure11': None, 'strMeasure12': None, 'strMeasure13': None, 'strMeasure14': None, 'strMeasure15': None, 'strImageSource': None, 'strImageAttribution': None, 'strCreativeCommonsConfirmed': 'Yes', 'dateModified': '2016-09-

02 11:16:58'}, {'idDrink': '15266', 'strDrink': 'Avalon', 'strDrinkAlternate': None, 'strTags': None, 'strVideo': None, 'strCategory': 'Ordinary Drink', 'strIBA': None, 'strAlcoholic': 'Alcoholic', 'strGlass': 'Highball glass', 'strInstructions': 'Fill a tall glass with ice. Layer the Finlandia Vodka, lemon and apple juices, Pisang Ambon, and top up with lemonade. Stir slightly and garnish with a spiralled cucumber skin and a red cherry. The cucumber provides zest and looks attractive. This drink, created by Timo Haimi, took first prize in the 1991 Finlandia Vodka Long Drink Competition.', 'strInstructionsES': 'Llena un vaso alto con hielo. Agregue el vodka, los jugos de limón y manzana, el licor de plátano Pisang Ambon y complete con limonada. Revuelva ligeramente y adorne con una cáscara de pepino en espiral y una cereza roja. El pepino proporciona entusiasmo y se ve atractivo. Esta bebida, creada por Timo Haimi, ganó el primer premio en el Concurso de Tragos Largos de Vodka de Finlandia de 1991.', 'strInstructionsDE': 'Füllen Sie ein hohes Glas mit Eis. Legen Sie den Finlandia Wodka, Zitronen- und Apfelsäfte, Pisang Ambon und geben Sie Limonade dazu. Leicht umrühren und mit einer spiralförmigen Gurkenhaut und einer roten Kirsche garnieren. Die Gurke sorgt für Schärfe und sieht attraktiv aus. Dieses von Timo Haimi kreierte Getränk erhielt 1991 den ersten Preis beim Finlandia Wodka Long Drink Wettbewerb.', 'strInstructionsFR': None, 'strInstructionsIT': 'Riempi un bicchiere alto di ghiaccio.\r\nVersare la Vodka, succhi di limone, mela, Pisang Ambon o un liquore alla banana e completare con la limonata.\r\nMescolare leggermente e guarnire con una buccia di cetriolo a spirale e una ciliegia rossa.\r\nIl cetriolo fornisce la scorza e sembra attraente.\r\nQuesta bevanda, creata da Timo Haimi, ha vinto il primo premio nel 1991 Finlandia Vodka Long Drink Competition.', 'strInstructionsZH-HANS': None, 'strInstructionsZH-HANT': None, 'strDrinkThumb': 'https://www.thecocktaildb.com/images/media/drink/3k9qic1493068931.jpg', 'strIngredient1': 'Vodka', 'strIngredient2': 'Pisang Ambon', 'strIngredient3': 'Apple juice', 'strIngredient4': 'Lemon juice', 'strIngredient5': 'Lemonade', 'strIngredient6': None, 'strIngredient7': None, 'strIngredient8': None, 'strIngredient9': None, 'strIngredient10': None, 'strIngredient11': None, 'strIngredient12': None, 'strIngredient13': None, 'strIngredient14': None, 'strIngredient15': None, 'strMeasure1': '3 parts', 'strMeasure2': '1 part ', 'strMeasure3': '6 parts ', 'strMeasure4': '1 1/2 part ', 'strMeasure5': None, 'strMeasure6': None, 'strMeasure7': None, 'strMeasure8': None, 'strMeasure9': None, 'strMeasure10': None, 'strMeasure11': None, 'strMeasure12': None, 'strMeasure13': None, 'strMeasure14': None, 'strMeasure15': None, 'strImageSource': None, 'strImageAttribution': None, 'strCreativeCommonsConfirmed': 'No', 'dateModified': '2017-04-24 22:22:11'}, {'idDrink': '15106', 'strDrink': 'Apello', 'strDrinkAlternate': None, 'strTags': None, 'strVideo': None, 'strCategory': 'Other / Unknown', 'strIBA': None, 'strAlcoholic': 'Non alcoholic', 'strGlass': 'Collins Glass', 'strInstructions': 'Stirr. Grnish with maraschino cherry.', 'strInstructionsES': 'Mezclar. Decorar con cereza marrasquino.',

'strInstructionsDE': 'Rühren. Mit Maraschinokirsche garnieren.', 'strInstructionsFR': None, 'strInstructionsIT': 'Versare tutti gli ingredienti in un mixing glass. Mescolate.\r\nGuarnire con una ciliegia al maraschino.', 'strInstructionsZH-HANS': None, 'strInstructionsZH-HANT': None, 'strDrinkThumb': 'https://www.thecocktaildb.com/images/media/drink/uptxtv1468876415.jpg', 'strIngredient1': 'Orange juice', 'strIngredient2': 'Grapefruit juice', 'strIngredient3': 'Apple juice', 'strIngredient4': 'Maraschino cherry', 'strIngredient5': None, 'strIngredient6': None, 'strIngredient7': None, 'strIngredient8': None, 'strIngredient9': None, 'strIngredient10': None, 'strIngredient11': None, 'strIngredient12': None, 'strIngredient13': None, 'strIngredient14': None, 'strIngredient15': None, 'strMeasure1': '4 cl ', 'strMeasure2': '3 cl ', 'strMeasure3': '1 cl ', 'strMeasure4': '1 ', 'strMeasure5': None, 'strMeasure6': None, 'strMeasure7': None, 'strMeasure8': None, 'strMeasure9': None, 'strMeasure10': None, 'strMeasure11': None, 'strMeasure12': None, 'strMeasure13': None, 'strMeasure14': None, 'strMeasure15': None, 'strImageSource': None, 'strImageAttribution': None, 'strCreativeCommonsConfirmed': 'No', 'dateModified': '2016-07-18 22:13:35'}, {'idDrink': '17839', 'strDrink': 'Affair', 'strDrinkAlternate': None, 'strTags': None, 'strVideo': None, 'strCategory': 'Ordinary Drink', 'strIBA': None, 'strAlcoholic': 'Alcoholic', 'strGlass': 'Highball glass', 'strInstructions': 'Pour schnapps, orange juice, and cranberry juice over ice in a highball glass. Top with club soda and serve.', 'strInstructionsES': 'Vierta el Schnapps, el jugo de naranja y el jugo de arándano sobre hielo en un vaso alto. Complete con el agua carbonatada y sirva.', 'strInstructionsDE': 'Gießen Sie Schnaps, Orangensaft und Cranberrysaft über Eis in ein Highball-Glas. Mit Club-Soda überziehen und servieren.', 'strInstructionsFR': None, 'strInstructionsIT': "Versare la grappa, il succo d'arancia e il succo di mirtillo rosso sul ghiaccio in un bicchiere highball. Completare con la soda club e servire.", 'strInstructionsZH-HANS': None, 'strInstructionsZH-HANT': None, 'strDrinkThumb': 'https://www.thecocktaildb.com/images/media/drink/h5za6y1582477994.jpg', 'strIngredient1': 'Strawberry schnapps', 'strIngredient2': 'Orange juice', 'strIngredient3': 'Cranberry juice', 'strIngredient4': 'Club soda', 'strIngredient5': None, 'strIngredient6': None, 'strIngredient7': None, 'strIngredient8': None, 'strIngredient9': None, 'strIngredient10': None, 'strIngredient11': None, 'strIngredient12': None, 'strIngredient13': None, 'strIngredient14': None, 'strIngredient15': None, 'strMeasure1': '2 oz ', 'strMeasure2': '2 oz ', 'strMeasure3': '2 oz ', 'strMeasure4': None, 'strMeasure5': None, 'strMeasure6': None, 'strMeasure7': None, 'strMeasure8': None, 'strMeasure9': None, 'strMeasure10': None, 'strMeasure11': None, 'strMeasure12': None, 'strMeasure13': None, 'strMeasure14': None, 'strMeasure15': None, 'strImageSource': None, 'strImageAttribution': None, 'strCreativeCommonsConfirmed': 'Yes', 'dateModified': '2016-09-01 10:05:34'}, {'idDrink': '17835', 'strDrink': 'Abilene', 'strDrinkAlternate': None, 'strTags': None, 'strVideo': None,

'strCategory': 'Ordinary Drink', 'strIBA': None, 'strAlcoholic': 'Alcoholic', 'strGlass': 'Highball glass', 'strInstructions': 'Pour all of the ingredients into a highball glass almost filled with ice cubes. Stir well.', 'strInstructionsES': 'Coloque todos los ingredientes en un vaso alto casi lleno de cubitos de hielo. Sacudir bien.', 'strInstructionsDE': None, 'strInstructionsFR': None, 'strInstructionsIT': 'Versare tutti gli ingredienti in un bicchiere highball riempito di cubetti di ghiaccio. Mescolare bene.', 'strInstructionsZH-HANS': None, 'strInstructionsZH-HANT': None, 'strDrinkThumb': 'https://www.thecocktaildb.com/images/media/drink/smb2oe1582479072.jpg', 'strIngredient1': 'Dark rum', 'strIngredient2': 'Peach nectar', 'strIngredient3': 'Orange juice', 'strIngredient4': None, 'strIngredient5': None, 'strIngredient6': None, 'strIngredient7': None, 'strIngredient8': None, 'strIngredient9': None, 'strIngredient10': None, 'strIngredient11': None, 'strIngredient12': None, 'strIngredient13': None, 'strIngredient14': None, 'strIngredient15': None, 'strMeasure1': '1 1/2 oz ', 'strMeasure2': '2 oz ', 'strMeasure3': '3 oz ', 'strMeasure4': None, 'strMeasure5': None, 'strMeasure6': None, 'strMeasure7': None, 'strMeasure8': None, 'strMeasure9': None, 'strMeasure10': None, 'strMeasure11': None, 'strMeasure12': None, 'strMeasure13': None, 'strMeasure14': None, 'strMeasure15': None, 'strImageSource': None, 'strImageAttribution': None, 'strCreativeCommonsConfirmed': 'Yes', 'dateModified': '2016-04-29 09:44:25'}, {'idDrink': '11023', 'strDrink': 'Almeria', 'strDrinkAlternate': None, 'strTags': None, 'strVideo': None, 'strCategory': 'Ordinary Drink', 'strIBA': None, 'strAlcoholic': 'Alcoholic', 'strGlass': 'Cocktail glass', 'strInstructions': 'In a shaker half-filled with ice cubes, combine all of the ingredients. Shake well. Strain into a cocktail glass.', 'strInstructionsES': 'En una coctelera llena hasta la mitad con cubitos de hielo, mezcla todos los ingredientes. Agitar bien. Colar el contenido en un vaso de cóctel.', 'strInstructionsDE': 'In einem Shaker, der halb mit Eiswürfeln gefüllt ist, alle Zutaten vermengen. Gut schütteln. In ein Cocktailglas abseihen.', 'strInstructionsFR': None, 'strInstructionsIT': 'In uno shaker riempito a metà con cubetti di ghiaccio, unire tutti gli ingredienti.Agitare bene.Filtrare in un bicchiere da cocktail.', 'strInstructionsZH-HANS': None, 'strInstructionsZH-HANT': None, 'strDrinkThumb': 'https://www.thecocktaildb.com/images/media/drink/rwsyyu1483388181.jpg', 'strIngredient1': 'Dark rum', 'strIngredient2': 'Kahlua', 'strIngredient3': 'Egg white', 'strIngredient4': None, 'strIngredient5': None, 'strIngredient6': None, 'strIngredient7': None, 'strIngredient8': None, 'strIngredient9': None, 'strIngredient10': None, 'strIngredient11': None, 'strIngredient12': None, 'strIngredient13': None, 'strIngredient14': None, 'strIngredient15': None, 'strMeasure1': '2 oz ', 'strMeasure2': '1 oz ', 'strMeasure3': '1 ', 'strMeasure4': None, 'strMeasure5': None, 'strMeasure6': None, 'strMeasure7': None, 'strMeasure8': None, 'strMeasure9': None, 'strMeasure10': None, 'strMeasure11': None,

'strMeasure12': None, 'strMeasure13': None, 'strMeasure14': None, 'strMeasure15': None, 'strImageSource': None, 'strImageAttribution': None, 'strCreativeCommonsConfirmed': 'No', 'dateModified': '2017-01-02 20:16:21'}, {'idDrink': '17228', 'strDrink': 'Addison', 'strDrinkAlternate': None, 'strTags': None, 'strVideo': None, 'strCategory': 'Cocktail', 'strIBA': None, 'strAlcoholic': 'Alcoholic', 'strGlass': 'Martini Glass', 'strInstructions': 'Shake together all the ingredients and strain into a cold glass.', 'strInstructionsES': 'Agitar todos los ingredientes y colar la mezcla en un vaso enfriado.', 'strInstructionsDE': 'Alle Zutaten vermengen und in ein kaltes Glas abseihen.', 'strInstructionsFR': None, 'strInstructionsIT': 'Shakerare insieme tutti gli ingredienti e filtrare in un bicchiere freddo.', 'strInstructionsZH-HANS': None, 'strInstructionsZH-HANT': None, 'strDrinkThumb': 'https://www.thecocktaildb.com/images/media/drink/yzva7x1504820300.jpg', 'strIngredient1': 'Gin', 'strIngredient2': 'Vermouth', 'strIngredient3': None, 'strIngredient4': None, 'strIngredient5': None, 'strIngredient6': None, 'strIngredient7': None, 'strIngredient8': None, 'strIngredient9': None, 'strIngredient10': None, 'strIngredient11': None, 'strIngredient12': None, 'strIngredient13': None, 'strIngredient14': None, 'strIngredient15': None, 'strMeasure1': '1 1/2 shot ', 'strMeasure2': '1 1/2 shot ', 'strMeasure3': None, 'strMeasure4': None, 'strMeasure5': None, 'strMeasure6': None, 'strMeasure7': None, 'strMeasure8': None, 'strMeasure9': None, 'strMeasure10': None, 'strMeasure11': None, 'strMeasure12': None, 'strMeasure13': None, 'strMeasure14': None, 'strMeasure15': None, 'strImageSource': None, 'strImageAttribution': None, 'strCreativeCommonsConfirmed': 'No', 'dateModified': '2017-09-07 22:38:20'}, {'idDrink': '11046', 'strDrink': 'Applecar', 'strDrinkAlternate': None, 'strTags': None, 'strVideo': None, 'strCategory': 'Ordinary Drink', 'strIBA': None, 'strAlcoholic': 'Alcoholic', 'strGlass': 'Cocktail glass', 'strInstructions': 'Shake all ingredients with ice, strain into a cocktail glass, and serve.', 'strInstructionsES': 'Agitar todos los ingredientes con hielo, colar en un vaso de cóctel y servir.', 'strInstructionsDE': 'Alle Zutaten mit Eis schütteln, in ein Cocktailglas abseihen und servieren.', 'strInstructionsFR': None, 'strInstructionsIT': 'Shakerare tutti gli ingredienti con ghiaccio, filtrare in una coppetta da cocktail e servire.', 'strInstructionsZH-HANS': None, 'strInstructionsZH-HANT': None, 'strDrinkThumb': 'https://www.thecocktaildb.com/images/media/drink/sbffau1504389764.jpg', 'strIngredient1': 'Applejack', 'strIngredient2': 'Triple sec', 'strIngredient3': 'Lemon juice', 'strIngredient4': None, 'strIngredient5': None, 'strIngredient6': None, 'strIngredient7': None, 'strIngredient8': None, 'strIngredient9': None, 'strIngredient10': None, 'strIngredient11': None, 'strIngredient12': None, 'strIngredient13': None, 'strIngredient14': None, 'strIngredient15': None, 'strMeasure1': '1 oz ', 'strMeasure2': '1 oz ', 'strMeasure3': '1 oz ', 'strMeasure4': None, 'strMeasure5': None, 'strMeasure6': None, 'strMeasure7': None, 'strMeasure8': None,

'strMeasure9': None, 'strMeasure10': None, 'strMeasure11': None, 'strMeasure12': None, 'strMeasure13': None, 'strMeasure14': None, 'strMeasure15': None, 'strImageSource': None, 'strImageAttribution': None, 'strCreativeCommonsConfirmed': 'No', 'dateModified': '2017-09-02 23:02:44'}, {'idDrink': '17836', 'strDrink': 'Acapulco', 'strDrinkAlternate': None, 'strTags': None, 'strVideo': None, 'strCategory': 'Ordinary Drink', 'strIBA': None, 'strAlcoholic': 'Alcoholic', 'strGlass': 'Old-fashioned glass', 'strInstructions': 'Combine and shake all ingredients (except mint) with ice and strain into an old-fashioned glass over ice cubes. Add the sprig of mint and serve.', 'strInstructionsES': 'Mezcle y agite todos los ingredientes (excepto la menta) con hielo y cuélelos en un vaso de rocas sobre cubitos de hielo. Añadir una ramita de menta y servir.', 'strInstructionsDE': 'Alle Zutaten (außer Minze) mit Eis mischen und schütteln und in ein old-fashioned Glas über Eiswürfel abseihen. Den Minzzweig dazugeben und servieren.', 'strInstructionsFR': None, 'strInstructionsIT': 'Unire e scuotere tutti gli ingredienti (tranne la menta) con ghiaccio e filtrare in un bicchiere vecchio stile su cubetti di ghiaccio.Aggiungere il rametto di menta e servire.', 'strInstructionsZH-HANS': None, 'strInstructionsZH-HANT': None, 'strDrinkThumb': 'https://www.thecocktaildb.com/images/media/drink/il9e0r1582478841.jpg', 'strIngredient1': 'Light rum', 'strIngredient2': 'Triple sec', 'strIngredient3': 'Lime juice', 'strIngredient4': 'Sugar', 'strIngredient5': 'Egg white', 'strIngredient6': 'Mint', 'strIngredient7': None, 'strIngredient8': None, 'strIngredient9': None, 'strIngredient10': None, 'strIngredient11': None, 'strIngredient12': None, 'strIngredient13': None, 'strIngredient14': None, 'strIngredient15': None, 'strMeasure1': '1 1/2 oz ', 'strMeasure2': '1 1/2 tsp ', 'strMeasure3': '1 tblsp ', 'strMeasure4': '1 tsp ', 'strMeasure5': '1 ', 'strMeasure6': '1 ', 'strMeasure7': None, 'strMeasure8': None, 'strMeasure9': None, 'strMeasure10': None, 'strMeasure11': None, 'strMeasure12': None, 'strMeasure13': None, 'strMeasure14': None, 'strMeasure15': None, 'strImageSource': None, 'strImageAttribution': None, 'strCreativeCommonsConfirmed': 'Yes', 'dateModified': '2016-09-02 11:26:16'}, {'idDrink': '17840', 'strDrink': 'Affinity', 'strDrinkAlternate': None, 'strTags': None, 'strVideo': None, 'strCategory': 'Ordinary Drink', 'strIBA': None, 'strAlcoholic': 'Alcoholic', 'strGlass': 'Cocktail glass', 'strInstructions': 'In a mixing glass half-filled with ice cubes, combine all of the ingredients. Stir well. Strain into a cocktail glass.', 'strInstructionsES': 'En un vaso mezclador lleno hasta la mitad con cubitos de hielo, combine todos los ingredientes. Sacudir bien. Colar en un vaso de cóctel.', 'strInstructionsDE': 'In einem Mischglas, das halb mit Eiswürfeln gefüllt ist, alle Zutaten vermengen. Gut umrühren. In ein Cocktailglas abseihen.', 'strInstructionsFR': None, 'strInstructionsIT': 'In un mixing glass riempito a metà con cubetti di ghiaccio, unire tutti gli ingredienti. Filtrare in un bicchiere da cocktail. Mescolare bene.', 'strInstructionsZH-HANS': None, 'strInstructionsZH-HANT': None,

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e del giorno.", 'strInstructionsZH-HANS': None, 'strInstructionsZH-HANT': None, 'strDrinkThumb': 'https://www.thecocktaildb.com/images/media/drink/7dkf0i1487602928.jpg', 'strIngredient1': 'Vodka', 'strIngredient2': 'Lime juice', 'strIngredient3': 'Soda water', 'strIngredient4': None, 'strIngredient5': None, 'strIngredient6': None, 'strIngredient7': None, 'strIngredient8': None, 'strIngredient9': None, 'strIngredient10': None, 'strIngredient11': None, 'strIngredient12': None, 'strIngredient13': None, 'strIngredient14': None, 'strIngredient15': None, 'strMeasure1': '4 cl ', 'strMeasure2': '1 dash ', 'strMeasure3': None, 'strMeasure4': None, 'strMeasure5': None, 'strMeasure6': None, 'strMeasure7': None, 'strMeasure8': None, 'strMeasure9': None, 'strMeasure10': None, 'strMeasure11': None, 'strMeasure12': None, 'strMeasure13': None, 'strMeasure14': None, 'strMeasure15': None, 'strImageSource': None, 'strImageAttribution': None, 'strCreativeCommonsConfirmed': 'No', 'dateModified': '2017-02-20 15:02:08'}, {'idDrink': '11021', 'strDrink': 'Allegheny', 'strDrinkAlternate': None, 'strTags': None, 'strVideo': None, 'strCategory': 'Ordinary Drink', 'strIBA': None, 'strAlcoholic': 'Alcoholic', 'strGlass': 'Cocktail glass', 'strInstructions': 'Shake all ingredients (except lemon peel) with ice and strain into a cocktail glass. Top with the twist of lemon peel and serve.', 'strInstructionsES': 'Agite todos los ingredientes (excepto el twist de limón) con hielo y cuele en un vaso de cóctel. Cubra con el twist de limón y sirva.', 'strInstructionsDE': 'Alle Zutaten (außer Zitronenschale) mit Eis schütteln und in ein Cocktailglas abseihen. Mit der Drehung der Zitronenschale garnieren und servieren.', 'strInstructionsFR': None, 'strInstructionsIT': 'Shakerare tutti gli ingredienti (tranne la buccia di limone) con ghiaccio e filtrare in una coppetta da cocktail.Completare con la scorza di limone e servire.', 'strInstructionsZH-HANS': None, 'strInstructionsZH-HANT': None, 'strDrinkThumb': 'https://www.thecocktaildb.com/images/media/drink/uwvyts1483387934.jpg', 'strIngredient1': 'Dry Vermouth', 'strIngredient2': 'Bourbon', 'strIngredient3': 'Blackberry brandy', 'strIngredient4': 'Lemon juice', 'strIngredient5': 'Lemon peel', 'strIngredient6': None, 'strIngredient7': None, 'strIngredient8': None, 'strIngredient9': None, 'strIngredient10': None, 'strIngredient11': None, 'strIngredient12': None, 'strIngredient13': None, 'strIngredient14': None, 'strIngredient15': None, 'strMeasure1': '1 oz ', 'strMeasure2': '1 oz ', 'strMeasure3': '1 1/2 tsp ', 'strMeasure4': '1 1/2 tsp ', 'strMeasure5': '1 twist of ', 'strMeasure6': None, 'strMeasure7': None, 'strMeasure8': None, 'strMeasure9': None, 'strMeasure10': None, 'strMeasure11': None, 'strMeasure12': None, 'strMeasure13': None, 'strMeasure14': None, 'strMeasure15': None, 'strImageSource': None, 'strImageAttribution': None, 'strCreativeCommonsConfirmed': 'No', 'dateModified': '2017-01-02 20:12:14'}, {'idDrink': '15941', 'strDrink': 'Americano', 'strDrinkAlternate': None, 'strTags': 'IBA,Classic', 'strVideo': 'https://www.youtube.com/watch?v=TmeUJ2g3ogM', 'strCategory': 'Ordinary Drink', 'strIBA':

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de naranja.', 'strInstructionsDE': 'Den Campari und den Wermut über
Eis in ein Glas gießen, einen Spritzer Sodawasser hinzufügen und mit
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```python
from IPython.display import Image, display

for drink in data['drinks']:
  print('---------------')
  print(drink['idDrink'])
  print(drink['strCategory'])
  print(drink['strInstructions'])
  print(drink['strDrinkThumb'])
  display(Image(url=drink['strDrinkThumb'], width=200))

  for key, value in drink.items():
    if key.startswith("strIngredient") and value is not None:
        print(f"{key}: {value}")
```

```
---------------
17222
Cocktail
Pour all ingredients into a cocktail shaker, mix and serve over ice
```

```
into a chilled glass.
https://www.thecocktaildb.com/images/media/drink/2x8thr1504816928.jpg

<IPython.core.display.Image object>

strIngredient1: Gin
strIngredient2: Grand Marnier
strIngredient3: Lemon Juice
strIngredient4: Grenadine
---------------
13501
Shot
Layered in a shot glass.
https://www.thecocktaildb.com/images/media/drink/tqpvqp1472668328.jpg

<IPython.core.display.Image object>

strIngredient1: Amaretto
strIngredient2: Baileys irish cream
strIngredient3: Cognac
---------------
17225
Cocktail
Shake all the ingredients in a cocktail shaker and ice then strain in
a cold glass.
https://www.thecocktaildb.com/images/media/drink/l3cd7f1504818306.jpg

<IPython.core.display.Image object>

strIngredient1: Gin
strIngredient2: Grenadine
strIngredient3: Heavy cream
strIngredient4: Milk
strIngredient5: Egg White
---------------
14610
Shot
Poor in the 151 first followed by the 101 served with a Coke or Dr
Pepper chaser.
https://www.thecocktaildb.com/images/media/drink/xuxpxt1479209317.jpg

<IPython.core.display.Image object>

strIngredient1: 151 proof rum
strIngredient2: Wild Turkey
---------------
17837
Ordinary Drink
In a shaker half-filled with ice cubes, combine all of the
ingredients. Shake well. Strain into a cocktail glass.
https://www.thecocktaildb.com/images/media/drink/v0at4i1582478473.jpg
```

```
<IPython.core.display.Image object>

strIngredient1: Dark rum
strIngredient2: Lemon juice
strIngredient3: Grenadine
---------------
13938
Ordinary Drink
Pour Vodka and Gin over ice, add Tonic and Stir
https://www.thecocktaildb.com/images/media/drink/rhhwmp1493067619.jpg

<IPython.core.display.Image object>

strIngredient1: Absolut Vodka
strIngredient2: Gin
strIngredient3: Tonic water
---------------
17833
Ordinary Drink
Shake ingredients with ice, strain into a cocktail glass, and serve.
https://www.thecocktaildb.com/images/media/drink/l74qo91582480316.jpg

<IPython.core.display.Image object>

strIngredient1: Applejack
strIngredient2: Grapefruit juice
---------------
15266
Ordinary Drink
Fill a tall glass with ice. Layer the Finlandia Vodka, lemon and apple
juices, Pisang Ambon, and top up with lemonade. Stir slightly and
garnish with a spiralled cucumber skin and a red cherry. The cucumber
provides zest and looks attractive. This drink, created by Timo Haimi,
took first prize in the 1991 Finlandia Vodka Long Drink Competition.
https://www.thecocktaildb.com/images/media/drink/3k9qic1493068931.jpg

<IPython.core.display.Image object>

strIngredient1: Vodka
strIngredient2: Pisang Ambon
strIngredient3: Apple juice
strIngredient4: Lemon juice
strIngredient5: Lemonade
---------------
15106
Other / Unknown
Stirr. Grnish with maraschino cherry.
https://www.thecocktaildb.com/images/media/drink/uptxtv1468876415.jpg

<IPython.core.display.Image object>
```

```
strIngredient1: Orange juice
strIngredient2: Grapefruit juice
strIngredient3: Apple juice
strIngredient4: Maraschino cherry
---------------
17839
Ordinary Drink
Pour schnapps, orange juice, and cranberry juice over ice in a
highball glass. Top with club soda and serve.
https://www.thecocktaildb.com/images/media/drink/h5za6y1582477994.jpg

<IPython.core.display.Image object>

strIngredient1: Strawberry schnapps
strIngredient2: Orange juice
strIngredient3: Cranberry juice
strIngredient4: Club soda
---------------
17835
Ordinary Drink
Pour all of the ingredients into a highball glass almost filled with
ice cubes. Stir well.
https://www.thecocktaildb.com/images/media/drink/smb2oe1582479072.jpg

<IPython.core.display.Image object>

strIngredient1: Dark rum
strIngredient2: Peach nectar
strIngredient3: Orange juice
---------------
11023
Ordinary Drink
In a shaker half-filled with ice cubes, combine all of the
ingredients. Shake well. Strain into a cocktail glass.
https://www.thecocktaildb.com/images/media/drink/rwsyyu1483388181.jpg

<IPython.core.display.Image object>

strIngredient1: Dark rum
strIngredient2: Kahlua
strIngredient3: Egg white
---------------
17228
Cocktail
Shake together all the ingredients and strain into a cold glass.
https://www.thecocktaildb.com/images/media/drink/yzva7x1504820300.jpg

<IPython.core.display.Image object>

strIngredient1: Gin
strIngredient2: Vermouth
```

```
---------------
11046
Ordinary Drink
Shake all ingredients with ice, strain into a cocktail glass, and
serve.
https://www.thecocktaildb.com/images/media/drink/sbffau1504389764.jpg

<IPython.core.display.Image object>

strIngredient1: Applejack
strIngredient2: Triple sec
strIngredient3: Lemon juice
---------------
17836
Ordinary Drink
Combine and shake all ingredients (except mint) with ice and strain
into an old-fashioned glass over ice cubes. Add the sprig of mint and
serve.
https://www.thecocktaildb.com/images/media/drink/il9e0r1582478841.jpg

<IPython.core.display.Image object>

strIngredient1: Light rum
strIngredient2: Triple sec
strIngredient3: Lime juice
strIngredient4: Sugar
strIngredient5: Egg white
strIngredient6: Mint
---------------
17840
Ordinary Drink
In a mixing glass half-filled with ice cubes, combine all of the
ingredients. Stir well. Strain into a cocktail glass.
https://www.thecocktaildb.com/images/media/drink/wzdtnn1582477684.jpg

<IPython.core.display.Image object>

strIngredient1: Scotch
strIngredient2: Sweet Vermouth
strIngredient3: Dry Vermouth
strIngredient4: Orange bitters
---------------
17180
Cocktail
Add all ingredients into cocktail shaker filled with ice. Shake well
and strain into cocktail glass. Garnish with a cherry.
https://www.thecocktaildb.com/images/media/drink/trbplb1606855233.jpg

<IPython.core.display.Image object>
```

```
strIngredient1: Gin
strIngredient2: lemon juice
strIngredient3: maraschino liqueur
---------------
15182
Ordinary Drink
Pour the vodka and creme over some ice cubes in a tall glass and fill
up with juice. to make it beuty full make the top of the glass with a
grenadine and sugar
https://www.thecocktaildb.com/images/media/drink/xrl66i1493068702.jpg

<IPython.core.display.Image object>

strIngredient1: Vodka
strIngredient2: Creme de Banane
strIngredient3: Orange juice
---------------
16311
Cocktail
Add all ingredients into mixing glass, chill and strain into cocktail
glass
https://www.thecocktaildb.com/images/media/drink/sutyqp1479209062.jpg

<IPython.core.display.Image object>

strIngredient1: Jack Daniels
strIngredient2: Midori melon liqueur
strIngredient3: Sour mix
---------------
12560
Cocktail
Mix. Serve over ice.
https://www.thecocktaildb.com/images/media/drink/vuquyv1468876052.jpg

<IPython.core.display.Image object>

strIngredient1: Grenadine
strIngredient2: Orange juice
strIngredient3: Pineapple juice
---------------
13162
Coffee / Tea
Build into a suiting glass, with no ice. Cream on top if wanted.
Served directly.
https://www.thecocktaildb.com/images/media/drink/vyrurp1472667777.jpg

<IPython.core.display.Image object>

strIngredient1: Kahlua
strIngredient2: Baileys irish cream
strIngredient3: Frangelico
```

```
strIngredient4: Coffee
strIngredient5: Cream
---------------
11014
Ordinary Drink
Shake all ingredients with ice and strain contents into a cocktail
glass. Sprinkle nutmeg on top and serve.
https://www.thecocktaildb.com/images/media/drink/0clus51606772388.jpg

<IPython.core.display.Image object>

strIngredient1: Gin
strIngredient2: Creme de Cacao
strIngredient3: Light cream
strIngredient4: Nutmeg
---------------
12756
Ordinary Drink
Mix and fill up with soda water. Drunk by finns on a sunny day any
time of the year and day.
https://www.thecocktaildb.com/images/media/drink/7dkf0i1487602928.jpg

<IPython.core.display.Image object>

strIngredient1: Vodka
strIngredient2: Lime juice
strIngredient3: Soda water
---------------
11021
Ordinary Drink
Shake all ingredients (except lemon peel) with ice and strain into a
cocktail glass. Top with the twist of lemon peel and serve.
https://www.thecocktaildb.com/images/media/drink/uwvyts1483387934.jpg

<IPython.core.display.Image object>

strIngredient1: Dry Vermouth
strIngredient2: Bourbon
strIngredient3: Blackberry brandy
strIngredient4: Lemon juice
strIngredient5: Lemon peel
---------------
15941
Ordinary Drink
Pour the Campari and vermouth over ice into glass, add a splash of
soda water and garnish with half orange slice.
https://www.thecocktaildb.com/images/media/drink/709s6m1613655124.jpg

<IPython.core.display.Image object>
```

```
strIngredient1: Campari
strIngredient2: Sweet Vermouth
strIngredient3: Lemon peel
strIngredient4: Orange peel
```

Creem o lista cu toate ingredientele disponibile

```
ingrediente = []
for drink in data['drinks']:
  for key, value in drink.items():
    if key.startswith("strIngredient") and value is not None:
        ingrediente.append(value)

print(ingrediente)
print(len(ingrediente))
unique_ingredients = set(ingrediente)
print(len(unique_ingredients))
```

```
['Gin', 'Grand Marnier', 'Lemon Juice', 'Grenadine', 'Amaretto',
'Baileys irish cream', 'Cognac', 'Gin', 'Grenadine', 'Heavy cream',
'Milk', 'Egg White', '151 proof rum', 'Wild Turkey', 'Dark rum',
'Lemon juice', 'Grenadine', 'Absolut Vodka', 'Gin', 'Tonic water',
'Applejack', 'Grapefruit juice', 'Vodka', 'Pisang Ambon', 'Apple
juice', 'Lemon juice', 'Lemonade', 'Orange juice', 'Grapefruit juice',
'Apple juice', 'Maraschino cherry', 'Strawberry schnapps', 'Orange
juice', 'Cranberry juice', 'Club soda', 'Dark rum', 'Peach nectar',
'Orange juice', 'Dark rum', 'Kahlua', 'Egg white', 'Gin', 'Vermouth',
'Applejack', 'Triple sec', 'Lemon juice', 'Light rum', 'Triple sec',
'Lime juice', 'Sugar', 'Egg white', 'Mint', 'Scotch', 'Sweet
Vermouth', 'Dry Vermouth', 'Orange bitters', 'Gin', 'lemon juice',
'maraschino liqueur', 'Vodka', 'Creme de Banane', 'Orange juice',
'Jack Daniels', 'Midori melon liqueur', 'Sour mix', 'Grenadine',
'Orange juice', 'Pineapple juice', 'Kahlua', 'Baileys irish cream',
'Frangelico', 'Coffee', 'Cream', 'Gin', 'Creme de Cacao', 'Light
cream', 'Nutmeg', 'Vodka', 'Lime juice', 'Soda water', 'Dry Vermouth',
'Bourbon', 'Blackberry brandy', 'Lemon juice', 'Lemon peel',
'Campari', 'Sweet Vermouth', 'Lemon peel', 'Orange peel']
89
59
```

Orase preluate de la API-ul geocities

```
accumulated_cities = []
url = "http://geodb-free-service.wirefreethought.com/v1/geo/places"
for offset in range(0, 100, 10):
    params = {"offset": offset, "limit": 10}
    response = requests.get(url, params=params)
    data = response.json()
```

```
    cities = [place for place in data["data"]]
    accumulated_cities.extend(cities)

for city in accumulated_cities[1:3]:
    print(city['population'])
```

```
490035
766936
```

```
city_with_max_population = max(accumulated_cities, key=lambda x:
x["population"])

city_with_min_population = min(accumulated_cities, key=lambda x:
x["population"])

total_population = sum([city["population"] for city in
accumulated_cities])

countries = set(city["country"] for city in accumulated_cities)

print("City with maximum population:")
print(city_with_max_population)

print("\nCity with minimum population:")
print(city_with_min_population)

print("\nTotal population:")
print(total_population)

print("\nTotal population:")
print(countries)
```

```
City with maximum population:
{'id': 216, 'wikiDataId': 'Q5838', 'type': 'CITY', 'name': 'Kabul',
'country': 'Afghanistan', 'countryCode': 'AF', 'region': 'Kabul',
'regionCode': 'KAB', 'regionWdId': 'Q188933', 'latitude':
34.532777777, 'longitude': 69.165833333, 'population': 4273156,
'distance': None, 'placeType': 'CITY'}

City with minimum population:
{'id': 12, 'wikiDataId': 'Q193226', 'type': 'CITY', 'name': 'Saranda',
'country': 'Albania', 'countryCode': 'AL', 'region': 'Vlorë County',
'regionCode': '12', 'regionWdId': 'Q192849', 'latitude': 39.883333333,
'longitude': 20.016666666, 'population': 41000, 'distance': None,
'placeType': 'CITY'}

Total population:
27474489

Total population:
```

```
{'Angola', 'Antigua and Barbuda', 'Armenia', 'Afghanistan', 'Albania',
'United Arab Emirates'}
```

Afisarea pe harta

```
!pip install cartopy
```

```
Collecting cartopy
  Downloading Cartopy-0.23.0-cp310-cp310-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl (11.6 MB)
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 11.6/11.6 MB 32.1 MB/s eta
0:00:00
ent already satisfied: numpy>=1.21 in /usr/local/lib/python3.10/dist-
packages (from cartopy) (1.25.2)
Requirement already satisfied: matplotlib>=3.5 in
/usr/local/lib/python3.10/dist-packages (from cartopy) (3.7.1)
Requirement already satisfied: shapely>=1.7 in
/usr/local/lib/python3.10/dist-packages (from cartopy) (2.0.3)
Requirement already satisfied: packaging>=20 in
/usr/local/lib/python3.10/dist-packages (from cartopy) (24.0)
Requirement already satisfied: pyshp>=2.3 in
/usr/local/lib/python3.10/dist-packages (from cartopy) (2.3.1)
Requirement already satisfied: pyproj>=3.3.1 in
/usr/local/lib/python3.10/dist-packages (from cartopy) (3.6.1)
Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.5-
>cartopy) (1.2.1)
Requirement already satisfied: cycler>=0.10 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.5-
>cartopy) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.5-
>cartopy) (4.51.0)
Requirement already satisfied: kiwisolver>=1.0.1 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.5-
>cartopy) (1.4.5)
Requirement already satisfied: pillow>=6.2.0 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.5-
>cartopy) (9.4.0)
Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.5-
>cartopy) (3.1.2)
Requirement already satisfied: python-dateutil>=2.7 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.5-
>cartopy) (2.8.2)
Requirement already satisfied: certifi in
/usr/local/lib/python3.10/dist-packages (from pyproj>=3.3.1->cartopy)
(2024.2.2)
Requirement already satisfied: six>=1.5 in
/usr/local/lib/python3.10/dist-packages (from python-dateutil>=2.7-
```

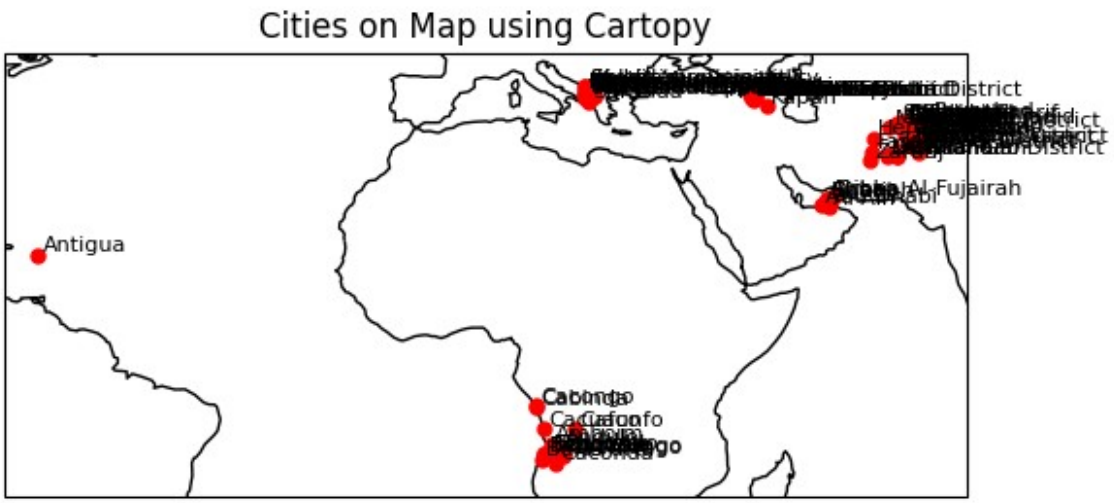
```
>matplotlib>=3.5->cartopy) (1.16.0)
Installing collected packages: cartopy
Successfully installed cartopy-0.23.0
```

```python
print(len(accumulated_cities))
```

```
100
```

```python
import matplotlib.pyplot as plt
import cartopy.crs as ccrs

lons = [city["longitude"] for city in accumulated_cities]
lats = [city["latitude"] for city in accumulated_cities]

# Calculate the extent of the map
min_lon, max_lon = min(lons), max(lons)
min_lat, max_lat = min(lats), max(lats)

# Create a map with a PlateCarree projection
fig, ax = plt.subplots(subplot_kw={'projection': ccrs.PlateCarree()})
ax.coastlines()

# Set the extent of the map
ax.set_extent([min_lon - 5, max_lon + 5, min_lat - 5, max_lat + 5],
crs=ccrs.PlateCarree())


# Plot the cities on the map
for city in accumulated_cities:
    lon, lat = city["longitude"], city["latitude"]
    ax.plot(lon, lat, 'ro', markersize=5,
transform=ccrs.PlateCarree())
    ax.text(lon + 0.6, lat + 0.6, city["name"], fontsize=8,
transform=ccrs.PlateCarree())

# Show the plot
plt.title('Cities on Map using Cartopy')
plt.show()
```

```
/usr/local/lib/python3.10/dist-packages/cartopy/io/__init__.py:241:
DownloadWarning: Downloading:
https://naturalearth.s3.amazonaws.com/110m_physical/ne_110m_coastline.
zip
  warnings.warn(f'Downloading: {url}', DownloadWarning)
```



Prelucrarea textului preluat din pagini WEB normale

Exemplu de izolare a textului dintr-o pagina web folosind justext

Justext este unul dintre proiectele care incearca izolarea textului util dintr-o pagina web. Procesul este cunoscut sub numele de "boilerplate removal".

justext

```
import sys
!{sys.executable} -m pip install justext

import justext
```

```
Collecting justext
  Downloading jusText-3.0.0-py2.py3-none-any.whl (837 kB)
━━━
```

```
paragraphs = justext.justext(response.content,
justext.get_stoplist("English"))
jstext = ''
for paragraph in paragraphs:
  if not paragraph.is_boilerplate:
    jstext = jstext + paragraph.text

print (jstext)
```

An overview of hierarchical storage managementHSM allows you to store data in a secure, cost-efficient manner. Here's what you need to know about it.Image: sdecoret/Adobe StockIf you're responsible for managing enterprise data, you know that storage can be both a blessing and a curse. On the one hand, having access to large amounts of data can be immensely helpful in making business decisions. On the other hand, storing all that data can be expensive and downright chaotic to manage.That's where hierarchical storage management comes in. HSM is a system for storing data in a secure, cost-efficient manner. In this guide, we give you a short crash course on HSM, what it is, how it works and some of the benefits it can offer your organization.What is HSM?HSM, or hierarchical storage management, is a system for storing data in a secure, cost-efficient manner. The basic idea behind HSM is to store data on the most appropriate type of storage media, depending on how frequently the data is accessed.For example, data that is accessed frequently can be stored on more expensive, higher-performance storage media such as solid-state drives, while data that is accessed less frequently can be stored on less expensive, lower-performance storage media such as hard disk drives.HSM, though a long-standing idea, has changed drastically since its conception due to the advancement of technological storage and communication methods. However, even as aspects such as size and access time are now unrecognizable compared to the past, many of the original concepts remain popular today – albeit on a much grander scale when dealing with big data.Must-read big data coverageHSM systems work by automatically moving data among different storage tiers based on how often that data needs to be accessed. Data that is accessed frequently will be stored on fast, expensive storage media like SSDs, while data that is accessed infrequently will be moved to slower, cheaper storage formats. This ensures that users always have quick access to the data they need while minimizing storage costs and power usage.HSM is comparable to the cache on most computer CPUs. Like the CPU cache, frequently used data is stored on small and fast SRAM memory while less frequently used data gets moved to slower but larger DRAM when new data needs to be loaded.HSM components and algorithmsHSM systems typically consist of three key components: A data migration policy, algorithms for managing data, and a mechanism for tiering or caching data. The data migration policy defines how data should be moved between different storage devices based on factors such as frequency of use or importance. The algorithms used by HSM systems help to determine which data should be stored on which device based on

criteria such as how often the data is accessed or its size.Some of these algorithms include the Least Recently Used replacement, which moves data that has not been accessed recently to lower-performing storage tiers. Size-Temperature Replacement is another commonly used algorithm that uses both temperature and size thresholds to determine when to migrate data. The Heuristic Threshold is a newer algorithm that uses machine learning technologies to predict more accurately when data should be migrated.Tiered and cached HSMThe debate regarding tiering versus caching is one that HSM system designers face when trying to determine the best way to utilize lower-performing storage tiers. While tiering offers better long-term performance, caching can provide better short-term performance.Regardless of whether you choose to tier or cache your storage, these mechanisms ensure that frequently accessed data is stored on faster, more accessible devices, while less frequently accessed data is stored on slower, cheaper devices.Increased security: Hierarchical storage management systems allow you to store sensitive data on more secure storage media such as SSDs or HDDs with encryption capabilities. These options can help increase your system’s overall security.Improved manageability: HSM systems allow you to store data on different types of storage media depending on how frequently it is accessed. They can help improve your system’s overall manageability by making it easier to find and retrieve specific files when needed.Top HSM solutionsThere are many different HSM solutions available on the market today. Some of the top HSM solutions for big data include:Note that this list is not exhaustive and is in no particular order. Many different HSM solutions are available on the market today, so make sure your storage engineers and other data professionals take some time to research several options to find the best solution for your company’s needs.Also Read"Kihara Kimachia is a seasoned technology writer and digital marketing consultant with over 15 years of experience. His expertise spans across a broad spectrum of topics including managed services, business software, systems and apps, artificial intelligence, machine learning, fintech, digital transformation, cloud computing, DeFi, SEO, IoT, HTML, CSS, and Python. His writings regularly feature in leading technology publications such as TechRepublic, Enterprise Networking Planet, IT Business Edge, Channel Insider, eSecurity Planet, Server Watch, Enterprise Storage Forum, and Makeuseof.

Before embarking on his writing career, Kihara earned a B.S. in International Business Administration, Marketing, and Management Information Systems from the United States International University - Africa. His academic background, coupled with his extensive professional experience, uniquely positions him at the intersection of technology and marketing.

In his spare time, Kihara enjoys decoding the latest trends in artificial intelligence and machine learning and is often found exploring the intricacies of fintech and digital transformation. Kihara's interest in fintech is fueled by his past work in the banking and insurance sectors, giving him a unique perspective on the convergence of finance and technology.You're All SetThanks for signing

```
up! Keep an eye out for a confirmation email from our team. To ensure
any newsletters you subscribed to hit your inbox, make sure to add
newsletters@nl.technologyadvice.com to your contacts list.
```

Exemplu de izolare a textului dintr-o pagina web folosind trafilatura

Trafilatura este un pachet Python (insotit de un utilitar la linia de comanda) care poate fi utilizat pentru acumularea textului de pe internet.

trafilatura

In primul rand, pentru cei care nu au instalat modulul trebuie rulata comanda pip corespunzatoare

```
!{sys.executable} -m pip install trafilatura
```

```
Collecting trafilatura
  Downloading trafilatura-1.8.1-py3-none-any.whl (1.0 MB)
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.0/1.0 MB 8.8 MB/s eta
0:00:00
ent already satisfied: certifi in /usr/local/lib/python3.10/dist-
packages (from trafilatura) (2024.2.2)
Collecting courlan>=1.0.0 (from trafilatura)
  Downloading courlan-1.0.0-py3-none-any.whl (35 kB)
Collecting htmldate>=1.8.0 (from trafilatura)
  Downloading htmldate-1.8.1-py3-none-any.whl (31 kB)
Requirement already satisfied: justext>=3.0.0 in
/usr/local/lib/python3.10/dist-packages (from trafilatura) (3.0.0)
Requirement already satisfied: lxml<5.2.0,>=4.9.4 in
/usr/local/lib/python3.10/dist-packages (from trafilatura) (4.9.4)
Requirement already satisfied: charset-normalizer>=3.2.0 in
/usr/local/lib/python3.10/dist-packages (from trafilatura) (3.3.2)
Requirement already satisfied: urllib3<3,>=1.26 in
/usr/local/lib/python3.10/dist-packages (from trafilatura) (2.0.7)
Requirement already satisfied: langcodes>=3.3.0 in
/usr/local/lib/python3.10/dist-packages (from courlan>=1.0.0-
>trafilatura) (3.3.0)
Collecting tld>=0.13 (from courlan>=1.0.0->trafilatura)
  Downloading tld-0.13-py2.py3-none-any.whl (263 kB)
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 263.8/263.8 kB 11.0 MB/s eta
0:00:00
 htmldate>=1.8.0->trafilatura)
  Downloading dateparser-1.2.0-py2.py3-none-any.whl (294 kB)
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 295.0/295.0 kB 12.9 MB/s eta
0:00:00
ent already satisfied: python-dateutil>=2.8.2 in
/usr/local/lib/python3.10/dist-packages (from htmldate>=1.8.0-
>trafilatura) (2.8.2)
Requirement already satisfied: pytz in /usr/local/lib/python3.10/dist-
packages (from dateparser>=1.1.2->htmldate>=1.8.0->trafilatura)
```

```
(2023.4)
Requirement already satisfied: regex!=2019.02.19,!=2021.8.27 in
/usr/local/lib/python3.10/dist-packages (from dateparser>=1.1.2-
>htmldate>=1.8.0->trafilatura) (2023.12.25)
Requirement already satisfied: tzlocal in
/usr/local/lib/python3.10/dist-packages (from dateparser>=1.1.2-
>htmldate>=1.8.0->trafilatura) (5.2)
Requirement already satisfied: six>=1.5 in
/usr/local/lib/python3.10/dist-packages (from python-dateutil>=2.8.2-
>htmldate>=1.8.0->trafilatura) (1.16.0)
Installing collected packages: tld, dateparser, courlan, htmldate,
trafilatura
Successfully installed courlan-1.0.0 dateparser-1.2.0 htmldate-1.8.1
tld-0.13 trafilatura-1.8.1
```

```python
from trafilatura import fetch_url, extract

surl = 'https://www.techrepublic.com/article/what-is-hierarchical-
storage-management/'

downloaded = fetch_url(surl)

downloaded is None  # assuming the download was successful
False

result = extract(downloaded)

print(result)
```

```
If you’re responsible for managing enterprise data, you know that
storage can be both a blessing and a curse. On the one hand, having
access to large amounts of data can be immensely helpful in making
business decisions. On the other hand, storing all that data can be
expensive and downright chaotic to manage.
SEE: Cloud data storage policy (TechRepublic Premium)
That’s where hierarchical storage management comes in. HSM is a system
for storing data in a secure, cost-efficient manner. In this guide, we
give you a short crash course on HSM, what it is, how it works and
some of the benefits it can offer your organization.
Jump to:
What is HSM?
HSM, or hierarchical storage management, is a system for storing data
in a secure, cost-efficient manner. The basic idea behind HSM is to
store data on the most appropriate type of storage media, depending on
how frequently the data is accessed.
For example, data that is accessed frequently can be stored on more
expensive, higher-performance storage media such as solid-state
drives, while data that is accessed less frequently can be stored on
less expensive, lower-performance storage media such as hard disk
drives.
SEE: Quick glossary: Data storage (TechRepublic Premium)
```

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How do HSM systems work?

HSM systems work by automatically moving data among different storage tiers based on how often that data needs to be accessed. Data that is accessed frequently will be stored on fast, expensive storage media like SSDs, while data that is accessed infrequently will be moved to slower, cheaper storage formats. This ensures that users always have quick access to the data they need while minimizing storage costs and power usage.

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HSM components and algorithms

HSM systems typically consist of three key components: A data migration policy, algorithms for managing data, and a mechanism for tiering or caching data. The data migration policy defines how data should be moved between different storage devices based on factors such as frequency of use or importance. The algorithms used by HSM systems help to determine which data should be stored on which device based on criteria such as how often the data is accessed or its size.

SEE: What is data migration? (TechRepublic)

Some of these algorithms include the Least Recently Used replacement, which moves data that has not been accessed recently to lower-performing storage tiers. Size-Temperature Replacement is another commonly used algorithm that uses both temperature and size thresholds to determine when to migrate data. The Heuristic Threshold is a newer algorithm that uses machine learning technologies to predict more accurately when data should be migrated.

Tiered and cached HSM

The debate regarding tiering versus caching is one that HSM system designers face when trying to determine the best way to utilize lower-performing storage tiers. While tiering offers better long-term performance, caching can provide better short-term performance. Regardless of whether you choose to tier or cache your storage, these mechanisms ensure that frequently accessed data is stored on faster, more accessible devices, while less frequently accessed data is stored on slower, cheaper devices.

Benefits of HSM

Cost savings: HSM systems allow users to store data on less expensive storage media whenever possible, allowing enterprises to save money on overall storage costs.

-

Increased performance: HSM systems allow you to store frequently accessed data on higher-performance storage media such as SSDs, thus

```
improving your system’s overall performance.
-
Increased security: Hierarchical storage management systems allow you
to store sensitive data on more secure storage media such as SSDs or
HDDs with encryption capabilities. These options can help increase
your system’s overall security.
-
Improved manageability: HSM systems allow you to store data on
different types of storage media depending on how frequently it is
accessed. They can help improve your system’s overall manageability by
making it easier to find and retrieve specific files when needed.
-
Top HSM solutions
There are many different HSM solutions available on the market today.
Some of the top HSM solutions for big data include:
IBM Spectrum Scale
-
EMC Celerra/VNX
-
NetApp FAS/AFF
-
HPE 3PAR StoreServe
-
Huawei OceanStor Dorado
-
Qumulo Core
-
Red Hat Ceph Storage
-
Read next: Top data integration tools (TechRepublic)
Note that this list is not exhaustive and is in no particular order.
Many different HSM solutions are available on the market today, so
make sure your storage engineers and other data professionals take
some time to research several options to find the best solution for
your company’s needs.
```

Analiza textului

variabila result contine un text cat de cat coerent in limba engleza.

Textele pot fi analizate folosind tehnici tip NLP (Natural Language Processing)

Una dintre tehnicile care fac parte din NLP este Named Entity Recognition (recunoasterea si extragerea numelor proprii). Eista mai multe proiecte care se ocupa de NER:

spacy, exemplu spacy

nltk

```
!{sys.executable} -m pip install spacy
```

```
Requirement already satisfied: spacy in
/usr/local/lib/python3.10/dist-packages (3.7.4)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in
/usr/local/lib/python3.10/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in
/usr/local/lib/python3.10/dist-packages (from spacy) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in
/usr/local/lib/python3.10/dist-packages (from spacy) (1.0.10)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in
/usr/local/lib/python3.10/dist-packages (from spacy) (2.0.8)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in
/usr/local/lib/python3.10/dist-packages (from spacy) (3.0.9)
Requirement already satisfied: thinc<8.3.0,>=8.2.2 in
/usr/local/lib/python3.10/dist-packages (from spacy) (8.2.3)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in
/usr/local/lib/python3.10/dist-packages (from spacy) (1.1.2)
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in
/usr/local/lib/python3.10/dist-packages (from spacy) (2.4.8)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in
/usr/local/lib/python3.10/dist-packages (from spacy) (2.0.10)
Requirement already satisfied: weasel<0.4.0,>=0.1.0 in
/usr/local/lib/python3.10/dist-packages (from spacy) (0.3.4)
Requirement already satisfied: typer<0.10.0,>=0.3.0 in
/usr/local/lib/python3.10/dist-packages (from spacy) (0.9.4)
Requirement already satisfied: smart-open<7.0.0,>=5.2.1 in
/usr/local/lib/python3.10/dist-packages (from spacy) (6.4.0)
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in
/usr/local/lib/python3.10/dist-packages (from spacy) (4.66.2)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in
/usr/local/lib/python3.10/dist-packages (from spacy) (2.31.0)
Requirement already satisfied: pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4 in
/usr/local/lib/python3.10/dist-packages (from spacy) (2.6.4)
Requirement already satisfied: jinja2 in
/usr/local/lib/python3.10/dist-packages (from spacy) (3.1.3)
Requirement already satisfied: setuptools in
/usr/local/lib/python3.10/dist-packages (from spacy) (67.7.2)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.10/dist-packages (from spacy) (24.0)
Requirement already satisfied: langcodes<4.0.0,>=3.2.0 in
/usr/local/lib/python3.10/dist-packages (from spacy) (3.3.0)
Requirement already satisfied: numpy>=1.19.0 in
/usr/local/lib/python3.10/dist-packages (from spacy) (1.25.2)
Requirement already satisfied: annotated-types>=0.4.0 in
/usr/local/lib/python3.10/dist-packages (from pydantic!=1.8,!
=1.8.1,<3.0.0,>=1.7.4->spacy) (0.6.0)
Requirement already satisfied: pydantic-core==2.16.3 in
/usr/local/lib/python3.10/dist-packages (from pydantic!=1.8,!
=1.8.1,<3.0.0,>=1.7.4->spacy) (2.16.3)
Requirement already satisfied: typing-extensions>=4.6.1 in
/usr/local/lib/python3.10/dist-packages (from pydantic!=1.8,!
```

```
=1.8.1,<3.0.0,>=1.7.4->spacy) (4.11.0)
Requirement already satisfied: charset-normalizer<4,>=2 in
/usr/local/lib/python3.10/dist-packages (from requests<3.0.0,>=2.13.0-
>spacy) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in
/usr/local/lib/python3.10/dist-packages (from requests<3.0.0,>=2.13.0-
>spacy) (3.6)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.10/dist-packages (from requests<3.0.0,>=2.13.0-
>spacy) (2.0.7)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.10/dist-packages (from requests<3.0.0,>=2.13.0-
>spacy) (2024.2.2)
Requirement already satisfied: blis<0.8.0,>=0.7.8 in
/usr/local/lib/python3.10/dist-packages (from thinc<8.3.0,>=8.2.2-
>spacy) (0.7.11)
Requirement already satisfied: confection<1.0.0,>=0.0.1 in
/usr/local/lib/python3.10/dist-packages (from thinc<8.3.0,>=8.2.2-
>spacy) (0.1.4)
Requirement already satisfied: click<9.0.0,>=7.1.1 in
/usr/local/lib/python3.10/dist-packages (from typer<0.10.0,>=0.3.0-
>spacy) (8.1.7)
Requirement already satisfied: cloudpathlib<0.17.0,>=0.7.0 in
/usr/local/lib/python3.10/dist-packages (from weasel<0.4.0,>=0.1.0-
>spacy) (0.16.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.10/dist-packages (from jinja2->spacy) (2.1.5)
```

Spacy are nevoie de descarcarea unor modele lingvistice

```
import sys
!{sys.executable} -m spacy download en_core_web_sm
```

```
Collecting en-core-web-sm==3.7.1
  Downloading
https://github.com/explosion/spacy-models/releases/download/en_core_we
b_sm-3.7.1/en_core_web_sm-3.7.1-py3-none-any.whl (12.8 MB)
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 22.0 MB/s eta
0:00:00
ent already satisfied: spacy<3.8.0,>=3.7.2 in
/usr/local/lib/python3.10/dist-packages (from en-core-web-sm==3.7.1)
(3.7.4)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.2->en-
core-web-sm==3.7.1) (3.0.12)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.2->en-
core-web-sm==3.7.1) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in
```

```
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.2->en-
core-web-sm==3.7.1) (1.0.10)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.2->en-
core-web-sm==3.7.1) (2.0.8)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.2->en-
core-web-sm==3.7.1) (3.0.9)
Requirement already satisfied: thinc<8.3.0,>=8.2.2 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.2->en-
core-web-sm==3.7.1) (8.2.3)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.2->en-
core-web-sm==3.7.1) (1.1.2)
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.2->en-
core-web-sm==3.7.1) (2.4.8)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.2->en-
core-web-sm==3.7.1) (2.0.10)
Requirement already satisfied: weasel<0.4.0,>=0.1.0 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.2->en-
core-web-sm==3.7.1) (0.3.4)
Requirement already satisfied: typer<0.10.0,>=0.3.0 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.2->en-
core-web-sm==3.7.1) (0.9.4)
Requirement already satisfied: smart-open<7.0.0,>=5.2.1 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.2->en-
core-web-sm==3.7.1) (6.4.0)
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.2->en-
core-web-sm==3.7.1) (4.66.2)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.2->en-
core-web-sm==3.7.1) (2.31.0)
Requirement already satisfied: pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.2->en-
core-web-sm==3.7.1) (2.6.4)
Requirement already satisfied: jinja2 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.2->en-
core-web-sm==3.7.1) (3.1.3)
Requirement already satisfied: setuptools in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.2->en-
core-web-sm==3.7.1) (67.7.2)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.2->en-
core-web-sm==3.7.1) (24.0)
Requirement already satisfied: langcodes<4.0.0,>=3.2.0 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.2->en-
```

```
core-web-sm==3.7.1) (3.3.0)
Requirement already satisfied: numpy>=1.19.0 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.2->en-
core-web-sm==3.7.1) (1.25.2)
Requirement already satisfied: annotated-types>=0.4.0 in
/usr/local/lib/python3.10/dist-packages (from pydantic!=1.8,!
=1.8.1,<3.0.0,>=1.7.4->spacy<3.8.0,>=3.7.2->en-core-web-sm==3.7.1)
(0.6.0)
Requirement already satisfied: pydantic-core==2.16.3 in
/usr/local/lib/python3.10/dist-packages (from pydantic!=1.8,!
=1.8.1,<3.0.0,>=1.7.4->spacy<3.8.0,>=3.7.2->en-core-web-sm==3.7.1)
(2.16.3)
Requirement already satisfied: typing-extensions>=4.6.1 in
/usr/local/lib/python3.10/dist-packages (from pydantic!=1.8,!
=1.8.1,<3.0.0,>=1.7.4->spacy<3.8.0,>=3.7.2->en-core-web-sm==3.7.1)
(4.11.0)
Requirement already satisfied: charset-normalizer<4,>=2 in
/usr/local/lib/python3.10/dist-packages (from requests<3.0.0,>=2.13.0-
>spacy<3.8.0,>=3.7.2->en-core-web-sm==3.7.1) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in
/usr/local/lib/python3.10/dist-packages (from requests<3.0.0,>=2.13.0-
>spacy<3.8.0,>=3.7.2->en-core-web-sm==3.7.1) (3.6)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.10/dist-packages (from requests<3.0.0,>=2.13.0-
>spacy<3.8.0,>=3.7.2->en-core-web-sm==3.7.1) (2.0.7)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.10/dist-packages (from requests<3.0.0,>=2.13.0-
>spacy<3.8.0,>=3.7.2->en-core-web-sm==3.7.1) (2024.2.2)
Requirement already satisfied: blis<0.8.0,>=0.7.8 in
/usr/local/lib/python3.10/dist-packages (from thinc<8.3.0,>=8.2.2-
>spacy<3.8.0,>=3.7.2->en-core-web-sm==3.7.1) (0.7.11)
Requirement already satisfied: confection<1.0.0,>=0.0.1 in
/usr/local/lib/python3.10/dist-packages (from thinc<8.3.0,>=8.2.2-
>spacy<3.8.0,>=3.7.2->en-core-web-sm==3.7.1) (0.1.4)
Requirement already satisfied: click<9.0.0,>=7.1.1 in
/usr/local/lib/python3.10/dist-packages (from typer<0.10.0,>=0.3.0-
>spacy<3.8.0,>=3.7.2->en-core-web-sm==3.7.1) (8.1.7)
Requirement already satisfied: cloudpathlib<0.17.0,>=0.7.0 in
/usr/local/lib/python3.10/dist-packages (from weasel<0.4.0,>=0.1.0-
>spacy<3.8.0,>=3.7.2->en-core-web-sm==3.7.1) (0.16.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.10/dist-packages (from jinja2-
>spacy<3.8.0,>=3.7.2->en-core-web-sm==3.7.1) (2.1.5)
✔ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart
Python in
order to load all the package's dependencies. You can do this by
```

```
selecting the
'Restart kernel' or 'Restart runtime' option.

import spacy
from spacy import displacy

NER = spacy.load("en_core_web_sm")

text1= NER(result)

for ent in text1.ents:
    print(ent.text, ent.start_char, ent.end_char, ent.label_)
```

```
TechRepublic Premium 344 364 ORG
HSM 421 424 ORG
HSM 541 544 ORG
Jump 628 632 PERSON
HSM 645 648 ORG
HSM 650 653 ORG
HSM 778 781 ORG
TechRepublic Premium 1206 1226 ORG
HSM 1228 1231 ORG
today 1526 1531 DATE
HSM 1600 1603 ORG
HSM 1618 1621 ORG
HSM 2047 2050 ORG
CPU 2110 2113 ORG
SRAM 2170 2174 ORG
DRAM 2246 2250 ORG
HSM 2285 2288 ORG
HSM 2315 2318 ORG
three 2348 2353 CARDINAL
HSM 2643 2646 ORG
TechRepublic 2817 2829 ORG
Size-Temperature Replacement 2989 3017 ORG
The Heuristic Threshold 3135 3158 ORG
HSM 3295 3298 ORG
HSM 3355 3358 ORG
HSM 3814 3817 ORG
HSM 3832 3835 ORG
HSM 4007 4010 ORG
HSM 4429 4432 ORG
HSM 4678 4681 ORG
HSM 4717 4720 ORG
today 4755 4760 DATE
HSM 4778 4781 ORG
IBM 4814 4817 ORG
NetApp 4853 4859 ORG
HPE 3PAR StoreServe 4870 4889 PRODUCT
Huawei 4892 4898 ORG
```

```
Qumulo Core 4918 4929 PERSON
TechRepublic 4994 5006 ORG
HSM 5092 5095 ORG
today 5134 5139 DATE
```

Articolele tehnice nu sunt teribil de utile in NLP. Sa incercam sa gasim o stire normala

https://www.bbc.com/future/article/20240412-the-secret-scandinavian-a-bomb-project

```
from trafilatura import fetch_url, extract

surl = 'https://edition.cnn.com/2024/04/13/tech/ai-jobs-threat-
guaranteed-income/index.html'

downloaded = fetch_url(surl)

bbcpage = extract(downloaded)

print(bbcpage)
```

```
Michael Tubbs was born and raised in Stockton, California, roughly a
one-hour drive from Silicon Valley, the birthplace of the AI
revolution that’s now forecast to forever change the way Americans
live and work.
But despite coming of age in Big Tech’s backyard, the America that
Tubbs grew up in was marked by “scarcity and poverty,” he told CNN.
Tubbs, 33, was born to a teenage mother, whom he says he never saw
when he was younger because “she was always working — and it was never
enough.”
His own experiences led him to think about different ways that the
wealthiest country in the world could help ameliorate poverty. When
Tubbs went on to become the first Black mayor of his hometown in 2016,
he spearheaded a guaranteed income pilot program in 2019 that did
something simple yet radical: Give out free money with no strings
attached.
That idea of guaranteed income is receiving renewed interest as AI
becomes an increasing threat to Americans’ livelihoods.
Global policymakers and business leaders are now increasingly warning
that the rise of artificial intelligence will likely have profound
impacts on the labor market and could put millions of people out of
work in the years ahead (while also creating new and different jobs in
the process). The International Monetary Fund warned earlier this year
that some 40% of jobs around the world could be affected by the rise
of AI, and that this trend will likely deepen the already cavernous
gulf between the haves and have-nots.
As more Americans’ jobs are increasingly at risk due to the threat of
AI, Tubbs and other proponents of guaranteed income say this could be
one solution to help provide a safety net and cushion the expected
blow AI will have on the labor market.
“We don’t really do a good job at designing policies or doing things
```

in times of crisis," Tubbs told CNN, saying it is urgent to start planning for guaranteed income programs before we see 40% of global jobs taken by AI.

For a period of two years starting in 2019, Stockton handed out to 125 randomly selected residents in low-income neighborhoods $500 a month with no conditions around how they used the funds or if they had employment. The initial results from the pilot program found that recipients had drastically improved their job prospects and financial stability and saw better physical and mental health outcomes.

"Let's get the guardrails in place now," he said. "Then, when we have to deal with that job displacement, we're better positioned to do so."

Silicon Valley's infatuation with guaranteed income

The idea of a guaranteed income is not new. Tubbs said he was inspired to pursue it after reading the works of Civil Rights leader Martin Luther King, Jr., who advocated for guaranteed income in his 1967 book, "Where Do We Go From Here: Chaos or Community?"

"I'm now convinced that the simplest approach will prove to be the most effective — the solution to poverty is to abolish it directly by a now widely discussed measure: the guaranteed income," King wrote at the time.

Decades after King's death, the idea of guaranteed income went on to see a resurgence of support emanating out of Silicon Valley. The concept emerged as a buzzword of sorts among many of Silicon Valley's elite — including Elon Musk, Mark Zuckerberg and Sam Altman — even before the public launch of ChatGPT in late 2022 re-upped a global debate about automation disrupting jobs.

"Universal income will be necessary over time if AI takes over most human jobs," Tesla CEO Musk tweeted back in 2018. Late last year, in an interview with UK Prime Minister Rishi Sunak, Musk said he thought AI would eventually bring about "universal high income," without sharing any details of what this could look like.

Meta CEO Mark Zuckerberg, meanwhile, called for the exploration of "ideas like universal basic income to make sure that everyone has a cushion to try new ideas," during a Harvard commencement speech in May 2017. In a Facebook post later that year, Zuckerberg celebrated Alaska's Permanent Fund Dividend — or the annual grants given to Alaska residents from a portion of the state's oil revenue — as a "novel approach to basic income" that "comes from conservative principles of smaller government, rather than progressive principles of a larger safety net."

Altman, CEO of one of the world's most powerful AI companies, OpenAI, has also been outspoken about what he sees as the need for some form of guaranteed income as many jobs are increasingly lost to automation. Back in 2016, when Altman was president of tech startup accelerator YCombinator, he announced he was seeking participants to help launch a study on basic income (or, as he described it at the time, "giving people enough money to live on with no strings attached.")

"I'm fairly confident that at some point in the future, as technology continues to eliminate traditional jobs and massive new wealth gets created, we're going to see some version of this at a national scale,"

Altman wrote in a 2016 blog post for YCombinator.
He has since left his post at YCombinator to focus on OpenAI, but Altman still chairs the board of OpenResearch, the nonprofit lab that is in the process of conducting this ongoing study on basic income that he helped launch.
Elizabeth Rhodes, research director at OpenResearch, told CNN earlier this year that it hopes to release initial findings this summer from a three-year study on unconditional income involving some 3,000 individuals in two states.
“We really see this as sort of a foundational exploratory study to understand what happens when you give individuals unconditional cash,” she told CNN.
While she stressed that she could not get into the specifics of her team’s research while the study is underway, she hopes that their findings can eventually provide some data that answers some of the most common questions surrounding how cash payments will impact people’s desire to work and its broader potential advantages or disadvantages within communities.
Other tech industry tycoons, including Twitter co-founder Jack Dorsey, have also thrown immense financial support behind guaranteed income programs. (In 2020, Dorsey donated some $18 million to Mayors for a Guaranteed Income, the organization that Tubbs founded).
Dozens of cities across the United States have already begun experimenting with guaranteed income programs in recent years, with most of them funded by nonprofit organizations but organized by local officials.
Tubbs said he ultimately thinks funding for these programs should come from the federal government but encouraged lawmakers to be creative about finding ways to raise revenue.
“For example, you could legalize cannabis federally and use that tax revenue, you could do a data dividend or some sort of robot tax or AI tax,” he suggested.
Opponents to guaranteed income programs, most of whom lean Republican, have argued that such efforts disincentivize work or that taxing successful tech companies can stifle innovation.
And in Texas, opponents of guaranteed income are taking their battle to court. Earlier this week, Texas Attorney General Ken Paxton sued Harris County over its guaranteed income program that is funded using federal money from the pandemic-era American Rescue Plan. “This scheme is plainly unconstitutional,” Paxton said in a statement. “I am suing to stop officials in Harris County from abusing public funds for political gain.”
In court documents, the attorney general goes on to slam the program as “illegal and illegitimate government overreach.”
‘It’s not just giving people money, it’s giving them opportunity’
Tomas Vargas Jr., a recipient of guaranteed income in the Stockton pilot program, told CNN that he heard critics saying that receiving the extra payments would make people “lazy.” But he says it ultimately gave him the opportunity to find better work.
“When I got the money, I was already in the mindset of hustling and

```
getting money. So, it just made me want to get more money," he said.
"The thing that I want people to understand about the guaranteed
income is it's not just giving people money, it's giving them
opportunity."
For years, Vargas said he woke up every day with the crippling anxiety
that comes with never quite knowing how he will be able to provide for
his family. He was juggling multiple jobs: working at UPS, repairing
cars, mowing lawns, delivering groceries and picking up any other work
he could find. He said he almost never saw his children and said he
briefly received food stamp assistance but was "instantly kicked off"
when he would pick up extra hours of work.
"There's one thing that I've always wanted as a father, and that's not
to make my kids go through the same things that I went through: having
no power, no water, or no food on the plates," he told CNN. "So I was
always trying to grind."
Vargas said the extra cash payments he received helped him focus and
apply for one full-time job, which he never had the time or energy to
do before. He now says he thinks guaranteed income could be one way to
provide a cushion for re-training or education programs for people
whose jobs are exposed to AI, the same way it helped him pivot to
better and more secure employment.
Vargas, like Tubbs, was born and raised in Stockton. Vargas said his
father was never around much growing up and he eventually moved in
with his grandmother when he was 12. Before participating in the
program, Vargas said he was "a really negative person" and that he
didn't look at himself as someone even worth investing in.
But the extra financial security allowed him to spend more time with
his children, and ultimately break the cycle of poverty he had seen in
his community his whole life.
"One of the biggest things that helped me realize my full potential
that I had in myself, and I was worth investing in, was seeing the
reaction from my kids," Vargas said, "and seeing the generational
trauma and healing in them."
```

Named Entity Regognition

Parte din NLP, identificarea tipurilor de numelor proprii presupune analiza textului si "ghicirea" tipului de entitate numită. Rezultatul este o listă de nume însoțită de tipul acestora (loc, persoana, etc.)

Spacy recunoaște următoarele entități

Entitate	Descriere
PERSON:	People, including fictional.
NORP sau NAT_REL_POL:	Nationalities or religious or political groups.
FAC:	Buildings, airports, highways, bridges, etc.
ORG:	Companies, agencies, institutions, etc.
GPE:	Countries, cities, states.

Entitate	Descriere
LOC:	Non-GPE locations, mountain ranges, bodies of water.
PRODUCT:	Objects, vehicles, foods, etc. (Not services.)
EVENT:	Named hurricanes, battles, wars, sports events, etc.
WORK_OF_ART:	Titles of books, songs, etc.
LAW:	Named documents made into laws.
LANGUAGE:	Any named language.
DATE:	Absolute or relative dates or periods.
TIME:	Times smaller than a day.
PERCENT:	Percentage, including ”%“.
MONEY:	Monetary values, including unit.
QUANTITY:	Measurements, as of weight or distance.
ORDINAL:	“first”, “second”, etc.
CARDINAL:	Numerals that do not fall under another type.

```
import spacy
from spacy import displacy

NER = spacy.load("en_core_web_sm")

text1= NER(bbcpage)

for ent in text1.ents:
    # if ent.label_ == 'PERSON':
        print(ent.text, ent.start_char, ent.end_char, ent.label_)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
<ipython-input-6-02d4d9c3f573> in <cell line: 6>()
      4 NER = spacy.load("en_core_web_sm")
      5
----> 6 text1= NER(bbcpage)
      7
      8 for ent in text1.ents:

NameError: name 'bbcpage' is not defined
```

Incercam un articol in limba romana

https://www.hotnews.ro/stiri-alegeri2024-27052128-exclusiv-bresa-securitate-autoritatea-electorala-permanenta-datele-personale-ale-expertilor-electorali-fost-dezvaluite-din-eroare.htm

```
hnurl = "https://www.hotnews.ro/stiri-alegeri2024-27052128-exclusiv-bresa-securitate-autoritatea-electorala-permanenta-datele-personale-
```

```
ale-expertilor-electorali-fost-dezvaluite-din-eroare.htm"

downloaded = fetch_url(hnurl)

hnpage = extract(downloaded)

print(hnpage)
```

Incidentul s-a produs joi, 11 aprilie, iar pentru câteva ore mai multe date personale, de exemplu codul numeric personal, ale experților electorali au putut să fie accesate pe un mail nesecurizat, a declarat, pentru HotNews.ro, președintele AEP, Toni Greblă.
Experții electorali sunt persoanele care pot deveni președinți ai birourilor electorale ale secțiilor de votare din țară sau locțiitorii lor.
Scurgerea de informații a fost raportată, vineri, Autorității Naționale de Supraveghere a Datelor cu caracter personal, a confirmat Toni Greblă pentru HotNews.ro.
Termenul legal pentru semnalarea scurgerilor de date este de 72 de ore. Potrivit informațiilor, codul numeric personal pentru 1.300 de persoane din corpul experților electorali a fost dezvăluit.
„Miercuri seara, până joi dimineața la ora 2, personalul AEP a fost angrenat în operațiunea de numărare și verificare a semnăturilor de la candidații pentru europarlamentare. Ultimul candidat a depus documentația la ora 23.00, miercuri noaptea. Joi dimineața, pentru că avem un calendar foarte strâns în care trebuie să facem o multitudine de activități, una din activități era să transmite în teritoriu lista experților electorali. Lista o trimitem în toate județele pe mailuri securizate”, a explicat Toni Greblă pentru HotNews.ro.
Președintele AEP susține că dintr-o „eroare” că într-un județ lista nu a fost trimisă pe mailul securizat, ci pe un alt mail.
Greblă: „A fost o suprasolicitare a personalului AEP. Regretăm acest incident”
„Timp de o oră a stat afișat și a putut să fie accesat de pe acest mail. Aproximativ 30 – 35 de persoane, unii au intrat de două ori, au putut să vadă lista. Nu știm dacă au fost luate capturi sau doar a fost accesată pagina”, a adăugat oficialul Autorității.
Potrivit acestuia, a fost șters de pe mailul destinat „publicului mai larg” și s-a făcut o anchetă internă.
Șeful AEP nu folosește termenul de „breșă de securitate”, care este sintagma clasică pentru definirea accesului neautorizat la date sau informații găzduite de computere, telefoane mobile sau alte device-uri.
„Nu este o breșă de securitate, numele sunt publice, doar datele cu caracter personal nu erau publice. Au apărut CNP-ul și domiciliul. Credem că acest mail a fost accesat de cei interesați de cei interesați să vadă dacă se regăsesc pe listă. Nu cred că cineva a făcut o captură ca să folosească datele cu caracter personal. Eu am obligația să anunț Autoritatea pentru Protecția Datelor cu caracter personal”, a fost poziția lui Toni Greblă pentru HotNews.ro.

```
La final, șeful AEP a spus că „regretă incidentul, dar oameni suntem
cu toții”.
„Dacă permitem unor competitori politici să îi așteptăm până la 12
noaptea să vină cu 200.000 de semnături și după aia o armată de 30–500
de oameni să verifice dacă acestea sunt în ordine... A fost o
suprasolicitare a personalului AEP, am stat până la 02.30 dimineața în
ultima zi de depunere a candidaturilor. Poate ar trebui depuse
candidaturile până la ora 17.00. Repet, regretăm acest incident. Este
o întâmplare nefericită, nu cred că are o mare amploare”, a spus Toni
Greblă.
Precizare: Într-o versiune anterioară a textului președintele AEP Toni
Greblă a definit experții electorali drept „persoanele care lucrează
în IT în procesul electoral, cei care la intrarea în secțiile de
votare preiau cărțile de identitate ale alegătorilor și verifică dacă
cetățenii au drept de vot”.
În realitate, conform informațiile primite de la AEP și contrare a
ceea ce a spus președintele instituției în discuția înregistrată cu
HotNews.ro, experții electorali sunt, așa cum apare corect în articol,
persoanele care pot deveni președinți ai birourilor electorale ale
secțiilor de votare din țară sau locțiitorii lor.
```

Alte modele spacy pot fi gasite la adresa:

https://spacy.io/models

Exista un model pentru limba romana, sa il testam

```
import sys
!{sys.executable} -m spacy download ro_core_news_lg

Collecting ro-core-news-lg==3.7.0
  Downloading
https://github.com/explosion/spacy-models/releases/download/ro_core_ne
ws_lg-3.7.0/ro_core_news_lg-3.7.0-py3-none-any.whl (568.5 MB)
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 568.5/568.5 MB 2.6 MB/s eta
0:00:00
ent already satisfied: spacy<3.8.0,>=3.7.0 in
/usr/local/lib/python3.10/dist-packages (from ro-core-news-lg==3.7.0)
(3.7.4)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.0->ro-
core-news-lg==3.7.0) (3.0.12)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.0->ro-
core-news-lg==3.7.0) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.0->ro-
core-news-lg==3.7.0) (1.0.10)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.0->ro-
```

```
core-news-lg==3.7.0) (2.0.8)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.0->ro-
core-news-lg==3.7.0) (3.0.9)
Requirement already satisfied: thinc<8.3.0,>=8.2.2 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.0->ro-
core-news-lg==3.7.0) (8.2.3)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.0->ro-
core-news-lg==3.7.0) (1.1.2)
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.0->ro-
core-news-lg==3.7.0) (2.4.8)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.0->ro-
core-news-lg==3.7.0) (2.0.10)
Requirement already satisfied: weasel<0.4.0,>=0.1.0 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.0->ro-
core-news-lg==3.7.0) (0.3.4)
Requirement already satisfied: typer<0.10.0,>=0.3.0 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.0->ro-
core-news-lg==3.7.0) (0.9.4)
Requirement already satisfied: smart-open<7.0.0,>=5.2.1 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.0->ro-
core-news-lg==3.7.0) (6.4.0)
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.0->ro-
core-news-lg==3.7.0) (4.66.2)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.0->ro-
core-news-lg==3.7.0) (2.31.0)
Requirement already satisfied: pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.0->ro-
core-news-lg==3.7.0) (2.6.4)
Requirement already satisfied: jinja2 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.0->ro-
core-news-lg==3.7.0) (3.1.3)
Requirement already satisfied: setuptools in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.0->ro-
core-news-lg==3.7.0) (67.7.2)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.0->ro-
core-news-lg==3.7.0) (24.0)
Requirement already satisfied: langcodes<4.0.0,>=3.2.0 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.0->ro-
core-news-lg==3.7.0) (3.3.0)
Requirement already satisfied: numpy>=1.19.0 in
/usr/local/lib/python3.10/dist-packages (from spacy<3.8.0,>=3.7.0->ro-
core-news-lg==3.7.0) (1.25.2)
```

```
Requirement already satisfied: annotated-types>=0.4.0 in
/usr/local/lib/python3.10/dist-packages (from pydantic!=1.8,!
=1.8.1,<3.0.0,>=1.7.4->spacy<3.8.0,>=3.7.0->ro-core-news-lg==3.7.0)
(0.6.0)
Requirement already satisfied: pydantic-core==2.16.3 in
/usr/local/lib/python3.10/dist-packages (from pydantic!=1.8,!
=1.8.1,<3.0.0,>=1.7.4->spacy<3.8.0,>=3.7.0->ro-core-news-lg==3.7.0)
(2.16.3)
Requirement already satisfied: typing-extensions>=4.6.1 in
/usr/local/lib/python3.10/dist-packages (from pydantic!=1.8,!
=1.8.1,<3.0.0,>=1.7.4->spacy<3.8.0,>=3.7.0->ro-core-news-lg==3.7.0)
(4.11.0)
Requirement already satisfied: charset-normalizer<4,>=2 in
/usr/local/lib/python3.10/dist-packages (from requests<3.0.0,>=2.13.0-
>spacy<3.8.0,>=3.7.0->ro-core-news-lg==3.7.0) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in
/usr/local/lib/python3.10/dist-packages (from requests<3.0.0,>=2.13.0-
>spacy<3.8.0,>=3.7.0->ro-core-news-lg==3.7.0) (3.6)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.10/dist-packages (from requests<3.0.0,>=2.13.0-
>spacy<3.8.0,>=3.7.0->ro-core-news-lg==3.7.0) (2.0.7)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.10/dist-packages (from requests<3.0.0,>=2.13.0-
>spacy<3.8.0,>=3.7.0->ro-core-news-lg==3.7.0) (2024.2.2)
Requirement already satisfied: blis<0.8.0,>=0.7.8 in
/usr/local/lib/python3.10/dist-packages (from thinc<8.3.0,>=8.2.2-
>spacy<3.8.0,>=3.7.0->ro-core-news-lg==3.7.0) (0.7.11)
Requirement already satisfied: confection<1.0.0,>=0.0.1 in
/usr/local/lib/python3.10/dist-packages (from thinc<8.3.0,>=8.2.2-
>spacy<3.8.0,>=3.7.0->ro-core-news-lg==3.7.0) (0.1.4)
Requirement already satisfied: click<9.0.0,>=7.1.1 in
/usr/local/lib/python3.10/dist-packages (from typer<0.10.0,>=0.3.0-
>spacy<3.8.0,>=3.7.0->ro-core-news-lg==3.7.0) (8.1.7)
Requirement already satisfied: cloudpathlib<0.17.0,>=0.7.0 in
/usr/local/lib/python3.10/dist-packages (from weasel<0.4.0,>=0.1.0-
>spacy<3.8.0,>=3.7.0->ro-core-news-lg==3.7.0) (0.16.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.10/dist-packages (from jinja2-
>spacy<3.8.0,>=3.7.0->ro-core-news-lg==3.7.0) (2.1.5)
Installing collected packages: ro-core-news-lg
Successfully installed ro-core-news-lg-3.7.0
✔ Download and installation successful
You can now load the package via spacy.load('ro_core_news_lg')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart
Python in
order to load all the package's dependencies. You can do this by
selecting the
'Restart kernel' or 'Restart runtime' option.
```

```python
import spacy
from spacy import displacy

NER = spacy.load("ro_core_news_lg")

text1= NER(hnpage)

for ent in text1.ents:
    print(ent.text, ent.start_char, ent.end_char, ent.label_)
```

```
joi, 11 aprilie 22 37 DATETIME
câteva ore 50 60 DATETIME
experților 126 136 PERSON
președintele 228 240 PERSON
AEP 241 244 ORGANIZATION
Toni Greblă 246 257 PERSON
Experții 259 267 PERSON
persoanele 284 294 PERSON
președinți 311 321 PERSON
locțiitorii 384 395 PERSON
vineri 443 449 DATETIME
Autorității Naționale de Supraveghere a Datelor cu caracter personal
451 519 ORGANIZATION
Toni Greblă 533 544 PERSON
72 de ore 625 634 DATETIME
1.300 690 695 NUMERIC_VALUE
persoane 699 707 PERSON
Miercuri seara, până joi dimineața la ora 2 760 803 DATETIME
personalul 805 815 PERSON
AEP 816 819 ORGANIZATION
candidații 898 908 PERSON
candidat 942 950 PERSON
ora 23.00 975 984 DATETIME
miercuri noaptea 986 1002 DATETIME
Joi dimineața 1004 1017 DATETIME
experților 1169 1179 PERSON
Toni Greblă 1262 1273 PERSON
Președintele 1293 1305 PERSON
AEP 1306 1309 ORGANIZATION
Greblă 1419 1425 PERSON
personalului 1455 1467 PERSON
AEP 1468 1471 ORGANIZATION
o oră 1507 1512 DATETIME
30 1583 1585 NUMERIC_VALUE
35 1588 1590 NUMERIC_VALUE
persoane 1594 1602 PERSON
două 1622 1626 NUMERIC_VALUE
oficialul 1735 1744 PERSON
Autorității 1745 1756 ORGANIZATION
Șeful 1866 1871 PERSON
```

```
AEP 1872 1875 ORGANIZATION
computere 2025 2034 PRODUCT
telefoane mobile 2036 2052 PRODUCT
Autoritatea pentru Protecția Datelor cu caracter personal” 2423 2481
ORGANIZATION
Toni Greblă 2502 2513 PERSON
HotNews.ro 2521 2531 LANGUAGE
șeful 2543 2548 PERSON
AEP 2549 2552 ORGANIZATION
oameni 2588 2594 PERSON
competitori 2633 2644 PERSON
12 noaptea 2677 2687 DATETIME
200.000 2699 2706 NUMERIC_VALUE
30–500 2744 2750 NUMERIC_VALUE
oameni 2754 2760 PERSON
personalului 2831 2843 PERSON
AEP 2844 2847 ORGANIZATION
02.30 dimineața 2865 2880 DATETIME
ultima zi 2884 2893 DATETIME
ora 17.00 2969 2978 DATETIME
Toni Greblă 3082 3093 PERSON
președintele 3144 3156 PERSON
AEP Toni Greblă 3157 3172 ORGANIZATION
persoanele 3210 3220 PERSON
secțiile de votare 3288 3306 FACILITY
alegătorilor 3340 3352 PERSON
cetățenii 3370 3379 PERSON
AEP 3447 3450 ORGANIZATION
președintele 3480 3492 PERSON
experții electorali 3545 3564 PERSON
persoanele 3604 3614 PERSON
președinți 3631 3641 PERSON
locțiitorii 3704 3715 PERSON
```

POS tagging

Part of Speech tagging implică marcarea tuturor elementelor de text cu informații despre valoarea morfologica. Aceste informații sunt atribute ale obiectului de tip "spacy.tokens.token.Token". Cateva atribute interesante (lista copleta aici)

Atribut	Tip	Descriere
text	str	continutul real textual al tokenului
text_with_ws	str	continutul textual cu spatiul ulterior, daca este prezent
ent_type_	str	Tipul named entity (daca este cazul)
lemma	str	Rădăcina cuvantului respectiv
is_alpha	bool	Tokenul este format din caractere alfanumerice?

Atribut	Tip	Descriere
is_ascii	bool	Tokenul este format din caractere ASCII
is_digit	bool	Tokenul este format din cifre
pos_	str	Denumirea pos din lista universala
dep_	str	Relatia de dependenta sintactica

```
for token in text1:
    print('Text: ' + token.text + ' Lema: ' + token.lemma_ + ' POS: '
+ token.pos_ + ' Tag: ' + token.tag_  + ' DEP: ' + token.dep_)
```

```
Text: Incidentul Lema: Incidentul POS: NOUN Tag: Ncmsry DEP: nsubj
Text: s- Lema: sine POS: PRON Tag: Px3--a--y-----w DEP: expl:pv
Text: a Lema: avea POS: AUX Tag: Va--3s DEP: aux
Text: produs Lema: produce POS: VERB Tag: Vmp--sm DEP: ROOT
Text: joi Lema: joi POS: NOUN Tag: Ncf--n DEP: obl
Text: , Lema: , POS: NOUN Tag: Ncf--n DEP: punct
Text: 11 Lema: 11 POS:  Tag: Mc-s-d DEP: nummod
Text: aprilie Lema: aprilie POS: NOUN Tag: Ncm--n DEP: nmod:tmod
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: iar Lema: iar POS: ADV Tag: Rc DEP: cc
Text: pentru Lema: pentru POS: ADP Tag: Spsa DEP: case
Text: câteva Lema: câtva POS: DET Tag: Di3fpr DEP: det
Text: ore Lema: oră POS: NOUN Tag: Ncfp-n DEP: nmod
Text: mai Lema: mai POS: ADV Tag: Rp DEP: advmod
Text: multe Lema: mult POS: DET Tag: Di3fp DEP: det
Text: date Lema: dată POS: NOUN Tag: Ncfp-n DEP: nsubj
Text: personale Lema: personal POS: ADJ Tag: Afpfp-n DEP: amod
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: de Lema: de POS: ADP Tag: Spsa DEP: advmod
Text: exemplu Lema: exemplu POS: NOUN Tag: Ncms-n DEP: fixed
Text: codul Lema: cod POS: NOUN Tag: Ncmsry DEP: nsubj
Text: numeric Lema: numeric POS: ADJ Tag: Afpms-n DEP: amod
Text: personal Lema: personal POS: ADJ Tag: Afpms-n DEP: amod
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: ale Lema: al POS: DET Tag: Tsfp DEP: det
Text: experților Lema: expert POS: NOUN Tag: Ncmpoy DEP: conj
Text: electorali Lema: electoral POS: ADJ Tag: Afpmp-n DEP: amod
Text: au Lema: avea POS: AUX Tag: Va--3p DEP: aux
Text: putut Lema: putea POS: VERB Tag: Vmp--sm DEP: conj
Text: să Lema: să POS: PART Tag: Qs DEP: mark
Text: fie Lema: fi POS: AUX Tag: Vasp3 DEP: aux:pass
Text: accesate Lema: accesa POS: VERB Tag: Vmp--pf DEP: ccomp
Text: pe Lema: pe POS: ADP Tag: Spsa DEP: case
Text: un Lema: un POS: DET Tag: Timsr DEP: det
Text: mail Lema: mail POS: NOUN Tag: Ncms-n DEP: obl
Text: nesecurizat Lema: nesecurizat POS: ADJ Tag: Afpms-n DEP: amod
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: a Lema: avea POS: AUX Tag: Va--3s DEP: aux
Text: declarat Lema: declara POS: VERB Tag: Vmp--sm DEP: conj
```

```
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: pentru Lema: pentru POS: ADP Tag: Spsa DEP: case
Text: HotNews.ro Lema: HotNews.ro POS: PROPN Tag: Np DEP: obl
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: președintele Lema: președinte POS: NOUN Tag: Ncmsry DEP:
nsubj:pass
Text: AEP Lema: AEP POS: PROPN Tag: Np DEP: nmod
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: Toni Lema: Toni POS: PROPN Tag: Np DEP: appos
Text: Greblă Lema: Greblă POS: PROPN Tag: Np DEP: flat
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text:
 Lema:
 POS: SPACE Tag: _SP DEP: dep
Text: Experții Lema: Experții POS: NOUN Tag: Ncmpry DEP: nsubj
Text: electorali Lema: electoral POS: ADJ Tag: Afpmp-n DEP: amod
Text: sunt Lema: fi POS: AUX Tag: Vaip3p DEP: cop
Text: persoanele Lema: persoană POS: NOUN Tag: Ncfpry DEP: ROOT
Text: care Lema: care POS: PRON Tag: Pw3--r DEP: nsubj
Text: pot Lema: putea POS: AUX Tag: Vmip3p DEP: acl
Text: deveni Lema: deveni POS: VERB Tag: Vmnp DEP: ccomp
Text: președinți Lema: președint POS: NOUN Tag: Ncmp-n DEP: xcomp
Text: ai Lema: al POS: DET Tag: Tsmp DEP: det
Text: birourilor Lema: birou POS: NOUN Tag: Ncfpoy DEP: nmod
Text: electorale Lema: electoral POS: ADJ Tag: Afpfp-n DEP: amod
Text: ale Lema: al POS: DET Tag: Tsfp DEP: det
Text: secțiilor Lema: secție POS: NOUN Tag: Ncfpoy DEP: nmod
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: votare Lema: votare POS: NOUN Tag: Ncfsrn DEP: nmod
Text: din Lema: din POS: ADP Tag: Spsa DEP: case
Text: țară Lema: țară POS: NOUN Tag: Ncfsrn DEP: nmod
Text: sau Lema: sau POS: CCONJ Tag: Ccssp DEP: cc
Text: locțiitorii Lema: locțiitor POS: NOUN Tag: Ncmpry DEP: conj
Text: lor Lema: lui POS: DET Tag: Ds3---p DEP: det
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text:
 Lema:
 POS: SPACE Tag: _SP DEP: dep
Text: Scurgerea Lema: scurgere POS: NOUN Tag: Ncfsry DEP: nsubj:pass
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: informații Lema: informație POS: NOUN Tag: Ncfp-n DEP: nmod
Text: a Lema: avea POS: AUX Tag: Va--3s DEP: aux
Text: fost Lema: fi POS: AUX Tag: Vap--sm DEP: aux:pass
Text: raportată Lema: raporta POS: VERB Tag: Vmp--sf DEP: ROOT
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: vineri Lema: vineri POS: NOUN Tag: Ncf--n DEP: advmod
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: Autorității Lema: Autorității POS: NOUN Tag: Ncfsoy DEP:
obl:agent
```

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Text: Naționale Lema: național POS: ADJ Tag: Afpfson DEP: amod
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: Supraveghere Lema: supraveghere POS: NOUN Tag: Ncfsrn DEP: nmod
Text: a Lema: al POS: DET Tag: Tsfs DEP: det
Text: Datelor Lema: Dată POS: NOUN Tag: Ncfpoy DEP: nmod
Text: cu Lema: cu POS: ADP Tag: Spsa DEP: case
Text: caracter Lema: caracter POS: NOUN Tag: Ncms-n DEP: nmod
Text: personal Lema: personal POS: ADJ Tag: Afpms-n DEP: amod
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: a Lema: avea POS: AUX Tag: Va--3s DEP: aux
Text: confirmat Lema: confirma POS: VERB Tag: Vmp--sm DEP: parataxis
Text: Toni Lema: Toni POS: PROPN Tag: Np DEP: nsubj
Text: Greblă Lema: Greblă POS: PROPN Tag: Np DEP: flat
Text: pentru Lema: pentru POS: ADP Tag: Spsa DEP: case
Text: HotNews.ro Lema: HotNews.ro POS: PROPN Tag: Np DEP: obl
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text:
 Lema:
 POS: SPACE Tag: _SP DEP: dep
Text: Termenul Lema: termen POS: NOUN Tag: Ncmsry DEP: nsubj
Text: legal Lema: legal POS: ADJ Tag: Afpms-n DEP: amod
Text: pentru Lema: pentru POS: ADP Tag: Spsa DEP: case
Text: semnalarea Lema: semnalare POS: NOUN Tag: Ncfsry DEP: nmod
Text: scurgerilor Lema: scurgerilor POS: NOUN Tag: Ncfpoy DEP: nmod
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: date Lema: dată POS: NOUN Tag: Ncfp-n DEP: nmod
Text: este Lema: fi POS: AUX Tag: Vaip3s DEP: cop
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: 72 Lema: 72 POS:  Tag: Mc-s-d DEP: nummod
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: ore Lema: oră POS: NOUN Tag: Ncfp-n DEP: ROOT
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text: Potrivit Lema: potrivit POS: ADP Tag: Spsd DEP: case
Text: informațiilor Lema: informație POS: NOUN Tag: Ncfpoy DEP: obl
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: codul Lema: cod POS: NOUN Tag: Ncmsry DEP: nsubj:pass
Text: numeric Lema: numeric POS: ADJ Tag: Afpms-n DEP: amod
Text: personal Lema: personal POS: ADJ Tag: Afpms-n DEP: amod
Text: pentru Lema: pentru POS: ADP Tag: Spsa DEP: case
Text: 1.300 Lema: 1.300 POS:  Tag: Mc-s-d DEP: nummod
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: persoane Lema: persoană POS: NOUN Tag: Ncfp-n DEP: nmod
Text: din Lema: din POS: ADP Tag: Spsa DEP: case
Text: corpul Lema: corp POS: NOUN Tag: Ncmsry DEP: nmod
Text: experților Lema: expert POS: NOUN Tag: Ncmpoy DEP: nmod
Text: electorali Lema: electoral POS: ADJ Tag: Afpmp-n DEP: amod
Text: a Lema: avea POS: AUX Tag: Va--3s DEP: aux
Text: fost Lema: fi POS: AUX Tag: Vap--sm DEP: aux:pass
Text: dezvăluit Lema: dezvălui POS: VERB Tag: Vmp--sm DEP: ROOT
```

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Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text:
 Lema:
 POS: SPACE Tag: _SP DEP: dep
Text: „ Lema: „ POS: PUNCT Tag: DBLQ DEP: punct
Text: Miercuri Lema: Miercuri POS: NOUN Tag: Ncf--n DEP: nsubj
Text: seara Lema: seara POS: ADV Tag: Rgp DEP: advmod
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: până Lema: până POS: SCONJ Tag: Csssp DEP: case
Text: joi Lema: joi POS: NOUN Tag: Ncf--n DEP: nmod:tmod
Text: dimineața Lema: dimineața POS: ADV Tag: Rgp DEP: advmod
Text: la Lema: la POS: ADP Tag: Spsa DEP: case
Text: ora Lema: oră POS: NOUN Tag: Ncfsry DEP: nmod:tmod
Text: 2 Lema: 2 POS:  Tag: Mc-s-d DEP: nummod
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: personalul Lema: personal POS: NOUN Tag: Ncmsry DEP: nsubj:pass
Text: AEP Lema: AEP POS: PROPN Tag: Np DEP: nmod
Text: a Lema: avea POS: AUX Tag: Va--3s DEP: aux
Text: fost Lema: fi POS: AUX Tag: Vap--sm DEP: aux:pass
Text: angrenat Lema: angrena POS: VERB Tag: Vmp--sm DEP: ROOT
Text: în Lema: în POS: ADP Tag: Spsa DEP: case
Text: operațiunea Lema: operațiune POS: NOUN Tag: Ncfsry DEP: obl
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: numărare Lema: numărare POS: NOUN Tag: Ncfsrn DEP: nmod
Text: și Lema: și POS: CCONJ Tag: Crssp DEP: cc
Text: verificare Lema: verificare POS: NOUN Tag: Ncfsrn DEP: conj
Text: a Lema: al POS: DET Tag: Tsfs DEP: det
Text: semnăturilor Lema: semnăturilor POS: NOUN Tag: Ncfpoy DEP: nmod
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: la Lema: la POS: ADP Tag: Spsa DEP: fixed
Text: candidații Lema: candidat POS: NOUN Tag: Ncmpry DEP: nmod
Text: pentru Lema: pentru POS: ADP Tag: Spsa DEP: case
Text: europarlamentare Lema: europarlamentare POS: NOUN Tag: Ncfsrn
DEP: nmod
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text: Ultimul Lema: ultim POS: NUM Tag: Momsrly DEP: nummod
Text: candidat Lema: candidat POS: NOUN Tag: Ncms-n DEP: nsubj
Text: a Lema: avea POS: AUX Tag: Va--3s DEP: aux
Text: depus Lema: depune POS: VERB Tag: Vmp--sm DEP: ROOT
Text: documentația Lema: documentație POS: NOUN Tag: Ncfsry DEP: obj
Text: la Lema: la POS: ADP Tag: Spsa DEP: case
Text: ora Lema: oră POS: NOUN Tag: Ncfsry DEP: obl
Text: 23.00 Lema: 23.00 POS:  Tag: Mc-s-d DEP: nummod
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: miercuri Lema: miercuri POS: NOUN Tag: Ncf--n DEP: conj
Text: noaptea Lema: noaptea POS: ADV Tag: Rgp DEP: advmod
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text: Joi Lema: joi POS: ADV Tag: Rgp DEP: xcomp
Text: dimineața Lema: dimineața POS: ADV Tag: Rgp DEP: advmod
```

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Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: pentru Lema: pentru POS: ADP Tag: Spsa DEP: mark
Text: că Lema: că POS: SCONJ Tag: Csssp DEP: fixed
Text: avem Lema: avea POS: VERB Tag: Vmip1p DEP: advcl
Text: un Lema: un POS: DET Tag: Timsr DEP: det
Text: calendar Lema: calendar POS: NOUN Tag: Ncms-n DEP: obj
Text: foarte Lema: foarte POS: ADV Tag: Rp DEP: advmod
Text: strâns Lema: strâns POS: ADV Tag: Rgp DEP: amod
Text: în Lema: în POS: ADP Tag: Spsa DEP: case
Text: care Lema: care POS: PRON Tag: Pw3--r DEP: obl
Text: trebuie Lema: trebui POS: VERB Tag: Vmip3 DEP: acl
Text: să Lema: să POS: PART Tag: Qs DEP: mark
Text: facem Lema: face POS: VERB Tag: Vmip1p DEP: csubj
Text: o Lema: un POS: DET Tag: Tifsr DEP: det
Text: multitudine Lema: multitudine POS: NOUN Tag: Ncfsrn DEP: obj
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: activități Lema: activitate POS: NOUN Tag: Ncfp-n DEP: nmod
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: una Lema: unul POS: PRON Tag: Pi3fsr DEP: appos
Text: din Lema: din POS: ADP Tag: Spsa DEP: case
Text: activități Lema: activitate POS: NOUN Tag: Ncfp-n DEP: nmod
Text: era Lema: fi POS: AUX Tag: Vaii3s DEP: aux
Text: să Lema: să POS: PART Tag: Qs DEP: mark
Text: transmite Lema: transmite POS: AUX Tag: Vmip3s DEP: ROOT
Text: în Lema: în POS: ADP Tag: Spsa DEP: case
Text: teritoriu Lema: teritoriu POS: NOUN Tag: Ncms-n DEP: obl
Text: lista Lema: listă POS: NOUN Tag: Ncfsry DEP: obj
Text: experților Lema: expert POS: NOUN Tag: Ncmpoy DEP: nmod
Text: electorali Lema: electoral POS: ADJ Tag: Afpmp-n DEP: amod
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text: Lista Lema: listă POS: NOUN Tag: Ncfsry DEP: nsubj
Text: o Lema: el POS: PRON Tag: Pp3fsa--------w DEP: obj
Text: trimitem Lema: trimite POS: VERB Tag: Vmip1p DEP: ROOT
Text: în Lema: în POS: ADP Tag: Spsa DEP: case
Text: toate Lema: tot POS: DET Tag: Di3fpr DEP: det
Text: județele Lema: județ POS: NOUN Tag: Ncfpry DEP: obl
Text: pe Lema: pe POS: ADP Tag: Spsa DEP: case
Text: mailuri Lema: mail POS: NOUN Tag: Ncfp-n DEP: obl
Text: securizate Lema: securizat POS: ADJ Tag: Afpfp-n DEP: amod
Text: ” Lema: ” POS: PUNCT Tag: DBLQ DEP: punct
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: a Lema: avea POS: AUX Tag: Va--3s DEP: aux
Text: explicat Lema: explica POS: VERB Tag: Vmp--sm DEP: parataxis
Text: Toni Lema: Toni POS: PROPN Tag: Np DEP: nsubj
Text: Greblă Lema: Greblă POS: PROPN Tag: Np DEP: flat
Text: pentru Lema: pentru POS: ADP Tag: Spsa DEP: case
Text: HotNews.ro Lema: HotNews.ro POS: PROPN Tag: Np DEP: obl
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text:
```

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 Lema:
 POS: SPACE Tag: _SP DEP: dep
Text: Președintele Lema: președinte POS: NOUN Tag: Ncmsry DEP: nsubj
Text: AEP Lema: AEP POS: PROPN Tag: Np DEP: nmod
Text: susține Lema: susține POS: AUX Tag: Vmip3s DEP: ROOT
Text: că Lema: că POS: SCONJ Tag: Csssp DEP: mark
Text: dintr- Lema: dintru POS: ADP Tag: Spsay DEP: case
Text: o Lema: un POS: DET Tag: Tifsr DEP: det
Text: „ Lema: „ POS: PUNCT Tag: DBLQ DEP: punct
Text: eroare Lema: eroare POS: NOUN Tag: Ncfsrn DEP: ccomp
Text: ” Lema: ” POS: PUNCT Tag: DBLQ DEP: punct
Text: că Lema: că POS: SCONJ Tag: Csssp DEP: mark
Text: într- Lema: întru POS: ADP Tag: Spsay DEP: case
Text: un Lema: un POS: DET Tag: Timsr DEP: det
Text: județ Lema: județ POS: NOUN Tag: Ncms-n DEP: obl
Text: lista Lema: listă POS: NOUN Tag: Ncfsry DEP: nsubj:pass
Text: nu Lema: nu POS: PART Tag: Qz DEP: advmod
Text: a Lema: avea POS: AUX Tag: Va--3s DEP: aux
Text: fost Lema: fi POS: AUX Tag: Vap--sm DEP: aux:pass
Text: trimisă Lema: trimite POS: VERB Tag: Vmp--sf DEP: acl
Text: pe Lema: pe POS: ADP Tag: Spsa DEP: case
Text: mailul Lema: mail POS: NOUN Tag: Ncmsry DEP: obl
Text: securizat Lema: securizat POS: ADJ Tag: Afpms-n DEP: amod
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: ci Lema: ci POS: CCONJ Tag: Ccssp DEP: cc
Text: pe Lema: pe POS: ADP Tag: Spsa DEP: case
Text: un Lema: un POS: DET Tag: Timsr DEP: det
Text: alt Lema: alt POS: DET Tag: Di3ms----e DEP: det
Text: mail Lema: mail POS: NOUN Tag: Ncms-n DEP: conj
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text:
 Lema:
 POS: SPACE Tag: _SP DEP: dep
Text: Greblă Lema: Greblă POS: PROPN Tag: Np DEP: obl
Text: : Lema: : POS: PUNCT Tag: COLON DEP: punct
Text: „ Lema: „ POS: PUNCT Tag: DBLQ DEP: punct
Text: A Lema: avea POS: AUX Tag: Va--3s DEP: aux
Text: fost Lema: fi POS: AUX Tag: Vap--sm DEP: cop
Text: o Lema: un POS: DET Tag: Tifsr DEP: det
Text: suprasolicitare Lema: suprasolicitare POS: NOUN Tag: Ncfsrn DEP:
obj
Text: a Lema: al POS: DET Tag: Tsfs DEP: det
Text: personalului Lema: personal POS: NOUN Tag: Ncmsoy DEP: nmod
Text: AEP Lema: AEP POS: PROPN Tag: Np DEP: nmod
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text: Regretăm Lema: Regretăm POS: VERB Tag: Vmip1p DEP: ROOT
Text: acest Lema: acest POS: DET Tag: Dd3msr---e DEP: det
Text: incident Lema: incident POS: NOUN Tag: Ncms-n DEP: obj
Text: ” Lema: ” POS: PUNCT Tag: DBLQ DEP: punct
```

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Text:
 Lema:
 POS: SPACE Tag: _SP DEP: dep
Text: „ Lema: „ POS: PUNCT Tag: DBLQ DEP: punct
Text: Timp Lema: timp POS: NOUN Tag: Ncms-n DEP: nsubj
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: o Lema: unu POS:  Tag: Mcfsrl DEP: det
Text: oră Lema: oră POS: NOUN Tag: Ncfsrn DEP: nmod
Text: a Lema: avea POS: AUX Tag: Va--3s DEP: aux
Text: stat Lema: sta POS: VERB Tag: Vmp--sm DEP: conj
Text: afișat Lema: afișat POS: ADJ Tag: Afpms-n DEP: amod
Text: și Lema: și POS: CCONJ Tag: Crssp DEP: cc
Text: a Lema: avea POS: AUX Tag: Va--3s DEP: aux
Text: putut Lema: putea POS: VERB Tag: Vmp--sm DEP: conj
Text: să Lema: să POS: PART Tag: Qs DEP: mark
Text: fie Lema: fi POS: AUX Tag: Vasp3 DEP: aux:pass
Text: accesat Lema: accesa POS: VERB Tag: Vmp--sm DEP: ccomp
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: pe Lema: pe POS: ADP Tag: Spsa DEP: fixed
Text: acest Lema: acest POS: DET Tag: Dd3msr---e DEP: det
Text: mail Lema: mail POS: NOUN Tag: Ncms-n DEP: obl:agent
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text: Aproximativ Lema: aproximativ POS: ADV Tag: Rgp DEP: advmod
Text: 30 Lema: 30 POS:  Tag: Mc-s-d DEP: nummod
Text: – Lema: – POS: PUNCT Tag: DASH DEP: punct
Text: 35 Lema: 35 POS:  Tag: Mc-s-d DEP: nummod
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: persoane Lema: persoană POS: NOUN Tag: Ncfp-n DEP: obl
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: unii Lema: unul POS: PRON Tag: Pi3mpr DEP: nsubj
Text: au Lema: avea POS: AUX Tag: Va--3p DEP: aux
Text: intrat Lema: intra POS: VERB Tag: Vmp--sm DEP: ROOT
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: două Lema: doi POS: NUM Tag: Mcfp-l DEP: nummod
Text: ori Lema: oară POS: NOUN Tag: Ncfp-n DEP: obl
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: au Lema: avea POS: AUX Tag: Va--3p DEP: aux
Text: putut Lema: putea POS: VERB Tag: Vmp--sm DEP: conj
Text: să Lema: să POS: PART Tag: Qs DEP: mark
Text: vadă Lema: vedea POS: AUX Tag: Vmsp3 DEP: ccomp
Text: lista Lema: listă POS: NOUN Tag: Ncfsry DEP: obj
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text: Nu Lema: nu POS: PART Tag: Qz DEP: advmod
Text: știm Lema: ști POS: VERB Tag: Vmip1p DEP: ROOT
Text: dacă Lema: dacă POS: SCONJ Tag: Csssp DEP: mark
Text: au Lema: avea POS: AUX Tag: Va--3p DEP: aux
Text: fost Lema: fi POS: AUX Tag: Vap--sm DEP: aux:pass
Text: luate Lema: lua POS: VERB Tag: Vmp--pf DEP: ccomp
Text: capturi Lema: captură POS: NOUN Tag: Ncfp-n DEP: nsubj:pass
```

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Text: sau Lema: sau POS: CCONJ Tag: Ccssp DEP: cc
Text: doar Lema: doar POS: ADV Tag: Rgp DEP: advmod
Text: a Lema: avea POS: AUX Tag: Va--3s DEP: aux
Text: fost Lema: fi POS: AUX Tag: Vap--sm DEP: aux:pass
Text: accesată Lema: accesa POS: VERB Tag: Vmp--sf DEP: conj
Text: pagina Lema: pagină POS: NOUN Tag: Ncfsry DEP: nsubj:pass
Text: ” Lema: ” POS: PUNCT Tag: DBLQ DEP: punct
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: a Lema: avea POS: AUX Tag: Va--3s DEP: aux
Text: adăugat Lema: adăuga POS: VERB Tag: Vmp--sm DEP: conj
Text: oficialul Lema: oficial POS: NOUN Tag: Ncmsry DEP: obj
Text: Autorității Lema: autorități POS: NOUN Tag: Ncfsoy DEP: nmod
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text:
 Lema:
 POS: SPACE Tag: _SP DEP: dep
Text: Potrivit Lema: potrivit POS: ADP Tag: Spsd DEP: case
Text: acestuia Lema: acesta POS: PRON Tag: Pd3mso DEP: obl
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: a Lema: avea POS: AUX Tag: Va--3s DEP: aux
Text: fost Lema: fi POS: AUX Tag: Vap--sm DEP: aux:pass
Text: șters Lema: șterge POS: VERB Tag: Vmp--sm DEP: ROOT
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: pe Lema: pe POS: ADP Tag: Spsa DEP: fixed
Text: mailul Lema: mail POS: NOUN Tag: Ncmsry DEP: obl
Text: destinat Lema: destinat POS: ADJ Tag: Afpms-n DEP: amod
Text: „ Lema: „ POS: PUNCT Tag: DBLQ DEP: punct
Text: publicului Lema: public POS: NOUN Tag: Ncmsoy DEP: iobj
Text: mai Lema: mai POS: ADV Tag: Rp DEP: advmod
Text: larg Lema: larg POS: ADJ Tag: Afpms-n DEP: amod
Text: ” Lema: ” POS: PUNCT Tag: DBLQ DEP: punct
Text: și Lema: și POS: CCONJ Tag: Crssp DEP: cc
Text: s- Lema: sine POS: PRON Tag: Px3--a--y-----w DEP: expl:pass
Text: a Lema: avea POS: AUX Tag: Va--3s DEP: aux
Text: făcut Lema: face POS: VERB Tag: Vmp--sm DEP: conj
Text: o Lema: un POS: DET Tag: Tifsr DEP: det
Text: anchetă Lema: anchetă POS: NOUN Tag: Ncfsrn DEP: obj
Text: internă Lema: intern POS: ADJ Tag: Afpfsrn DEP: amod
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text:
 Lema:
 POS: SPACE Tag: _SP DEP: dep
Text: Șeful Lema: șef POS: NOUN Tag: Ncmsry DEP: nsubj
Text: AEP Lema: AEP POS: PROPN Tag: Np DEP: nmod
Text: nu Lema: nu POS: PART Tag: Qz DEP: advmod
Text: folosește Lema: folosi POS: AUX Tag: Vmip3s DEP: ROOT
Text: termenul Lema: termen POS: NOUN Tag: Ncmsry DEP: obj
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: „ Lema: „ POS: PUNCT Tag: DBLQ DEP: punct
```

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Text: breșă Lema: breșă POS: NOUN Tag: Ncfsrn DEP: nmod
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: securitate Lema: securitate POS: NOUN Tag: Ncfsrn DEP: nmod
Text: ” Lema: ” POS: PUNCT Tag: DBLQ DEP: punct
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: care Lema: care POS: PRON Tag: Pw3--r DEP: nsubj
Text: este Lema: fi POS: AUX Tag: Vaip3s DEP: cop
Text: sintagma Lema: sintagme POS: NOUN Tag: Ncfsry DEP: acl
Text: clasică Lema: clasic POS: ADJ Tag: Afpfsrn DEP: amod
Text: pentru Lema: pentru POS: ADP Tag: Spsa DEP: case
Text: definirea Lema: definire POS: NOUN Tag: Ncfsry DEP: nmod
Text: accesului Lema: acces POS: NOUN Tag: Ncmsoy DEP: nmod
Text: neautorizat Lema: neautorizat POS: ADJ Tag: Afpms-n DEP: amod
Text: la Lema: la POS: ADP Tag: Spsa DEP: case
Text: date Lema: dată POS: NOUN Tag: Ncfp-n DEP: obl
Text: sau Lema: sau POS: CCONJ Tag: Ccssp DEP: cc
Text: informații Lema: informație POS: NOUN Tag: Ncfp-n DEP: conj
Text: găzduite Lema: găzdui POS: VERB Tag: Vmp--pf DEP: amod
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: computere Lema: computere POS: NOUN Tag: Ncfp-n DEP: obl:agent
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: telefoane Lema: telefoan POS: NOUN Tag: Ncfp-n DEP: conj
Text: mobile Lema: mobil POS: ADJ Tag: Afpfp-n DEP: amod
Text: sau Lema: sau POS: CCONJ Tag: Ccssp DEP: cc
Text: alte Lema: alt POS: DET Tag: Di3fpr---e DEP: det
Text: device Lema: device POS: NOUN Tag: Ncms-n DEP: conj
Text: -uri Lema: -uri POS: DET Tag: Tifp-y DEP: det
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text:
 Lema:
 POS: SPACE Tag: _SP DEP: dep
Text: „ Lema: „ POS: PUNCT Tag: DBLQ DEP: punct
Text: Nu Lema: nu POS: PART Tag: Qz DEP: advmod
Text: este Lema: fi POS: AUX Tag: Vaip3s DEP: cop
Text: o Lema: un POS: DET Tag: Tifsr DEP: det
Text: breșă Lema: breșă POS: NOUN Tag: Ncfsrn DEP: obl
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: securitate Lema: securitate POS: NOUN Tag: Ncfsrn DEP: nmod
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: numele Lema: nume POS: NOUN Tag: Ncmsry DEP: nsubj
Text: sunt Lema: fi POS: AUX Tag: Vaip3p DEP: cop
Text: publice Lema: public POS: ADJ Tag: Afpfp-n DEP: conj
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: doar Lema: doar POS: ADV Tag: Rgp DEP: advmod
Text: datele Lema: dată POS: NOUN Tag: Ncfpry DEP: nsubj
Text: cu Lema: cu POS: ADP Tag: Spsa DEP: case
Text: caracter Lema: caracter POS: NOUN Tag: Ncms-n DEP: nmod
Text: personal Lema: personal POS: ADJ Tag: Afpms-n DEP: amod
Text: nu Lema: nu POS: PART Tag: Qz DEP: advmod
```

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Text: erau Lema: fi POS: AUX Tag: Vaii3p DEP: cop
Text: publice Lema: public POS: ADJ Tag: Afpfp-n DEP: ROOT
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text: Au Lema: avea POS: AUX Tag: Va--3p DEP: aux
Text: apărut Lema: apărea POS: VERB Tag: Vmp--sm DEP: ROOT
Text: CNP Lema: CNP POS: PROPN Tag: Np DEP: nsubj
Text: -ul Lema: -ul POS: DET Tag: Tfmsry DEP: det
Text: și Lema: și POS: CCONJ Tag: Crssp DEP: cc
Text: domiciliul Lema: domiciliu POS: NOUN Tag: Ncmsry DEP: conj
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text: Credem Lema: Credem POS: VERB Tag: Vmip1p DEP: ROOT
Text: că Lema: că POS: SCONJ Tag: Csssp DEP: mark
Text: acest Lema: acest POS: DET Tag: Dd3msr---e DEP: det
Text: mail Lema: mail POS: NOUN Tag: Ncms-n DEP: nsubj:pass
Text: a Lema: avea POS: AUX Tag: Va--3s DEP: aux
Text: fost Lema: fi POS: AUX Tag: Vap--sm DEP: aux:pass
Text: accesat Lema: accesa POS: VERB Tag: Vmp--sm DEP: ccomp
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: cei Lema: acela POS: PRON Tag: Pd3mpr DEP: obl:agent
Text: interesați Lema: interesat POS: ADJ Tag: Afpmp-n DEP: amod
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: cei Lema: acela POS: PRON Tag: Pd3mpr DEP: obl:pmod
Text: interesați Lema: interesat POS: ADJ Tag: Afpmp-n DEP: amod
Text: să Lema: să POS: PART Tag: Qs DEP: mark
Text: vadă Lema: vedea POS: AUX Tag: Vmsp3 DEP: advcl
Text: dacă Lema: dacă POS: SCONJ Tag: Csssp DEP: mark
Text: se Lema: sine POS: PRON Tag: Px3--a--------w DEP: expl:pass
Text: regăsesc Lema: regăsi POS: AUX Tag: Vmip3p DEP: ccomp
Text: pe Lema: pe POS: ADP Tag: Spsa DEP: case
Text: listă Lema: listă POS: NOUN Tag: Ncfsrn DEP: obl
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text: Nu Lema: nu POS: PART Tag: Qz DEP: advmod
Text: cred Lema: crede POS: AUX Tag: Vmip1s DEP: ROOT
Text: că Lema: că POS: SCONJ Tag: Csssp DEP: mark
Text: cineva Lema: cineva POS: PRON Tag: Pi3-sr DEP: nsubj
Text: a Lema: avea POS: AUX Tag: Va--3s DEP: aux
Text: făcut Lema: face POS: VERB Tag: Vmp--sm DEP: ccomp
Text: o Lema: un POS: DET Tag: Tifsr DEP: det
Text: captură Lema: captură POS: NOUN Tag: Ncfsrn DEP: obj
Text: ca Lema: ca POS: ADV Tag: Rc DEP: mark
Text: să Lema: să POS: PART Tag: Qs DEP: mark
Text: folosească Lema: folosi POS: AUX Tag: Vmsp3 DEP: advcl
Text: datele Lema: dată POS: NOUN Tag: Ncfpry DEP: obj
Text: cu Lema: cu POS: ADP Tag: Spsa DEP: case
Text: caracter Lema: caracter POS: NOUN Tag: Ncms-n DEP: nmod
Text: personal Lema: personal POS: ADJ Tag: Afpms-n DEP: amod
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text: Eu Lema: eu POS: PRON Tag: Pp1-sn--------s DEP: nsubj
Text: am Lema: avea POS: AUX Tag: Va--1 DEP: ROOT
```

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Text: obligația Lema: obligație POS: NOUN Tag: Ncfsry DEP: obj
Text: să Lema: să POS: PART Tag: Qs DEP: mark
Text: anunț Lema: anunța POS: AUX Tag: Vmip1s DEP: acl
Text: Autoritatea Lema: autoritate POS: NOUN Tag: Ncfsry DEP: obj
Text: pentru Lema: pentru POS: ADP Tag: Spsa DEP: case
Text: Protecția Lema: protecție POS: NOUN Tag: Ncfsry DEP: nmod
Text: Datelor Lema: Dată POS: NOUN Tag: Ncfpoy DEP: nmod
Text: cu Lema: cu POS: ADP Tag: Spsa DEP: case
Text: caracter Lema: caracter POS: NOUN Tag: Ncms-n DEP: nmod
Text: personal Lema: personal POS: ADJ Tag: Afpms-n DEP: amod
Text: ” Lema: ” POS: PUNCT Tag: DBLQ DEP: punct
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: a Lema: avea POS: AUX Tag: Va--3s DEP: aux
Text: fost Lema: fi POS: AUX Tag: Vap--sm DEP: conj
Text: poziția Lema: poziție POS: NOUN Tag: Ncfsry DEP: nsubj
Text: lui Lema: lui POS: DET Tag: Tf-so DEP: det
Text: Toni Lema: Toni POS: PROPN Tag: Np DEP: nmod
Text: Greblă Lema: Greblă POS: PROPN Tag: Np DEP: flat
Text: pentru Lema: pentru POS: ADP Tag: Spsa DEP: case
Text: HotNews.ro Lema: HotNews.ro POS: PROPN Tag: Np DEP: nmod
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text:
 Lema:
 POS: SPACE Tag: _SP DEP: dep
Text: La Lema: la POS: ADP Tag: Spsa DEP: case
Text: final Lema: final POS: NOUN Tag: Ncms-n DEP: obl
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: șeful Lema: șef POS: NOUN Tag: Ncmsry DEP: nsubj
Text: AEP Lema: AEP POS: PROPN Tag: Np DEP: nmod
Text: a Lema: avea POS: AUX Tag: Va--3s DEP: aux
Text: spus Lema: spune POS: VERB Tag: Vmp--sm DEP: ROOT
Text: că Lema: că POS: SCONJ Tag: Csssp DEP: mark
Text: „ Lema: „ POS: PUNCT Tag: DBLQ DEP: punct
Text: regretă Lema: regreta POS: VERB Tag: Vmis3s DEP: ccomp
Text: incidentul Lema: incident POS: NOUN Tag: Ncmsry DEP: obj
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: dar Lema: dar POS: CCONJ Tag: Ccssp DEP: cc
Text: oameni Lema: om POS: NOUN Tag: Ncmp-n DEP: nsubj
Text: suntem Lema: fi POS: VERB Tag: Vmip1p DEP: dep
Text: cu Lema: cu POS: ADP Tag: Spsa DEP: case
Text: toții Lema: tot POS: PRON Tag: Pi3mpr DEP: obl
Text: ” Lema: ” POS: PUNCT Tag: DBLQ DEP: punct
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text:
 Lema:
 POS: SPACE Tag: _SP DEP: dep
Text: „ Lema: „ POS: PUNCT Tag: DBLQ DEP: punct
Text: Dacă Lema: dacă POS: SCONJ Tag: Csssp DEP: mark
Text: permitem Lema: permite POS: VERB Tag: Vmip1p DEP: ROOT
Text: unor Lema: un POS: DET Tag: Ti-po DEP: det
```

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Text: competitori Lema: competitor POS: NOUN Tag: Ncmp-n DEP: iobj
Text: politici Lema: politic POS: ADJ Tag: Afpmp-n DEP: amod
Text: să Lema: să POS: PART Tag: Qs DEP: mark
Text: îi Lema: el POS: PRON Tag: Pp3-sd--------w DEP: iobj
Text: așteptăm Lema: aștepta POS: VERB Tag: Vmip1p DEP: ccomp
Text: până Lema: până POS: ADP Tag: Spsa DEP: case
Text: la Lema: la POS: ADP Tag: Spsa DEP: fixed
Text: 12 Lema: 12 POS:  Tag: Mc-s-d DEP: nummod
Text: noaptea Lema: noaptea POS: ADV Tag: Rgp DEP: obl
Text: să Lema: să POS: PART Tag: Qs DEP: mark
Text: vină Lema: veni POS: AUX Tag: Vmsp3 DEP: advcl
Text: cu Lema: cu POS: ADP Tag: Spsa DEP: case
Text: 200.000 Lema: 200.000 POS:  Tag: Mc-s-d DEP: nummod
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: semnături Lema: semnătură POS: NOUN Tag: Ncfp-n DEP: obl
Text: și Lema: și POS: CCONJ Tag: Crssp DEP: cc
Text: după Lema: după POS: ADP Tag: Spsa DEP: case
Text: aia Lema: acela POS: PRON Tag: Pd3fsr DEP: conj
Text: o Lema: un POS: DET Tag: Tifsr DEP: det
Text: armată Lema: armată POS: NOUN Tag: Ncfsrn DEP: nsubj
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: 30–500 Lema: 30–500 POS:  Tag: Mc-s-d DEP: nummod
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: oameni Lema: om POS: NOUN Tag: Ncmp-n DEP: obj
Text: să Lema: să POS: PART Tag: Qs DEP: mark
Text: verifice Lema: verifica POS: AUX Tag: Vmsp3 DEP: ccomp
Text: dacă Lema: dacă POS: SCONJ Tag: Csssp DEP: mark
Text: acestea Lema: acesta POS: PRON Tag: Pd3fpr DEP: nsubj
Text: sunt Lema: fi POS: AUX Tag: Vaip3p DEP: cop
Text: în Lema: în POS: ADP Tag: Spsa DEP: case
Text: ordine Lema: ordine POS: NOUN Tag: Ncfsrn DEP: obl
Text: ... Lema: ... POS: PUNCT Tag: HELLIP DEP: punct
Text: A Lema: avea POS: AUX Tag: Va--3s DEP: aux
Text: fost Lema: fi POS: AUX Tag: Vap--sm DEP: cop
Text: o Lema: un POS: DET Tag: Tifsr DEP: det
Text: suprasolicitare Lema: suprasolicitare POS: NOUN Tag: Ncfsrn DEP:
nsubj
Text: a Lema: al POS: DET Tag: Tsfs DEP: det
Text: personalului Lema: personal POS: NOUN Tag: Ncmsoy DEP: nmod
Text: AEP Lema: AEP POS: PROPN Tag: Np DEP: nmod
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: am Lema: avea POS: AUX Tag: Va--1 DEP: aux
Text: stat Lema: sta POS: VERB Tag: Vmp--sm DEP: ROOT
Text: până Lema: până POS: ADP Tag: Spsa DEP: case
Text: la Lema: la POS: ADP Tag: Spsa DEP: fixed
Text: 02.30 Lema: 02.30 POS:  Tag: Mc-s-d DEP: nummod
Text: dimineața Lema: dimineața POS: ADV Tag: Rgp DEP: advmod
Text: în Lema: în POS: ADP Tag: Spsa DEP: case
Text: ultima Lema: ultim POS: NUM Tag: Mofsrly DEP: nummod
```

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Text: zi Lema: zi POS: NOUN Tag: Ncfsrn DEP: obl
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: depunere Lema: depunere POS: NOUN Tag: Ncfsrn DEP: nmod
Text: a Lema: al POS: DET Tag: Tsfs DEP: det
Text: candidaturilor Lema: candidature POS: NOUN Tag: Ncfpoy DEP: nmod
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text: Poate Lema: poate POS: ADV Tag: Rgp DEP: ROOT
Text: ar Lema: avea POS: AUX Tag: Va--3 DEP: aux
Text: trebui Lema: trebui POS: VERB Tag: Vmnp DEP: ccomp
Text: depuse Lema: depune POS: VERB Tag: Vmp--pf DEP: csubj
Text: candidaturile Lema: candidature POS: NOUN Tag: Ncfpry DEP:
nsubj:pass
Text: până Lema: până POS: SCONJ Tag: Csssp DEP: case
Text: la Lema: la POS: ADP Tag: Spsa DEP: fixed
Text: ora Lema: oră POS: NOUN Tag: Ncfsry DEP: nmod:tmod
Text: 17.00 Lema: 17.00 POS: NUM Tag: Mc-p-d DEP: nummod
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text: Repet Lema: repet POS: NOUN Tag: Ncms-n DEP: nsubj
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: regretăm Lema: regreta POS: VERB Tag: Vmip1p DEP: ROOT
Text: acest Lema: acest POS: DET Tag: Dd3msr---e DEP: det
Text: incident Lema: incident POS: NOUN Tag: Ncms-n DEP: obj
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text: Este Lema: fi POS: AUX Tag: Vaip3s DEP: cop
Text: o Lema: un POS: DET Tag: Tifsr DEP: det
Text: întâmplare Lema: întâmplare POS: NOUN Tag: Ncfsrn DEP: dep
Text: nefericită Lema: nefericit POS: ADJ Tag: Afpfsrn DEP: amod
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: nu Lema: nu POS: PART Tag: Qz DEP: advmod
Text: cred Lema: crede POS: AUX Tag: Vmip1s DEP: ROOT
Text: că Lema: că POS: SCONJ Tag: Csssp DEP: mark
Text: are Lema: avea POS: AUX Tag: Vmip3s DEP: ccomp
Text: o Lema: un POS: DET Tag: Tifsr DEP: det
Text: mare Lema: mare POS: ADJ Tag: Afpfsrn DEP: amod
Text: amploare Lema: amploare POS: NOUN Tag: Ncfsrn DEP: obj
Text: ” Lema: ” POS: PUNCT Tag: DBLQ DEP: punct
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: a Lema: avea POS: AUX Tag: Va--3s DEP: aux
Text: spus Lema: spune POS: VERB Tag: Vmp--sm DEP: parataxis
Text: Toni Lema: Toni POS: PROPN Tag: Np DEP: nsubj
Text: Greblă Lema: Greblă POS: PROPN Tag: Np DEP: flat
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text:
 Lema:
 POS: SPACE Tag: _SP DEP: dep
Text: Precizare Lema: precizare POS: NOUN Tag: Ncfsrn DEP: parataxis
Text: : Lema: : POS: PUNCT Tag: COLON DEP: punct
Text: Într- Lema: întru POS: ADP Tag: Spsay DEP: case
Text: o Lema: un POS: DET Tag: Tifsr DEP: det
```

```
Text: versiune Lema: versiune POS: NOUN Tag: Ncfsrn DEP: obl
Text: anterioară Lema: anterior POS: ADJ Tag: Afpfsrn DEP: amod
Text: a Lema: al POS: DET Tag: Tsfs DEP: det
Text: textului Lema: text POS: NOUN Tag: Ncmsoy DEP: nmod
Text: președintele Lema: președinte POS: NOUN Tag: Ncmsry DEP: nmod
Text: AEP Lema: AEP POS: PROPN Tag: Np DEP: nmod
Text: Toni Lema: Toni POS: PROPN Tag: Np DEP: nmod
Text: Greblă Lema: Greblă POS: PROPN Tag: Np DEP: nmod
Text: a Lema: avea POS: AUX Tag: Va--3s DEP: aux
Text: definit Lema: defini POS: VERB Tag: Vmp--sm DEP: ROOT
Text: experții Lema: expert POS: NOUN Tag: Ncmpry DEP: obj
Text: electorali Lema: electoral POS: ADJ Tag: Afpmp-n DEP: amod
Text: drept Lema: drept POS: ADJ Tag: Afpms-n DEP: case
Text: „ Lema: „ POS: PUNCT Tag: DBLQ DEP: punct
Text: persoanele Lema: persoană POS: NOUN Tag: Ncfpry DEP: obj
Text: care Lema: care POS: PRON Tag: Pw3--r DEP: nsubj
Text: lucrează Lema: lucra POS: VERB Tag: Vmip3 DEP: acl
Text: în Lema: în POS: ADP Tag: Spsa DEP: case
Text: IT Lema: IT POS: NOUN Tag: Yn DEP: obl
Text: în Lema: în POS: ADP Tag: Spsa DEP: case
Text: procesul Lema: proces POS: NOUN Tag: Ncmsry DEP: obl
Text: electoral Lema: electoral POS: ADJ Tag: Afpms-n DEP: amod
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: cei Lema: acela POS: PRON Tag: Pd3mpr DEP: appos
Text: care Lema: care POS: PRON Tag: Pw3--r DEP: nsubj
Text: la Lema: la POS: ADP Tag: Spsa DEP: case
Text: intrarea Lema: intrare POS: NOUN Tag: Ncfsry DEP: obl
Text: în Lema: în POS: ADP Tag: Spsa DEP: case
Text: secțiile Lema: secție POS: NOUN Tag: Ncfpry DEP: nmod
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: votare Lema: votare POS: NOUN Tag: Ncfsrn DEP: nmod
Text: preiau Lema: preiaue POS: AUX Tag: Vmip3p DEP: acl
Text: cărțile Lema: carte POS: NOUN Tag: Ncfpry DEP: obj
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: identitate Lema: identitate POS: NOUN Tag: Ncfsrn DEP: nmod
Text: ale Lema: al POS: DET Tag: Tsfp DEP: det
Text: alegătorilor Lema: alegător POS: NOUN Tag: Ncmpoy DEP: nmod
Text: și Lema: și POS: CCONJ Tag: Crssp DEP: cc
Text: verifică Lema: verifica POS: VERB Tag: Vmis3s DEP: conj
Text: dacă Lema: dacă POS: SCONJ Tag: Csssp DEP: mark
Text: cetățenii Lema: cetățean POS: NOUN Tag: Ncmpry DEP: nsubj
Text: au Lema: avea POS: AUX Tag: Va--3p DEP: ccomp
Text: drept Lema: drept POS: ADV Tag: Rgp DEP: obj
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: vot Lema: vot POS: NOUN Tag: Ncms-n DEP: obl:pmod
Text: ” Lema: ” POS: PUNCT Tag: DBLQ DEP: punct
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
Text:
 Lema:
```

```
 POS: SPACE Tag: _SP DEP: dep
Text: În Lema: în POS: ADP Tag: Spsa DEP: advmod
Text: realitate Lema: realitate POS: NOUN Tag: Ncfsrn DEP: fixed
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: conform Lema: conform POS: ADP Tag: Spsd DEP: case
Text: informațiile Lema: informație POS: NOUN Tag: Ncfpry DEP: obl
Text: primite Lema: primi POS: VERB Tag: Vmp--pf DEP: amod
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: la Lema: la POS: ADP Tag: Spsa DEP: fixed
Text: AEP Lema: AEP POS: PROPN Tag: Np DEP: obl
Text: și Lema: și POS: CCONJ Tag: Crssp DEP: cc
Text: contrare Lema: contrare POS: NOUN Tag: Ncfsrn DEP: conj
Text: a Lema: al POS: DET Tag: Tsfs DEP: det
Text: ceea Lema: acela POS: PRON Tag: Pd3fsr DEP: nsubj
Text: ce Lema: ce POS: PRON Tag: Pw3--r DEP: fixed
Text: a Lema: avea POS: AUX Tag: Va--3s DEP: aux
Text: spus Lema: spune POS: VERB Tag: Vmp--sm DEP: acl
Text: președintele Lema: președinte POS: NOUN Tag: Ncmsry DEP: nsubj
Text: instituției Lema: instituție POS: NOUN Tag: Ncfsoy DEP: nmod
Text: în Lema: în POS: ADP Tag: Spsa DEP: case
Text: discuția Lema: discuție POS: NOUN Tag: Ncfsry DEP: obl
Text: înregistrată Lema: înregistra POS: VERB Tag: Vmp--sf DEP: acl
Text: cu Lema: cu POS: ADP Tag: Spsa DEP: case
Text: HotNews.ro Lema: HotNews.ro POS: PROPN Tag: Np DEP: obl
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: experții Lema: expert POS: NOUN Tag: Ncmpry DEP: nsubj
Text: electorali Lema: electoral POS: ADJ Tag: Afpmp-n DEP: amod
Text: sunt Lema: fi POS: AUX Tag: Vaip3p DEP: cop
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: așa Lema: așa POS: ADV Tag: Rgp DEP: advmod
Text: cum Lema: cum POS: ADV Tag: Rw DEP: fixed
Text: apare Lema: apărea POS: AUX Tag: Vmip3s DEP: ROOT
Text: corect Lema: corect POS: ADV Tag: Rgp DEP: advmod
Text: în Lema: în POS: ADP Tag: Spsa DEP: case
Text: articol Lema: articol POS: NOUN Tag: Ncms-n DEP: obl
Text: , Lema: , POS: PUNCT Tag: COMMA DEP: punct
Text: persoanele Lema: persoană POS: NOUN Tag: Ncfpry DEP: nsubj
Text: care Lema: care POS: PRON Tag: Pw3--r DEP: nsubj
Text: pot Lema: putea POS: AUX Tag: Vmip3p DEP: acl
Text: deveni Lema: deveni POS: VERB Tag: Vmnp DEP: ccomp
Text: președinți Lema: președint POS: NOUN Tag: Ncmp-n DEP: xcomp
Text: ai Lema: al POS: DET Tag: Tsmp DEP: det
Text: birourilor Lema: birou POS: NOUN Tag: Ncfpoy DEP: nmod
Text: electorale Lema: electoral POS: ADJ Tag: Afpfp-n DEP: amod
Text: ale Lema: al POS: DET Tag: Tsfp DEP: det
Text: secțiilor Lema: secție POS: NOUN Tag: Ncfpoy DEP: nmod
Text: de Lema: de POS: ADP Tag: Spsa DEP: case
Text: votare Lema: votare POS: NOUN Tag: Ncfsrn DEP: nmod
Text: din Lema: din POS: ADP Tag: Spsa DEP: case
```

```
Text: țară Lema: țară POS: NOUN Tag: Ncfsrn DEP: nmod
Text: sau Lema: sau POS: CCONJ Tag: Ccssp DEP: cc
Text: locțiitorii Lema: locțiitor POS: NOUN Tag: Ncmpry DEP: conj
Text: lor Lema: lui POS: DET Tag: Ds3---p DEP: det
Text: . Lema: . POS: PUNCT Tag: PERIOD DEP: punct
```

Segmentarea imaginilor

Segmentarea imaginilor este o operatie destul de frecventa in analiza automata a imaginilor repetitive cu contrast similar. In masura in care avem o serie de imagini similare in care trebuie sa numaram automat obiectele, putem aplica un astfel de algoritm. Nu se urmareste precizia totala, incercam sa fim cat mai aproape de numarul real.

```
import numpy as np
import pandas as pd
from matplotlib import pyplot as plt

# Our image processing tools
import skimage.filters
import skimage.io
import skimage.morphology
import skimage.color
from skimage import data
```

Biblioteca skimage este insotita de o serie de imagini de test, una dintre ele fiind extrem de potrivita cu aceasta serie de algoritmi.

```
coins = data.coins()

print(type(coins))
print(coins.shape)
```

```
<class 'numpy.ndarray'>
(303, 384)
```

```
[[119 121 122 ...  68  70  75]
 [119 124 126 ...  64  65  69]
 [122 126 132 ...  63  62  63]
 ...
 [123 122 121 ...  63  62  61]
 [122 122 121 ...  61  61  61]
 [120 120 120 ...  60  61  63]]
```

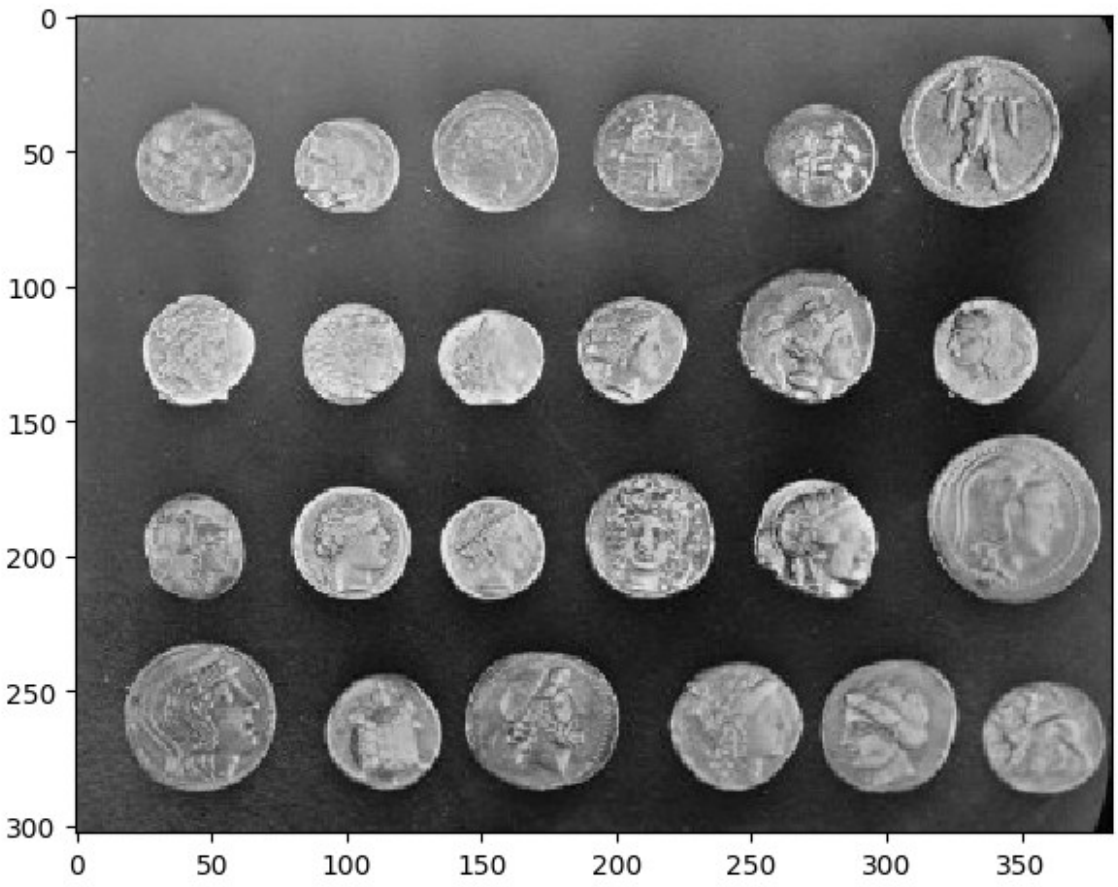
imaginile nu sunt stocate in format grafic ci in format numpy - ca matrice de trei strate.

```
print(coins[25:35])
```

```
[[119 121 122 ...  68  70  75]
 [119 124 126 ...  64  65  69]
 [122 126 132 ...  63  62  63]
 ...
 [123 122 121 ...  63  62  61]
 [122 122 121 ...  61  61  61]
 [120 120 120 ...  60  61  63]]
```

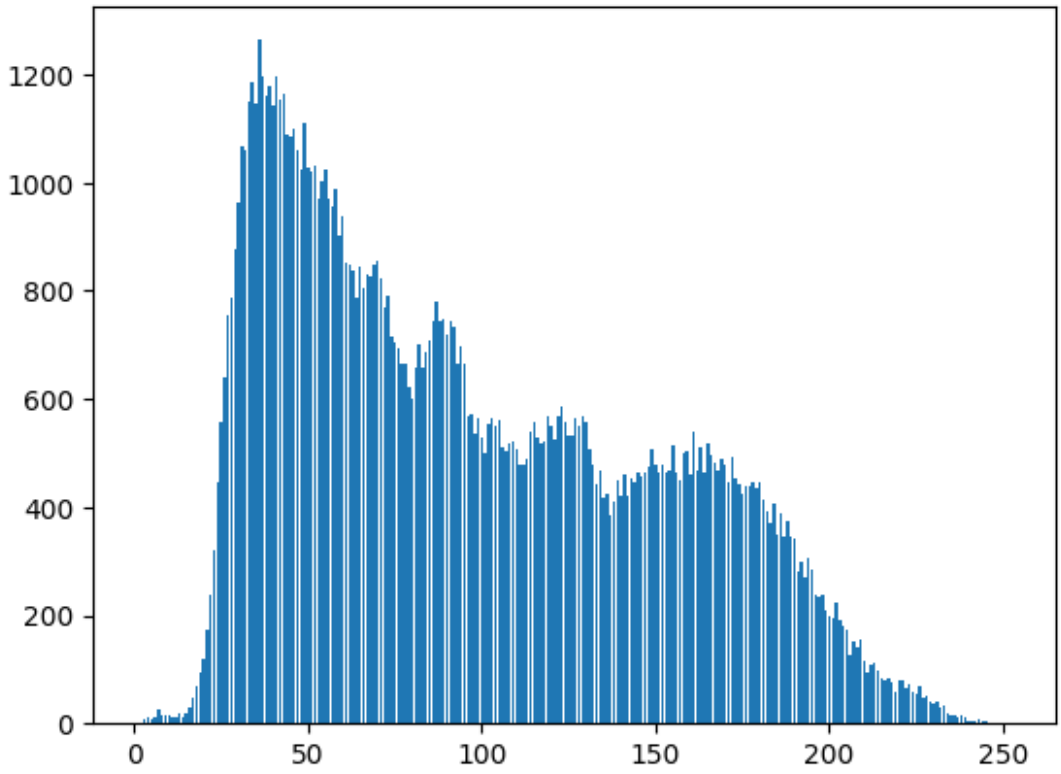
Putem folosi imshow pentru a vedea imaginea

```
skimage.io.imshow(coins)
skimage.io.show()
```



Putem afisa histograma imaginii (numarul de pixeli a fiecari intensitati de gri)

```
from skimage.exposure import histogram
hist, hist_centers = histogram(coins)
plt.bar(hist_centers, hist)
plt.show()
```

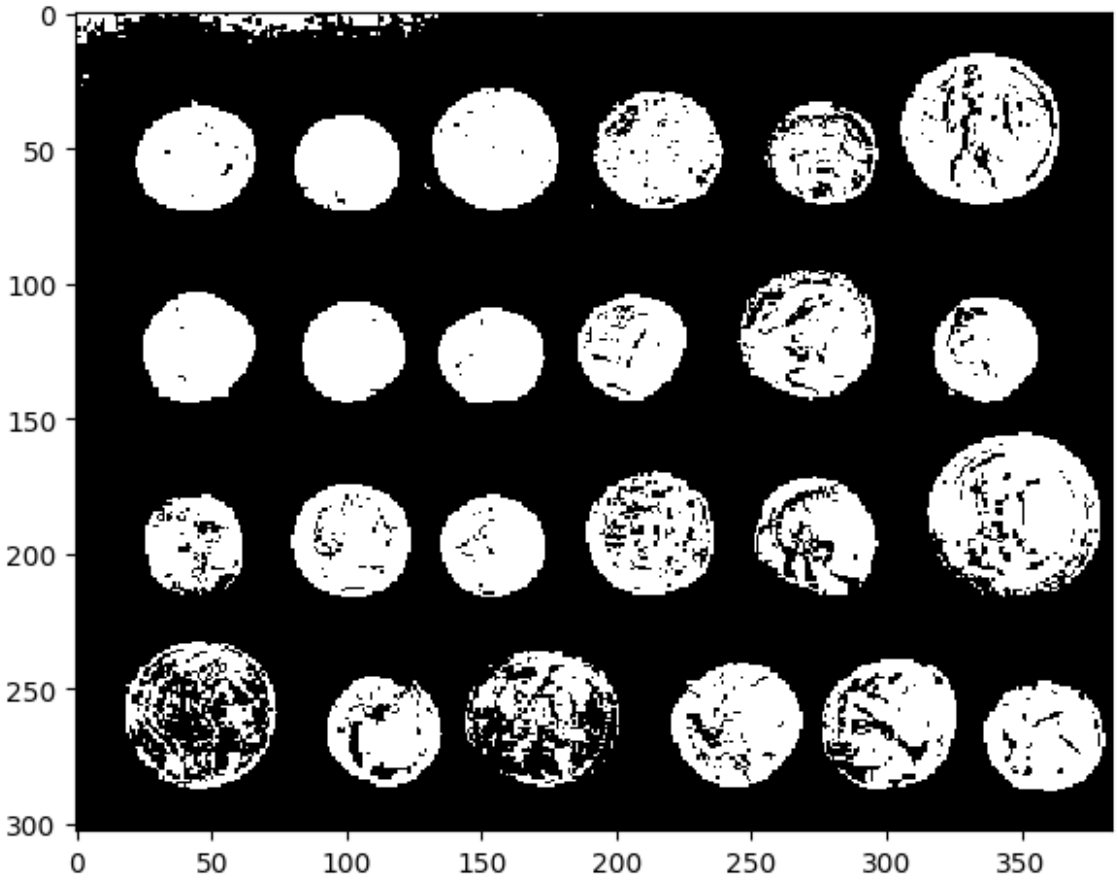


segmentare bazata pe valoare de prag (threshold)

Porneste de la ideea ca exista o valoare de prag, astfel incat toti pixelii care se gasesc deasupra acelei valori sunt parte din obiecte iar toti cei care se gasesc dedesubt sunt parte din fundal.

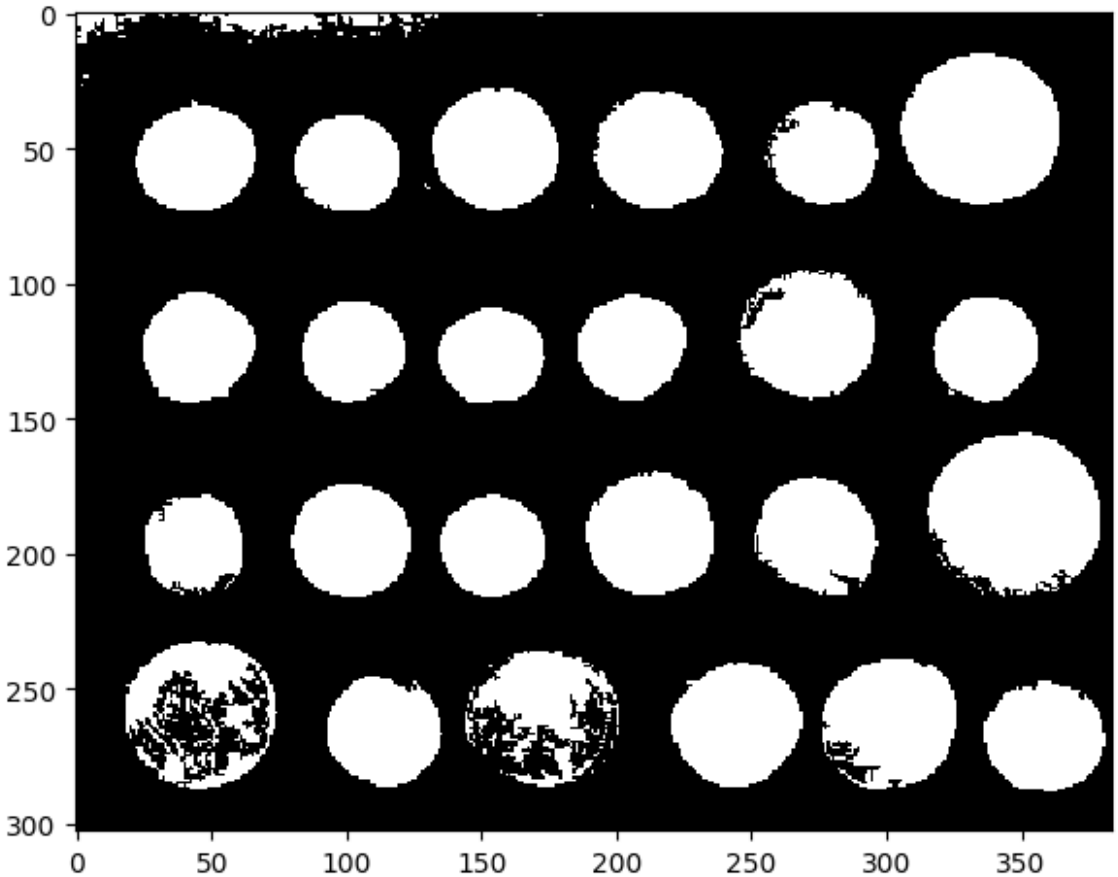
```
img120 = coins > 130

skimage.io.imshow(img120)
skimage.io.show()
```



Segmentarea este imperfecta, obiectele au ramas cu multe goluri. Exista algoritmi care permit umplerea golurilor, skimage ofera astfel de algoritmi.

```
from scipy import ndimage as ndi
fill_img120 = ndi.binary_fill_holes(img120)
skimage.io.imshow(fill_img120)
skimage.io.show()
```



ndi_label scaneaza imaginea si identifica ca obiecte separate toate regiunile cu pixeli conectati.

```
label_objects, nb_labels = ndi.label(fill_img120)
sizes = np.bincount(label_objects.ravel())
mask_sizes = sizes > 20
mask_sizes[0] = 0
coins_cleaned = mask_sizes[label_objects]

from skimage import data, filters, measure

labels = measure.label(coins_cleaned)
num_objects = labels.max()
print(num_objects)
```

```
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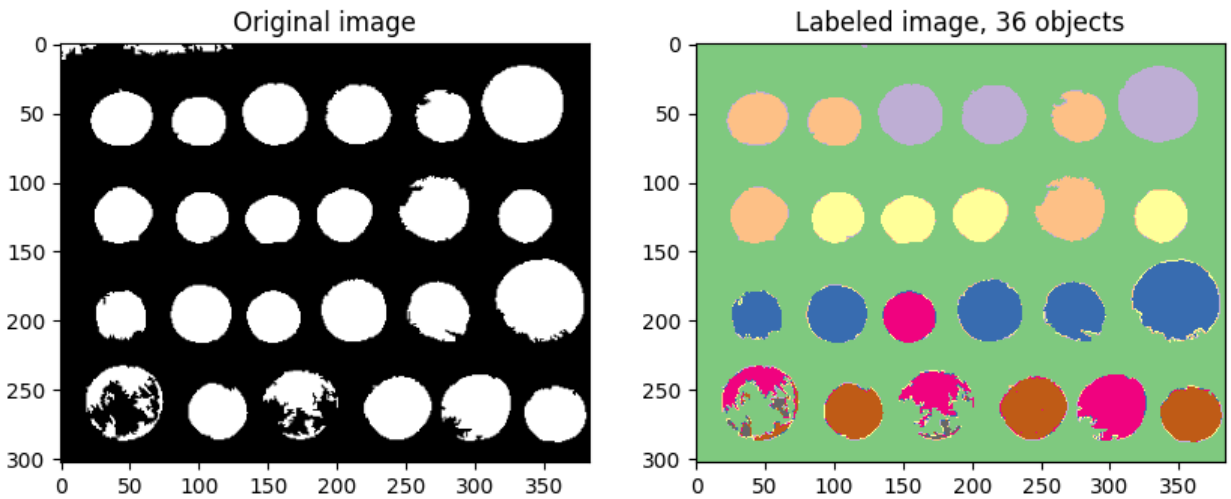
```

```
fig, ax = plt.subplots(ncols=2, figsize=(10, 5))
ax[0].imshow(coins_cleaned, cmap='gray')
ax[0].set_title('Original image')
ax[1].imshow(labels, cmap='Accent')
ax[1].set_title('Labeled image, %d objects' % num_objects)
plt.show()
```

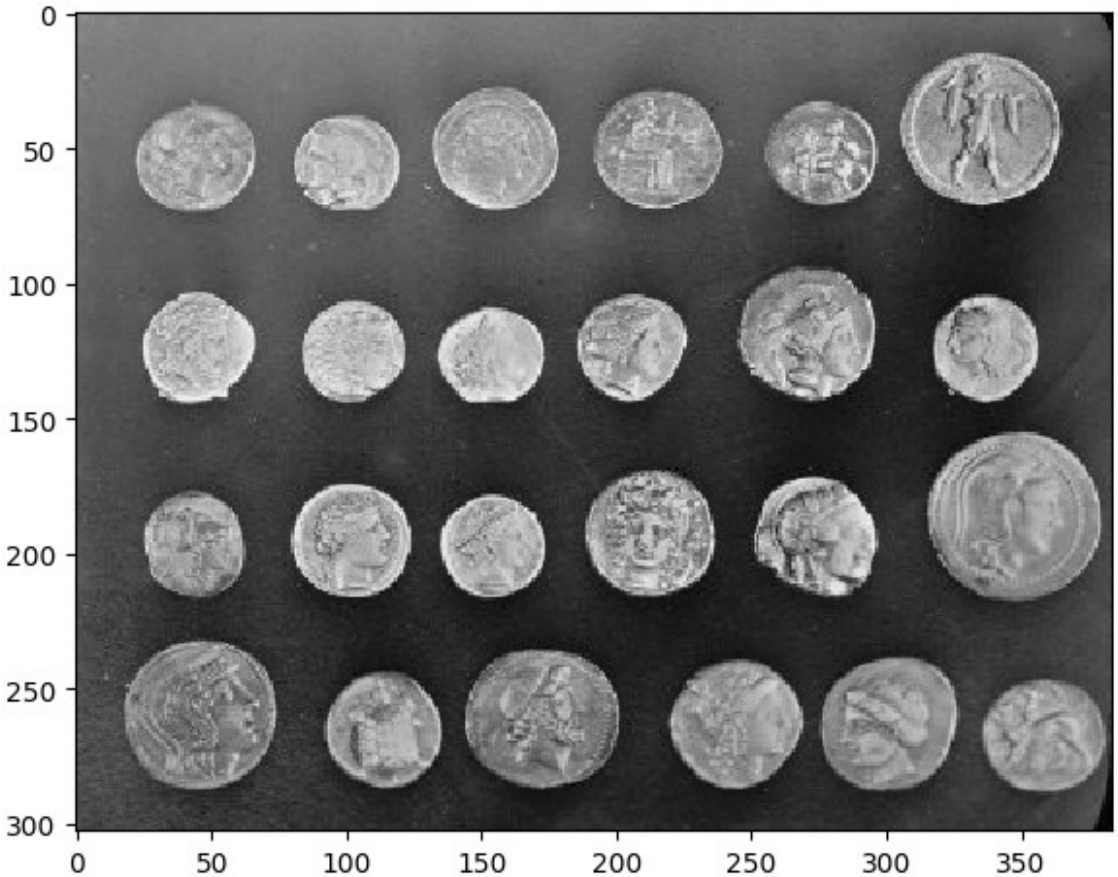


Segmentare bazata pe detectia muchiilor

Incepem din nou de la imaginea grayscale

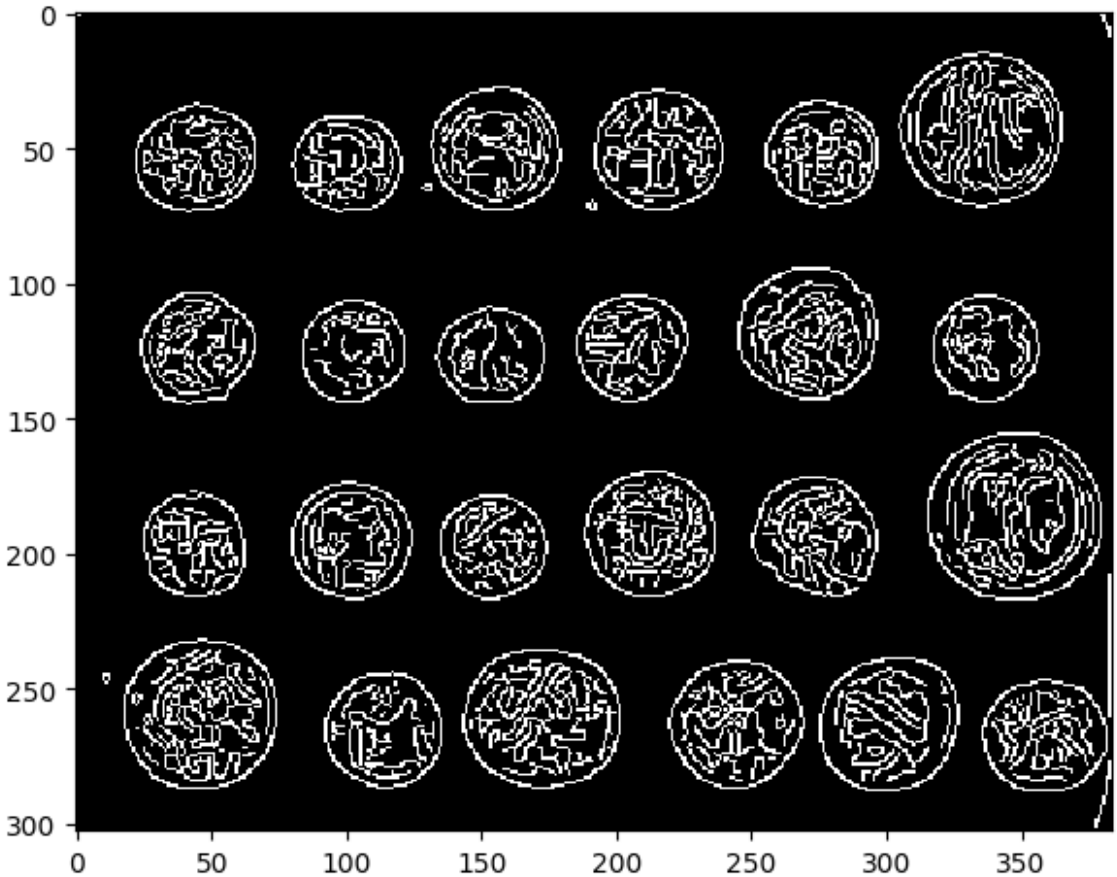
```
skimage.io.imshow(coins)
skimage.io.show()
```

```
/tmp/ipykernel_46346/863050958.py:1: FutureWarning: `imshow` is
deprecated since version 0.25 and will be removed in version 0.27.
Please use `matplotlib`, `napari`, etc. to visualize images.
  skimage.io.imshow(coins)
/tmp/ipykernel_46346/863050958.py:2: FutureWarning: `show` is
deprecated since version 0.25 and will be removed in version 0.27.
Please use `matplotlib`, `napari`, etc. to visualize images.
  skimage.io.show()
```



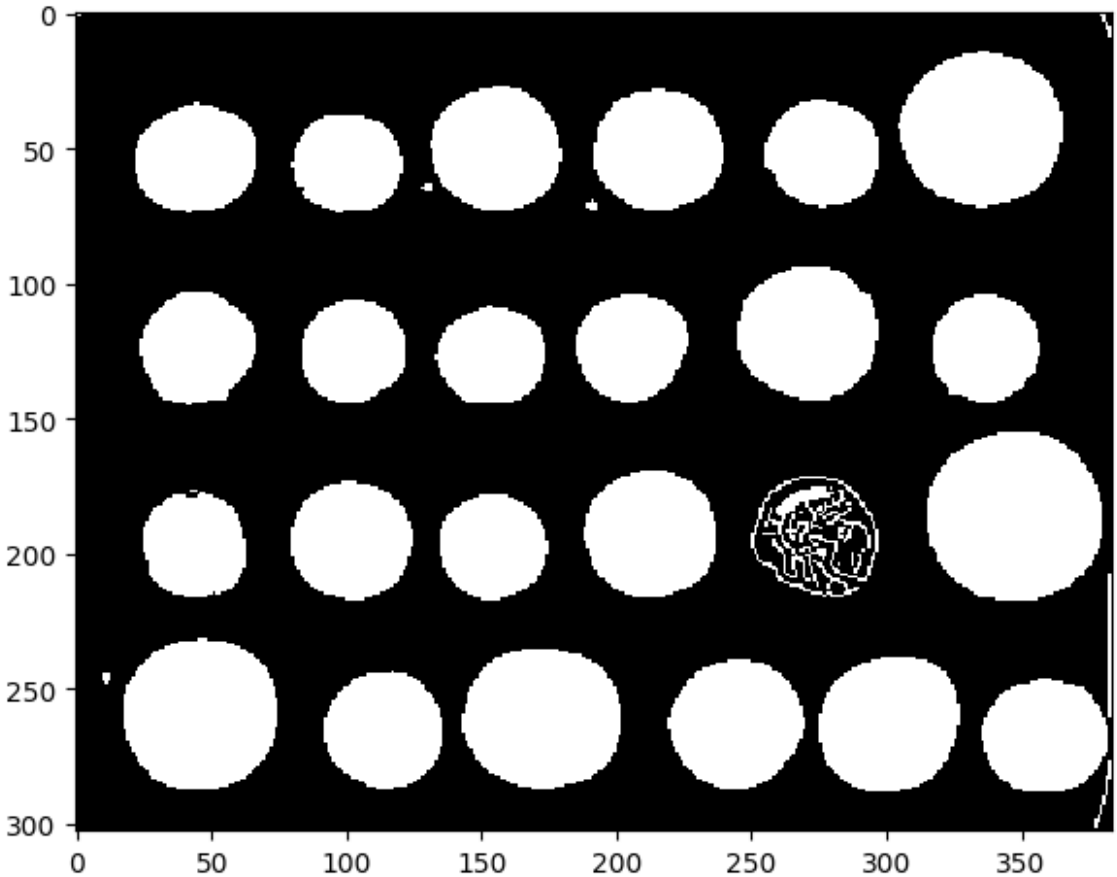

```
from skimage.feature import canny
edges = canny(coins/255.)
skimage.io.imshow(edges)
skimage.io.show()
```

```
/tmp/ipykernel_46346/51233653.py:3: FutureWarning: `imshow` is
deprecated since version 0.25 and will be removed in version 0.27.
Please use `matplotlib`, `napari`, etc. to visualize images.
  skimage.io.imshow(edges)
/tmp/ipykernel_46346/51233653.py:4: FutureWarning: `show` is
deprecated since version 0.25 and will be removed in version 0.27.
Please use `matplotlib`, `napari`, etc. to visualize images.
  skimage.io.show()
```




```
from scipy import ndimage as ndi
fill_holes = ndi.binary_fill_holes(edges)
skimage.io.imshow(fill_holes)
skimage.io.show()
```

```
/tmp/ipykernel_46346/3708109902.py:3: FutureWarning: `imshow` is
deprecated since version 0.25 and will be removed in version 0.27.
Please use `matplotlib`, `napari`, etc. to visualize images.
  skimage.io.imshow(fill_holes)
/tmp/ipykernel_46346/3708109902.py:4: FutureWarning: `show` is
deprecated since version 0.25 and will be removed in version 0.27.
Please use `matplotlib`, `napari`, etc. to visualize images.
  skimage.io.show()
```




```
label_objects, nb_labels = ndi.label(fill_holes)
sizes = np.bincount(label_objects.ravel())
mask_sizes = sizes > 20
mask_sizes[0] = 0
coins_cleaned = mask_sizes[label_objects]

from skimage import data, filters, measure

labels = measure.label(coins_cleaned)
num_objects = labels.max()
print(num_objects)
```

```
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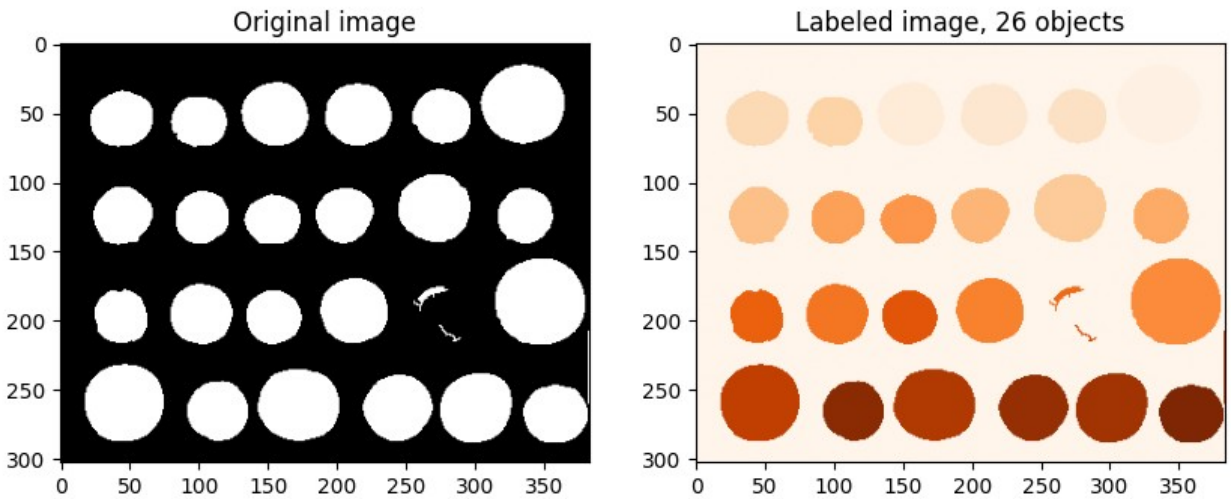
```

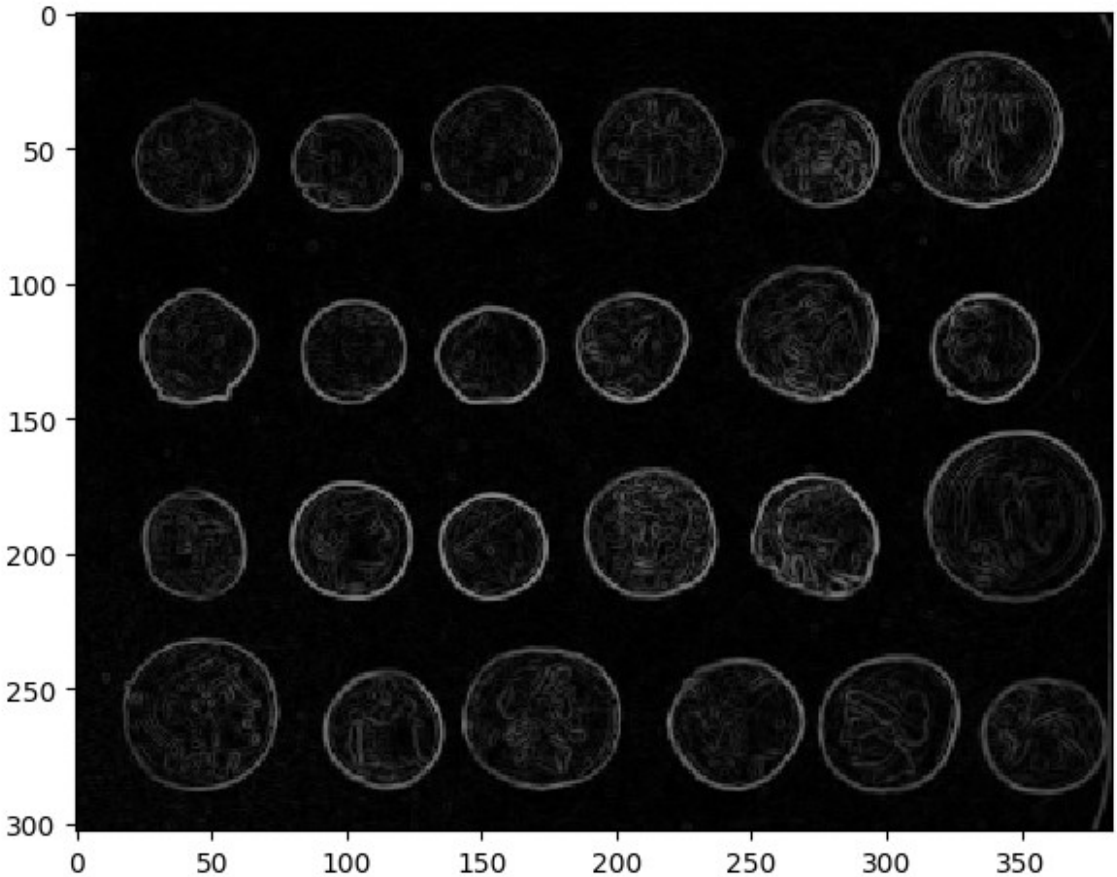
```
fig, ax = plt.subplots(ncols=2, figsize=(10, 5))
ax[0].imshow(coins_cleaned, cmap='gray')
ax[0].set_title('Original image')
ax[1].imshow(labels, cmap='Oranges')
ax[1].set_title('Labeled image, %d objects' % num_objects)
plt.show()
```



Watershed transform

```
from skimage.filters import sobel
elevation_map = sobel(coins)
skimage.io.imshow(elevation_map)
skimage.io.show()
```

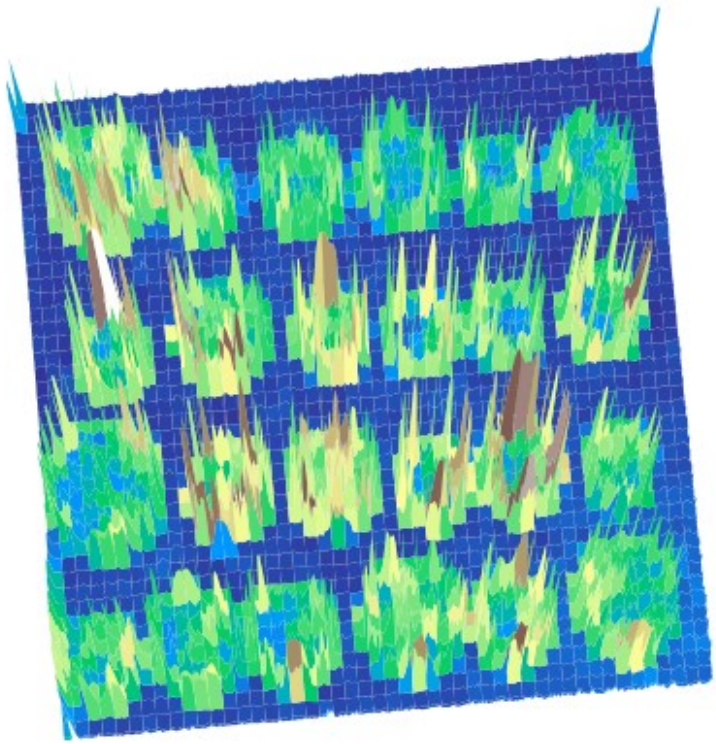
```
/tmp/ipykernel_46346/1119845918.py:3: FutureWarning: `imshow` is
deprecated since version 0.25 and will be removed in version 0.27.
Please use `matplotlib`, `napari`, etc. to visualize images.
  skimage.io.imshow(elevation_map)
/tmp/ipykernel_46346/1119845918.py:4: FutureWarning: `show` is
deprecated since version 0.25 and will be removed in version 0.27.
Please use `matplotlib`, `napari`, etc. to visualize images.
  skimage.io.show()
```




```
from mpl_toolkits.mplot3d import Axes3D

x = np.arange(elevation_map.shape[1])
y = np.arange(elevation_map.shape[0])
X, Y = np.meshgrid(x, y)
Z = elevation_map


fig = plt.figure(figsize=(8, 8))
ax = fig.add_subplot(111, projection='3d')
ax.plot_surface(X, Y, Z, cmap='terrain')
ax.view_init(elev=80, azim=85)
ax.set_proj_type('persp')
ax.set_axis_off()
plt.show()
```



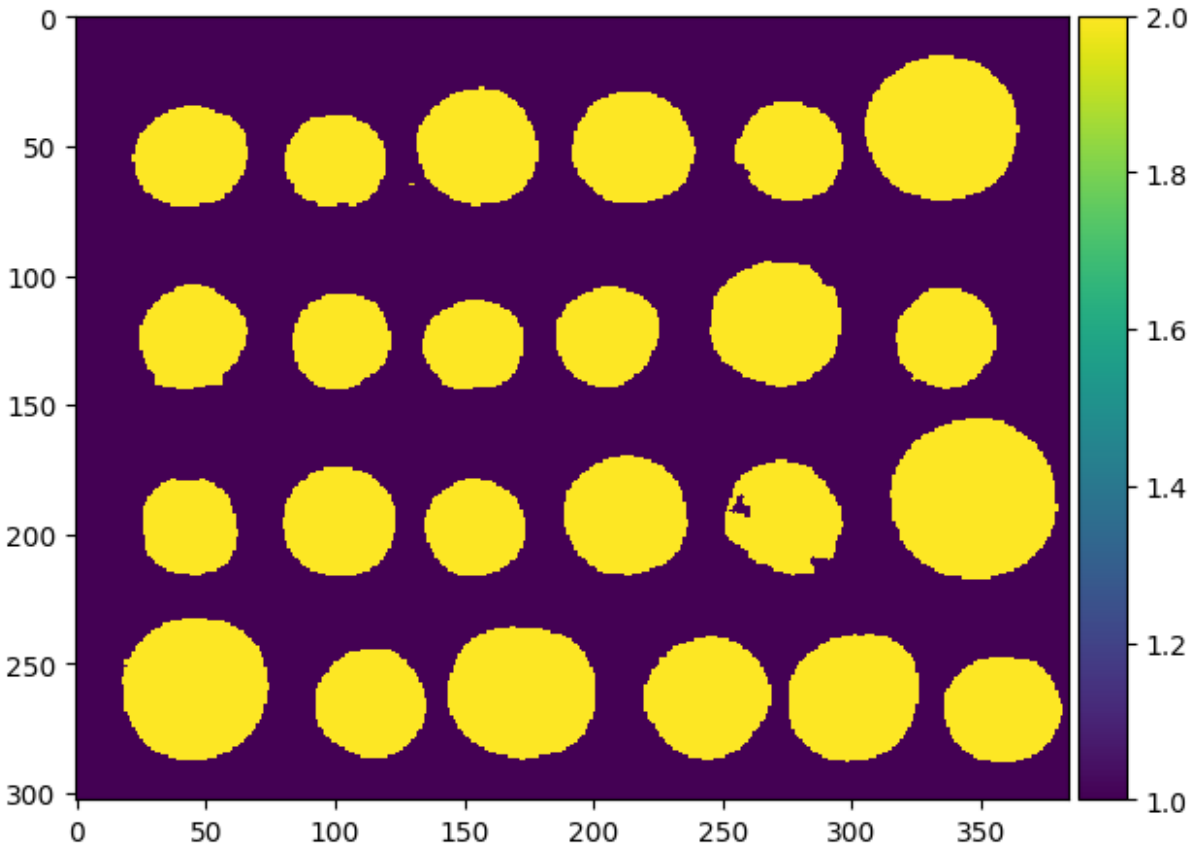
```
markers = np.zeros_like(coins)
markers[coins < 20] = 1
markers[coins > 150] = 2

from skimage.segmentation import watershed
segmentation = watershed(elevation_map, markers)

skimage.io.imshow(segmentation)
skimage.io.show()
```

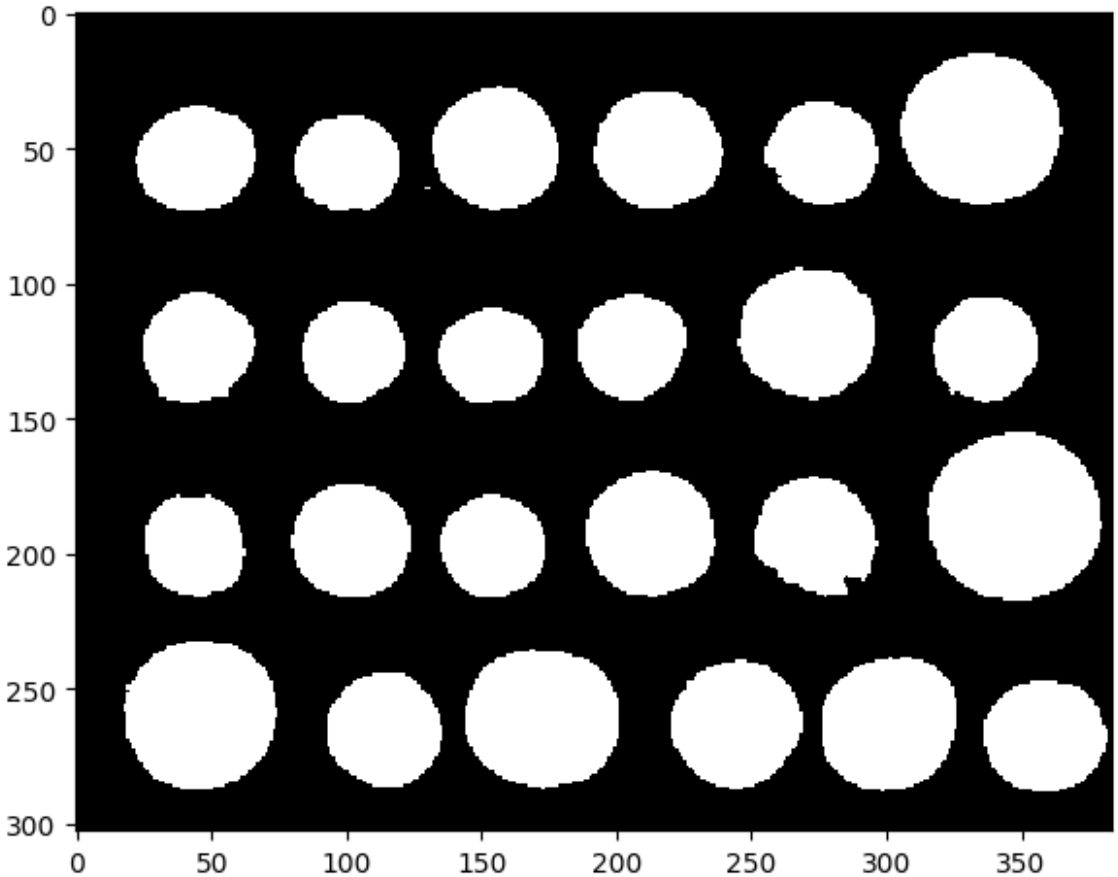
```
/tmp/ipykernel_46346/2516555142.py:1: FutureWarning: `imshow` is
deprecated since version 0.25 and will be removed in version 0.27.
Please use `matplotlib`, `napari`, etc. to visualize images.
```

```
  skimage.io.imshow(segmentation)
/usr/local/lib/python3.12/dist-packages/skimage/io/_plugins/matplotlib
_plugin.py:158: UserWarning: Low image data range; displaying image
with stretched contrast.
  lo, hi, cmap = _get_display_range(image)
/tmp/ipykernel_46346/2516555142.py:2: FutureWarning: `show` is
deprecated since version 0.25 and will be removed in version 0.27.
Please use `matplotlib`, `napari`, etc. to visualize images.
  skimage.io.show()
```




```
segmentation_filled = ndi.binary_fill_holes(segmentation-1)
skimage.io.imshow(segmentation_filled)
skimage.io.show()
```

```
/tmp/ipykernel_46346/2278032570.py:2: FutureWarning: `imshow` is
deprecated since version 0.25 and will be removed in version 0.27.
Please use `matplotlib`, `napari`, etc. to visualize images.
  skimage.io.imshow(segmentation_filled)
/tmp/ipykernel_46346/2278032570.py:3: FutureWarning: `show` is
deprecated since version 0.25 and will be removed in version 0.27.
Please use `matplotlib`, `napari`, etc. to visualize images.
  skimage.io.show()
```

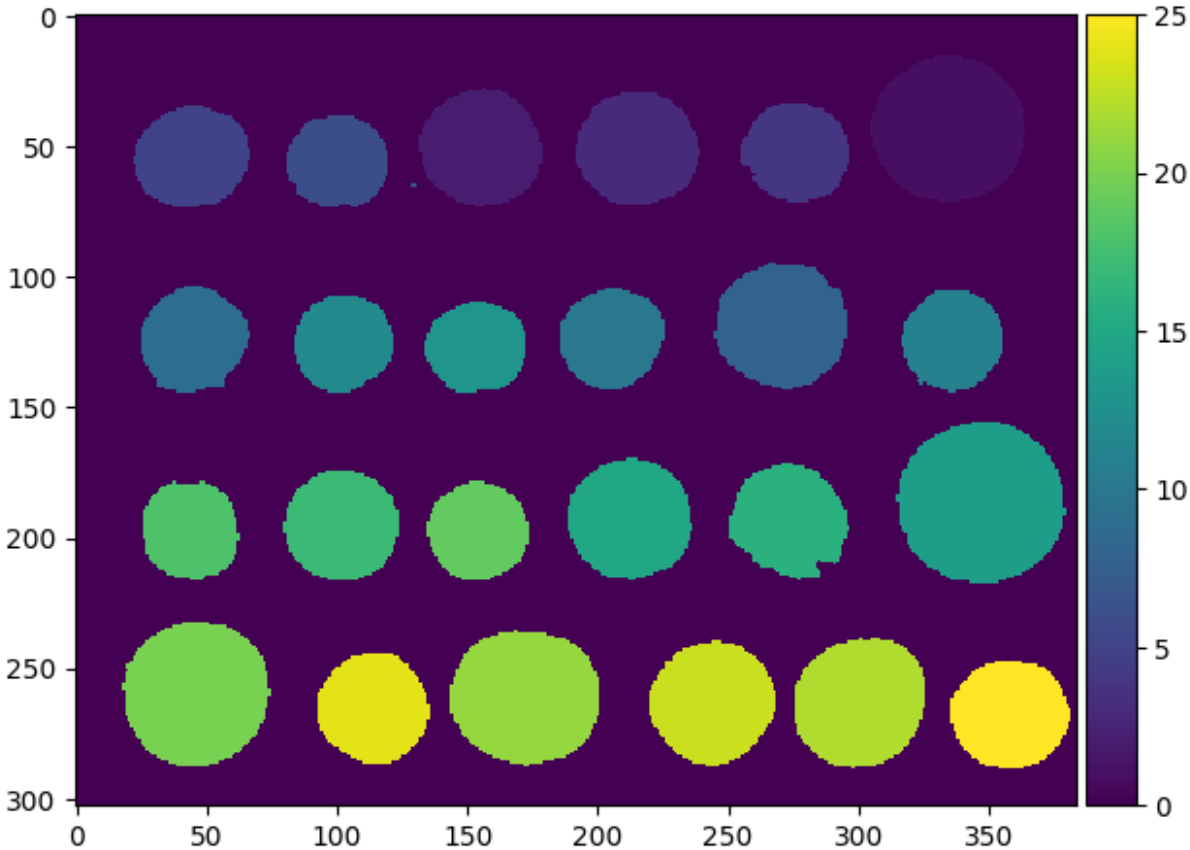



```
labels = measure.label(segmentation_filled)
num_objects = labels.max()
print(num_objects)
```

```
25
```

```
skimage.io.imshow(labels)
skimage.io.show()
```

```
/tmp/ipykernel_46346/1369540909.py:1: FutureWarning: `imshow` is
deprecated since version 0.25 and will be removed in version 0.27.
Please use `matplotlib`, `napari`, etc. to visualize images.
  skimage.io.imshow(labels)
/usr/local/lib/python3.12/dist-packages/skimage/io/_plugins/matplotlib
_plugin.py:158: UserWarning: Low image data range; displaying image
with stretched contrast.
  lo, hi, cmap = _get_display_range(image)
/tmp/ipykernel_46346/1369540909.py:2: FutureWarning: `show` is
deprecated since version 0.25 and will be removed in version 0.27.
Please use `matplotlib`, `napari`, etc. to visualize images.
  skimage.io.show()
```

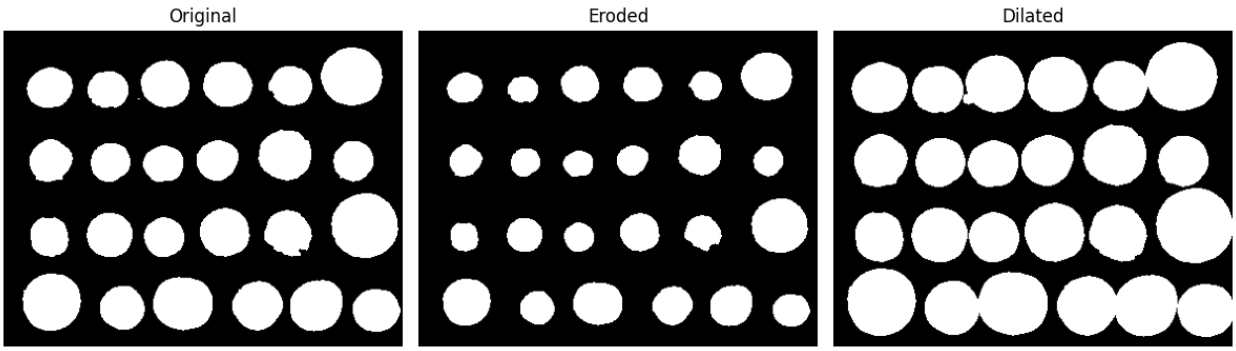


Operatii morfologice

```
from skimage import morphology

selem = morphology.disk(5)
eroded = morphology.binary_erosion(segmentation_filled, selem)
dilated = morphology.binary_dilation(segmentation_filled, selem)


fig, axes = plt.subplots(nrows=1, ncols=3, figsize=(12, 6))
ax = axes.ravel()
ax[0].imshow(segmentation_filled, cmap='gray')
ax[0].set_title('Original')
ax[1].imshow(eroded, cmap='gray')
ax[1].set_title('Eroded')
ax[2].imshow(dilated, cmap='gray')
ax[2].set_title('Dilated')
for a in ax:
    a.axis('off')
plt.tight_layout()
plt.show()
```

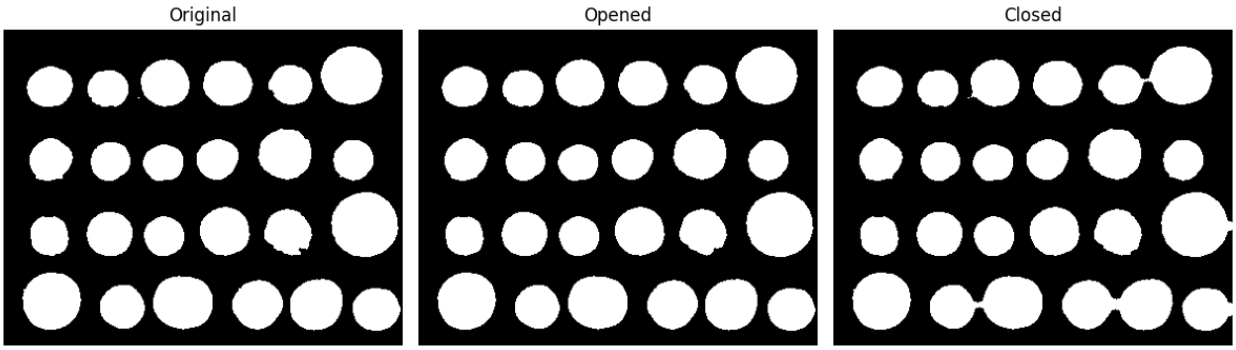


Closing = dilation + erosion

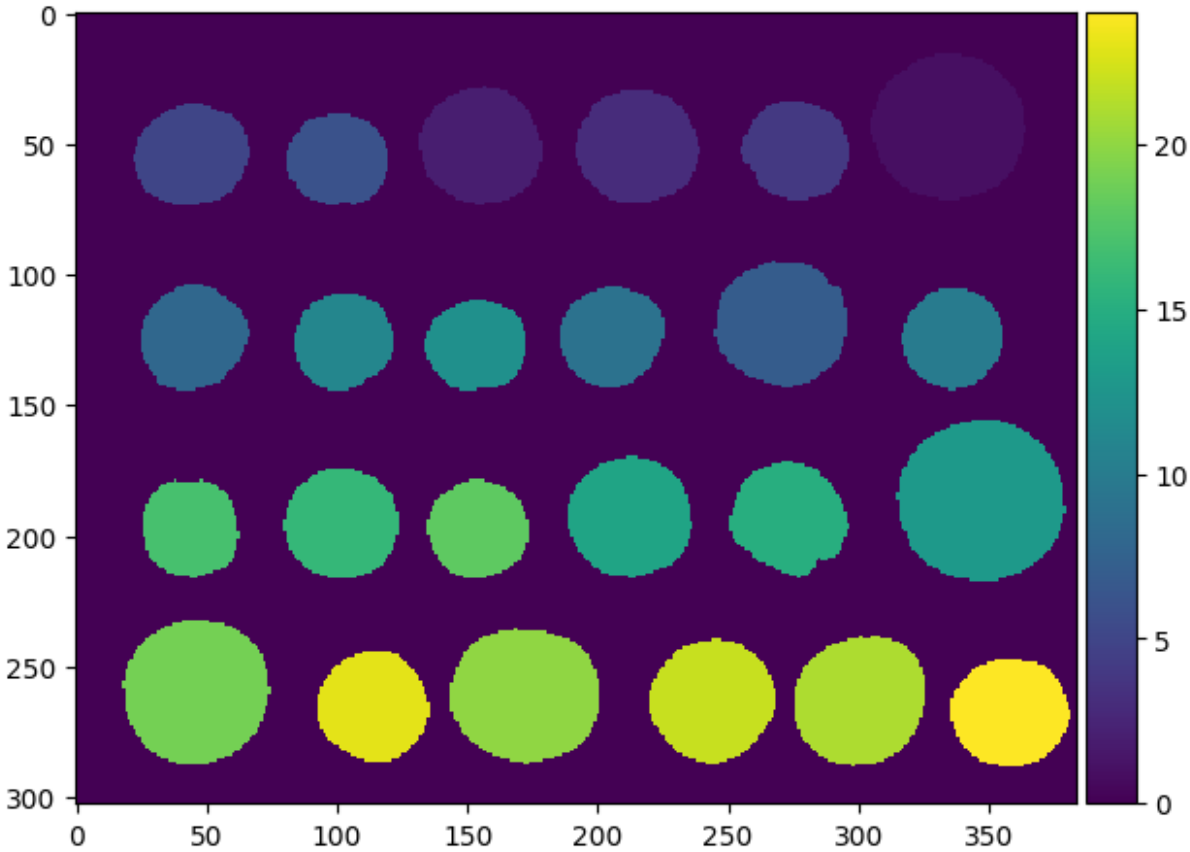
Opening = erosion + dilation

```
opened = morphology.binary_opening(segmentation_filled, selem)
closed = morphology.binary_closing(segmentation_filled, selem)

fig, axes = plt.subplots(nrows=1, ncols=3, figsize=(12, 6))
ax = axes.ravel()
ax[0].imshow(segmentation_filled, cmap='gray')
ax[0].set_title('Original')
ax[1].imshow(opened, cmap='gray')
ax[1].set_title('Opened')
ax[2].imshow(closed, cmap='gray')
ax[2].set_title('Closed')
for a in ax:
    a.axis('off')
plt.tight_layout()
plt.show()
```




```
labels_opened = measure.label(opened)
num_objects = labels.max()
print(num_objects)
```

```
25
```

```
skimage.io.imshow(labels_opened)
skimage.io.show()
```